

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                             feats_to_balance + feats_grouping +
                             feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df

def import_data_IR():
    if target_col[0] == "i_NH3":
```

```

        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

df = pd.read_csv(file_path, sep=";")
df["datetime"] = pd.to_datetime(df["datetime"])
df.sort_values("datetime").reset_index(drop=True)

if target_col[0] == "i_NH3":
    df["campaign"] = df["datetime"].apply(assign_campaign)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

```

```

# if camp == "2025_Jun":
#     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
# elif camp == "2025_Sep":
#     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
if camp == "2025_Apr":
    #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
    #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
    mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
elif camp == "2025_Jun":
    mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
elif camp == "2025_Sep":
    mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : campaign, column

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : campaign, column

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

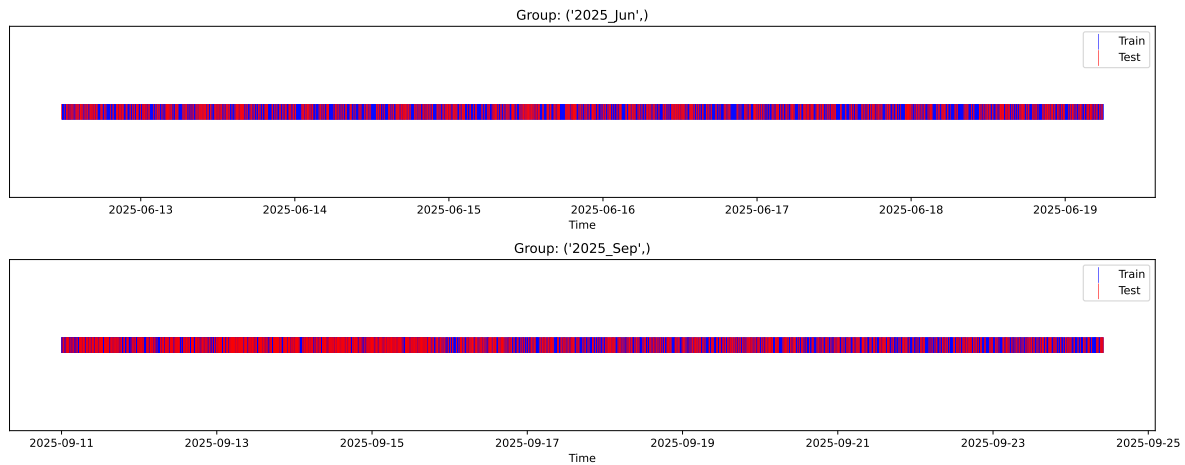
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

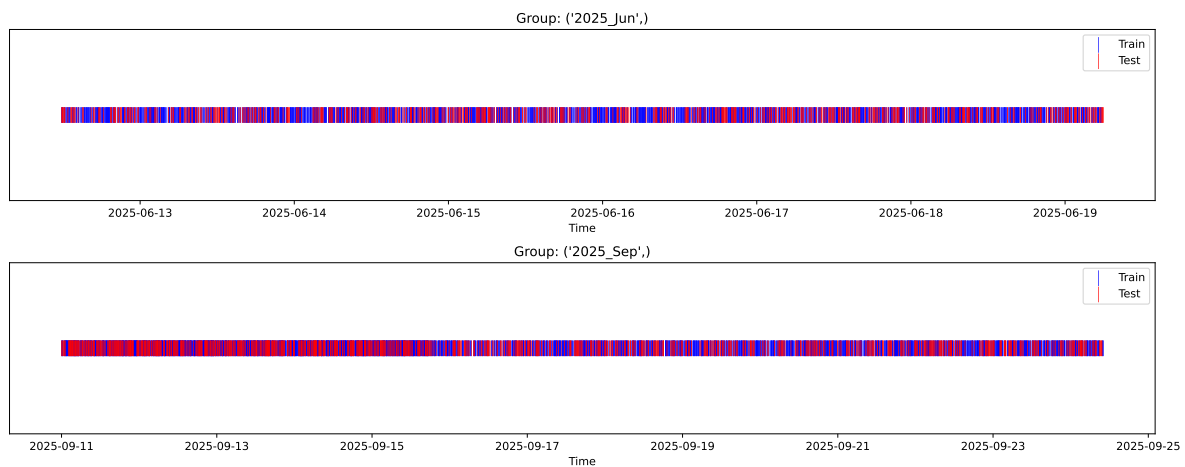
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

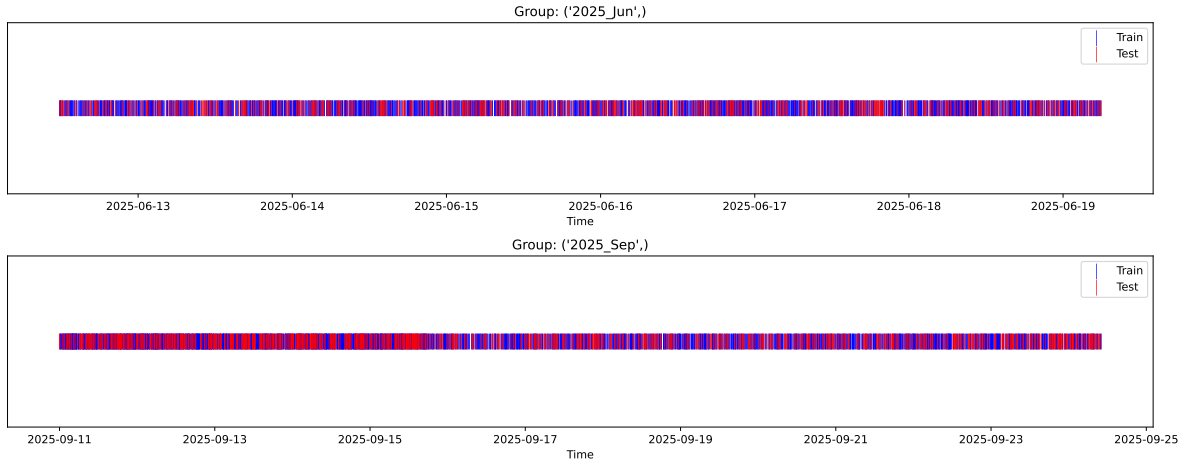
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



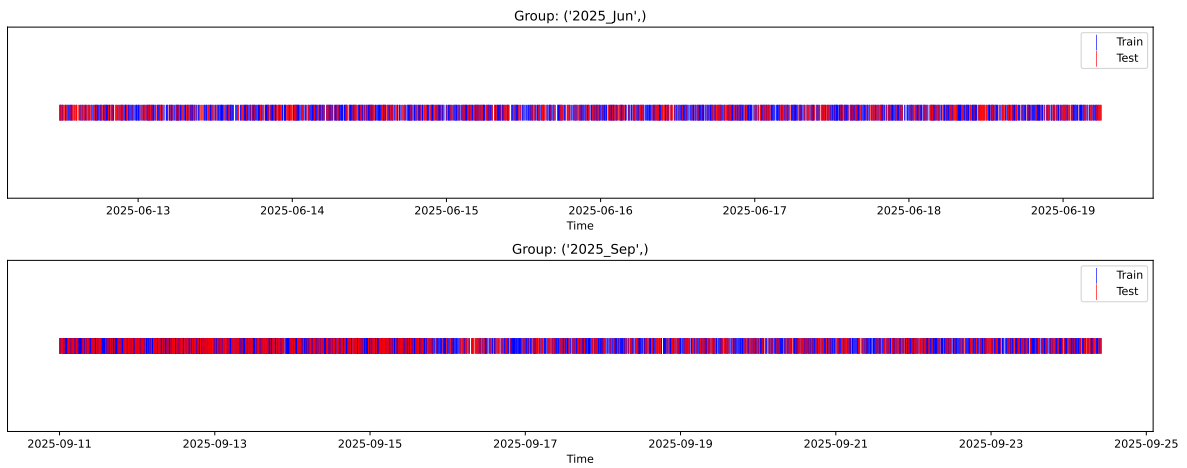
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



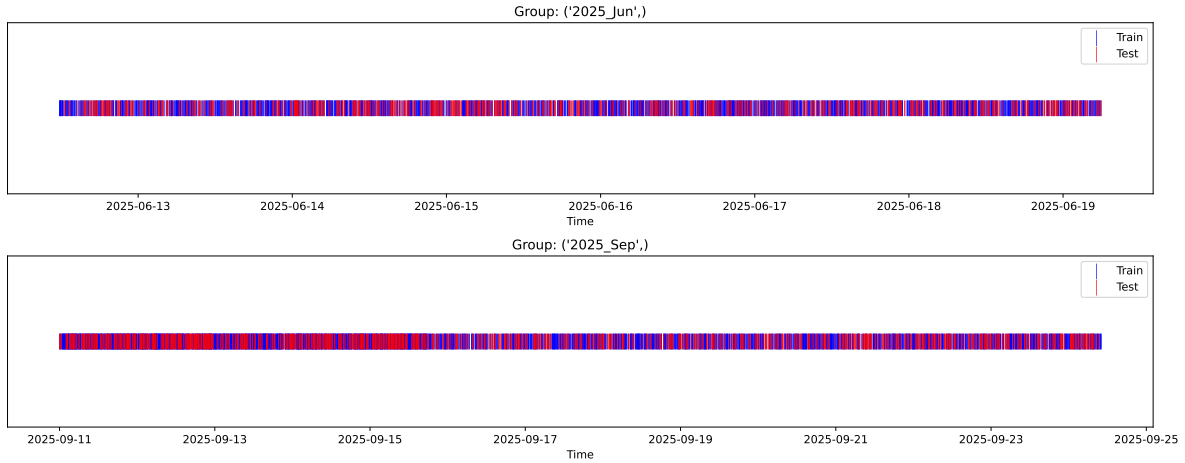
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



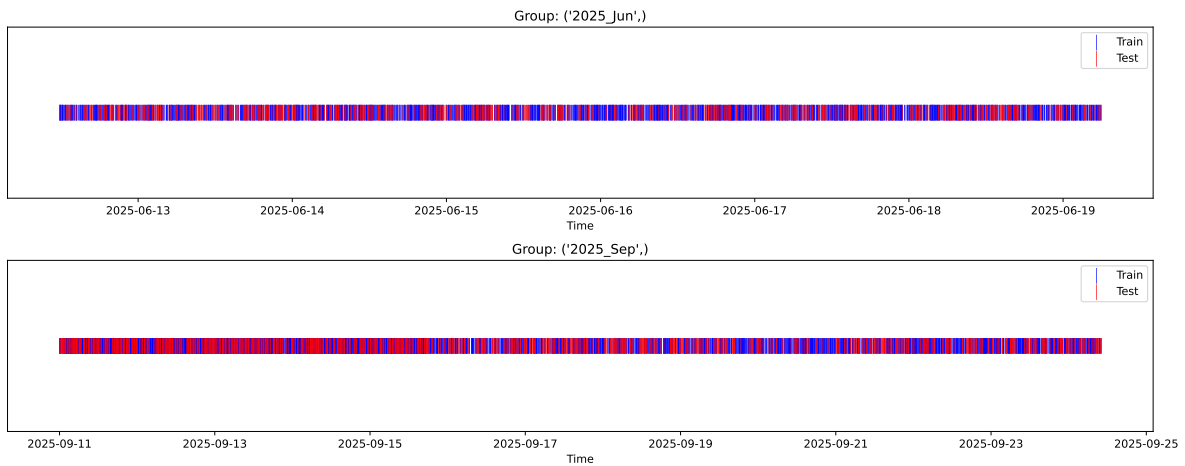
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 2025_Jun_1

Len(group) : 1585, Len(training) : 1110, Len(test) : 475

Distribution y_train vs y_test: y_train: mean=2.772, std=0.805, min=1.432, max=13.845
y_test: mean=2.807, std=0.764, min=1.373, max=8.682

Fold 0 — y_test: mean=2.75, std=0.70, y_train: mean=2.78, std=0.85

Fold 1 — y_test: mean=2.79, std=0.69, y_train: mean=2.77, std=0.85

Fold 2 — y_test: mean=2.77, std=0.74, y_train: mean=2.77, std=0.83

Fold 3 — y_test: mean=2.80, std=0.70, y_train: mean=2.76, std=0.85

Fold 4 — y_test: mean=2.77, std=0.77, y_train: mean=2.77, std=0.82

train scores CV: [0.33880003 0.37113528 0.3409849 0.35093461 0.3337788]

test scores CV: [0.45309325 0.33756815 0.43639596 0.41255846 0.44433302]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3471	0.4168	0.0418	0.3937	0.396	-0.0208	0.9501	0.8329

Coefficients:

coef_r_CO2 = -0.0889

coef_r_NH3 = -0.2349

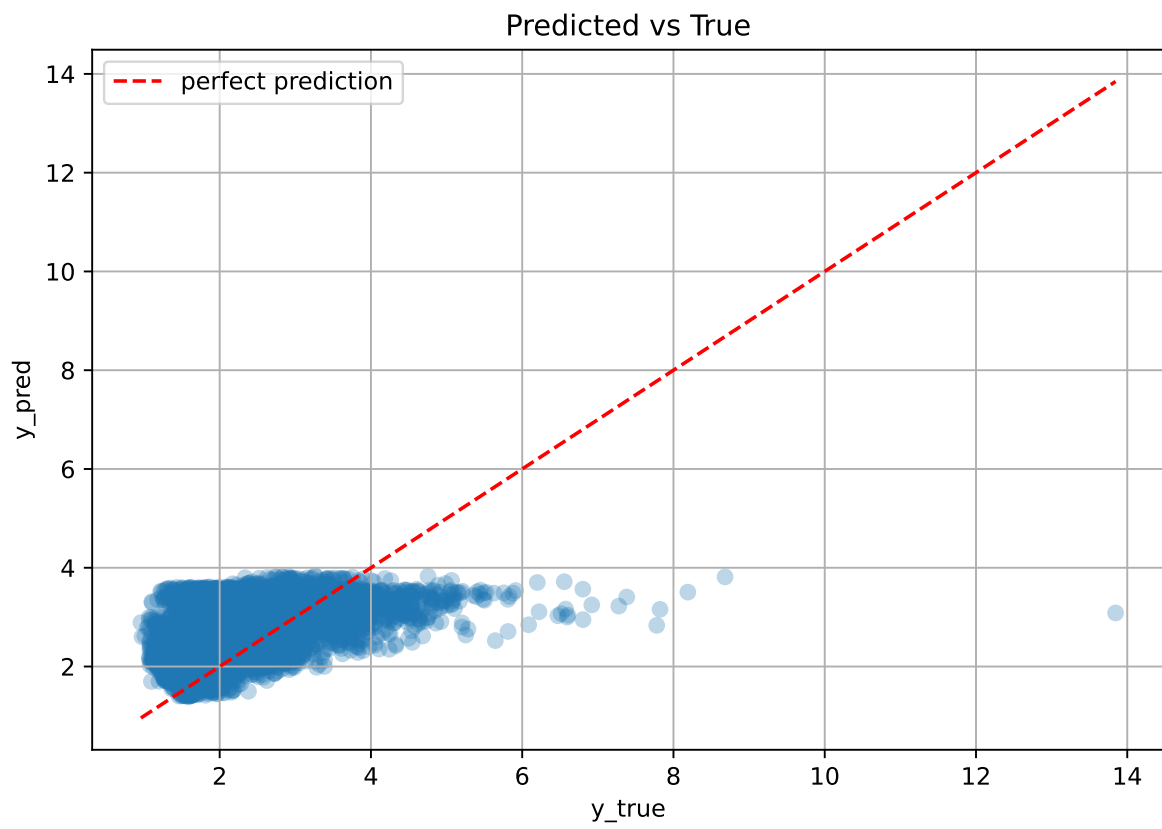
coef_r_THI = 0.8022

coef_r_temperature = -0.6449

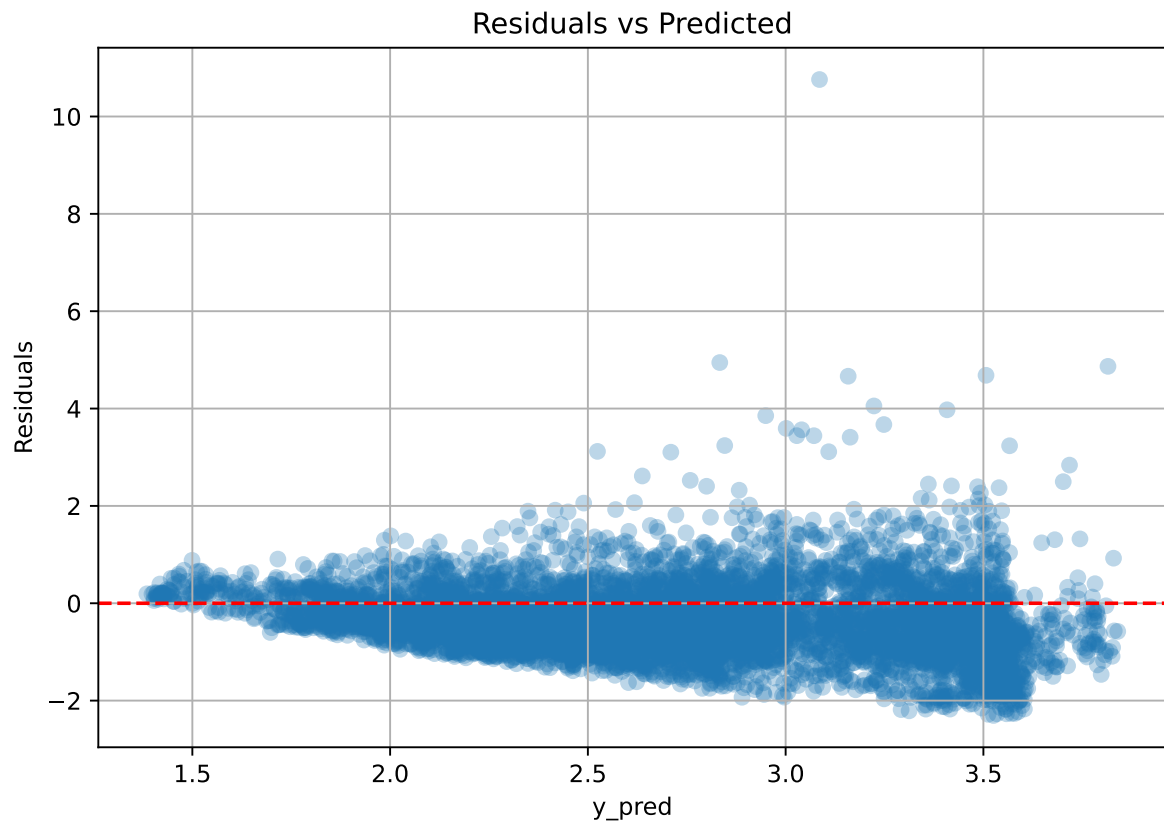
coef_r_umidity = -0.3798

intercept = 2.7723

2.3.2 Scatter



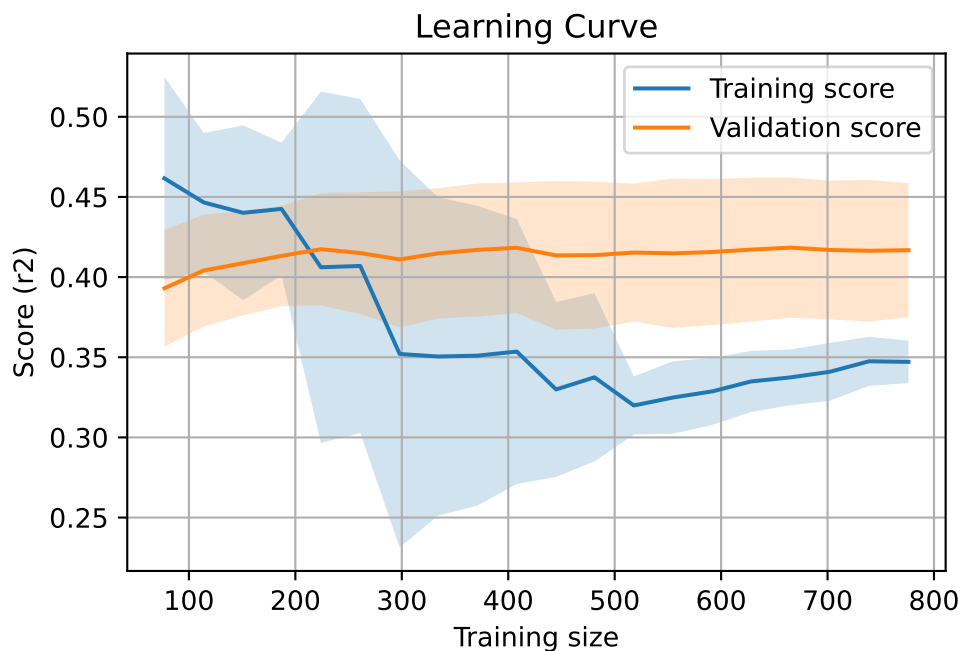
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 2025_Jun_1

`Len(group)` : 1585, `Len(training)` : 1110,

`Len(training_real)` : 777, `Len(test_of_training)` : 333, `Len(test)` : 475



2.3.5 Gridsearch — Group: 2025_Jun_2

Len(group) : 1587, Len(training) : 1111, Len(test) : 476

Distribution y_train vs y_test: y_train: mean=2.509, std=0.675, min=1.253, max=6.222
y_test: mean=2.528, std=0.697, min=1.305, max=7.383

Fold 0 — y_test: mean=2.46, std=0.68, y_train: mean=2.53, std=0.67

Fold 1 — y_test: mean=2.51, std=0.69, y_train: mean=2.51, std=0.67

Fold 2 — y_test: mean=2.56, std=0.67, y_train: mean=2.49, std=0.67

Fold 3 — y_test: mean=2.46, std=0.67, y_train: mean=2.53, std=0.67

Fold 4 — y_test: mean=2.50, std=0.68, y_train: mean=2.51, std=0.67

train scores CV: [0.22810899 0.20774064 0.22069569 0.22214819 0.21156576]

test scores CV: [0.17360107 0.23547914 0.19859637 0.1954499 0.22480749]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.2181	0.2056	0.0221	0.2163	0.2167	0.0111	1.054	1.0606

Coefficients:

coef_r_CO2 = -0.0258

coef_r_NH3 = 0.0226

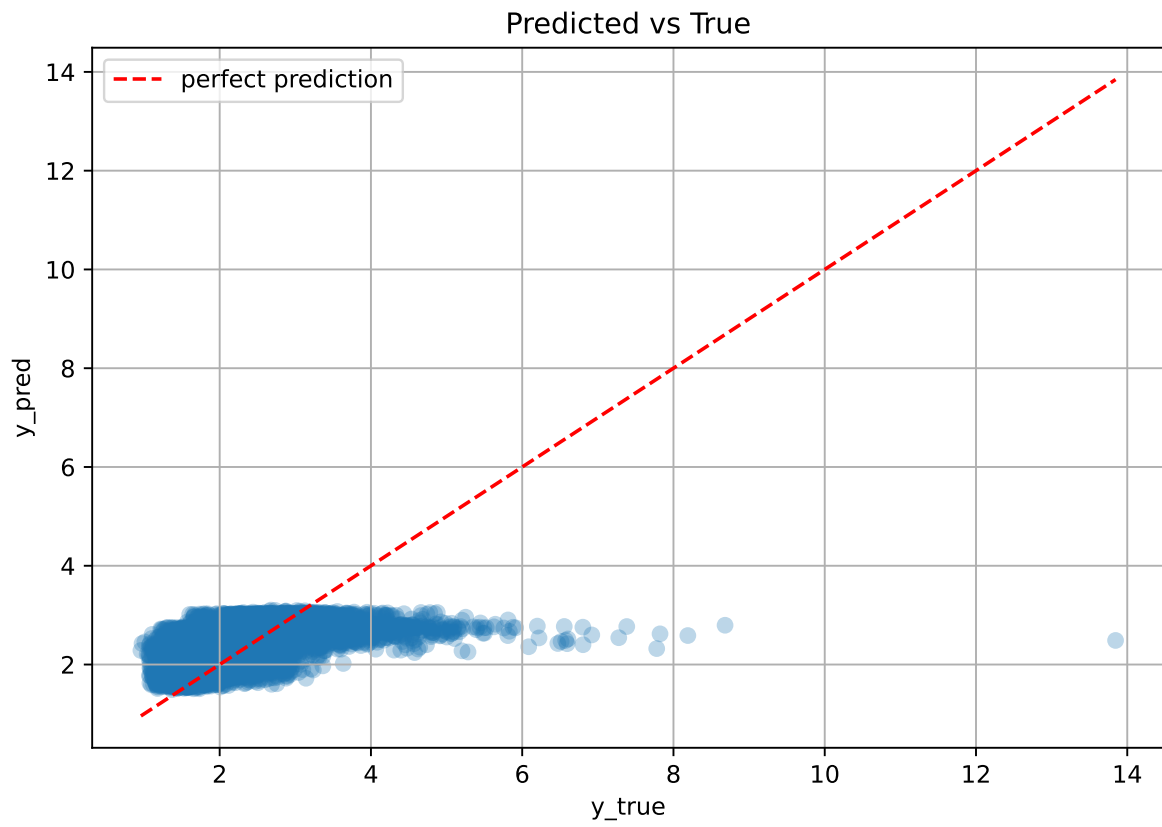
coef_r_THI = 1.0313

coef_r_temperature = -1.2395

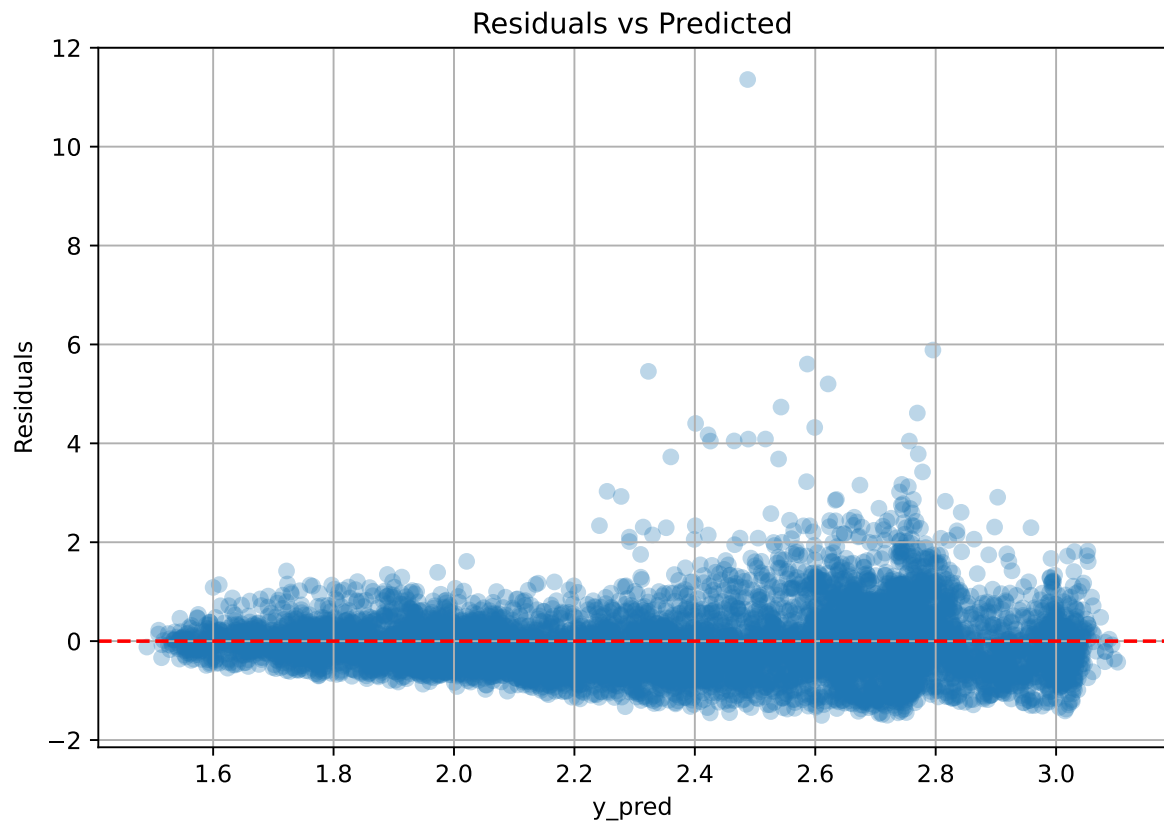
coef_r_umidity = -0.5903

intercept = 2.5090

2.3.6 Scatter



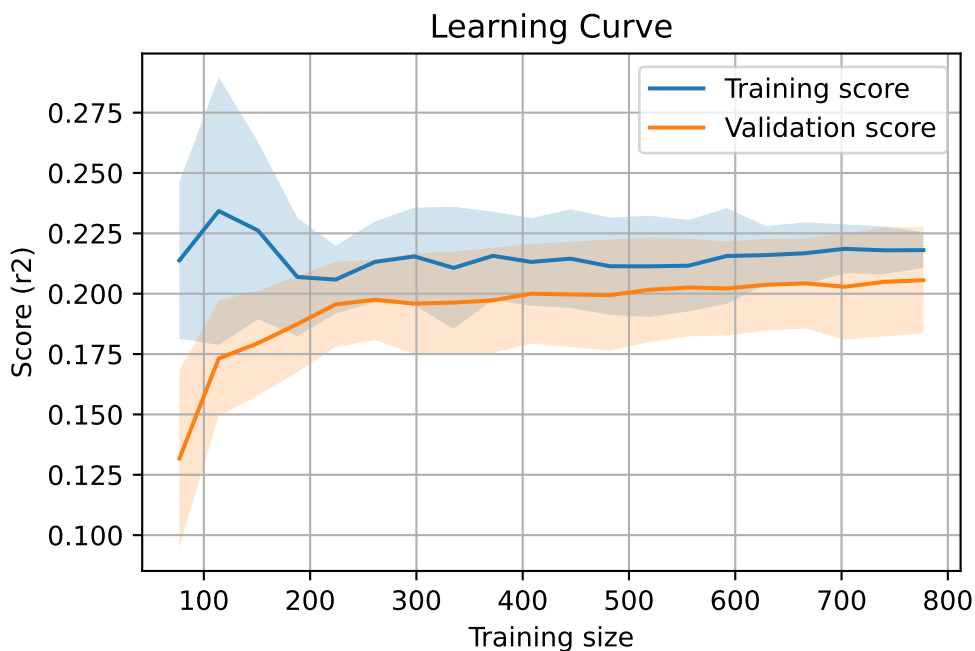
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2025_Jun_2

`Len(group)` : 1587, `Len(training)` : 1111,

`Len(training_real)` : 778, `Len(test_of_training)` : 333, `Len(test)` : 476



2.3.9 Gridsearch — Group: 2025_Sep_1

Len(group) : 3867, Len(training) : 2707, Len(test) : 1160

Distribution y_train vs y_test: y_train: mean=2.231, std=0.761, min=0.973, max=8.190
y_test: mean=2.216, std=0.723, min=1.110, max=6.920

Fold 0 — y_test: mean=2.22, std=0.76, y_train: mean=2.23, std=0.76

Fold 1 — y_test: mean=2.24, std=0.79, y_train: mean=2.23, std=0.75

Fold 2 — y_test: mean=2.24, std=0.77, y_train: mean=2.23, std=0.76

Fold 3 — y_test: mean=2.20, std=0.71, y_train: mean=2.25, std=0.78

Fold 4 — y_test: mean=2.25, std=0.84, y_train: mean=2.22, std=0.73

train scores CV: [0.41726736 0.44420839 0.41962722 0.41860333 0.42885578]

test scores CV: [0.45204131 0.39303028 0.44590827 0.44993845 0.4231198]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4257	0.4328	0.0224	0.4492	0.4498	0.017	1.0393	0.9836

Coefficients:

$\text{coef_r_CO2} = -0.0715$

$\text{coef_r_NH3} = -0.1378$

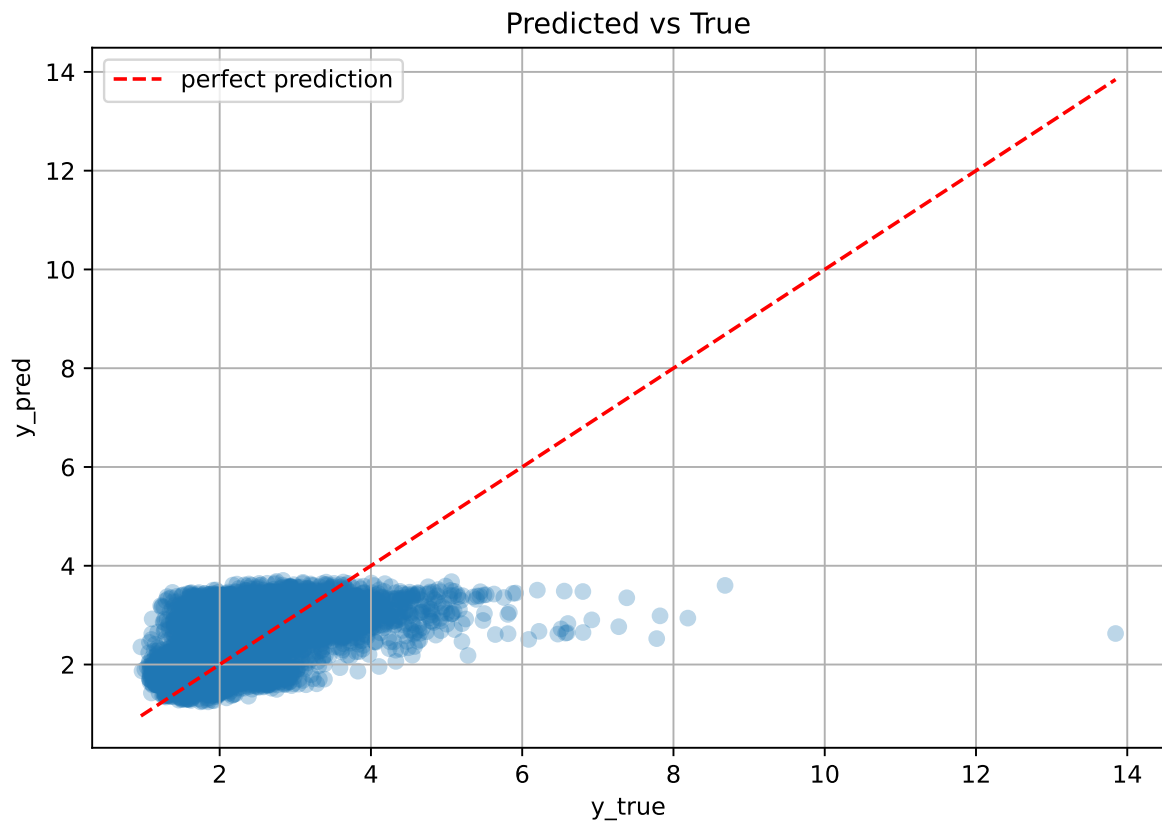
$\text{coef_r_THI} = 0.5493$

$\text{coef_r_temperature} = -0.6546$

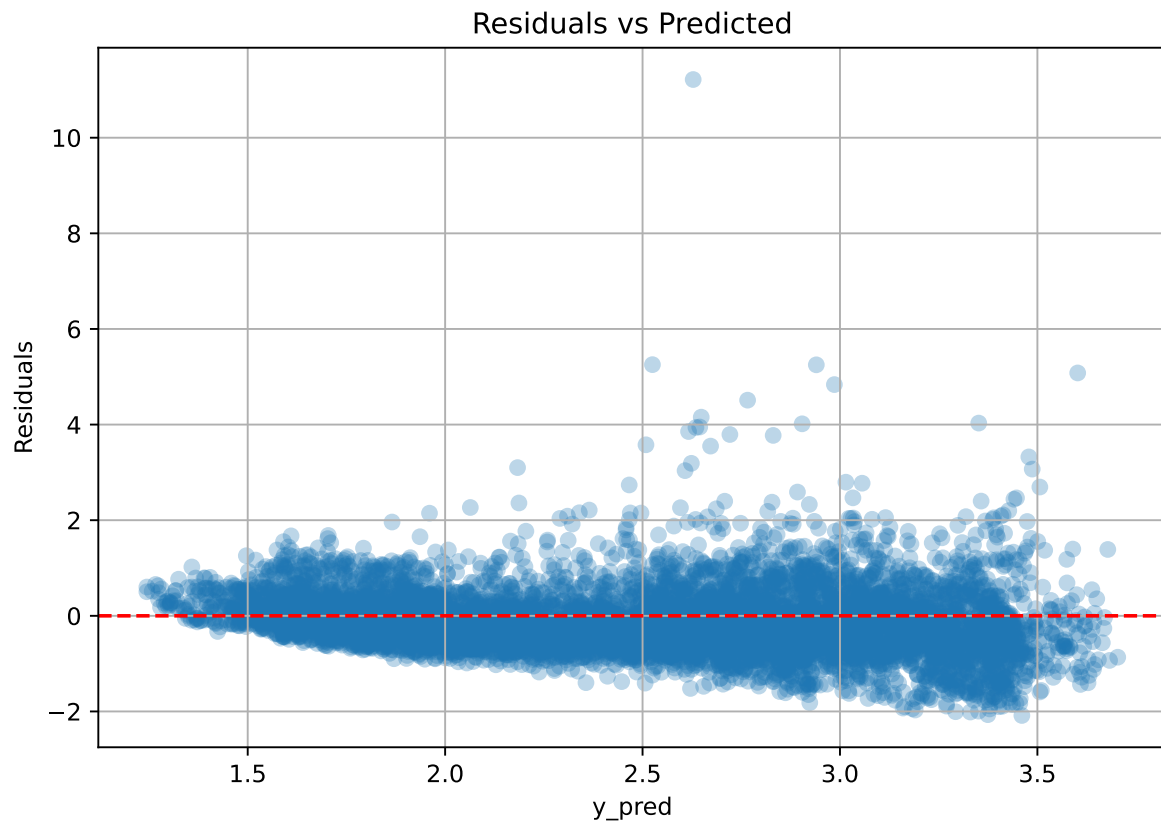
$\text{coef_r_umidity} = -0.6480$

$\text{intercept} = 2.2310$

2.3.10 Scatter



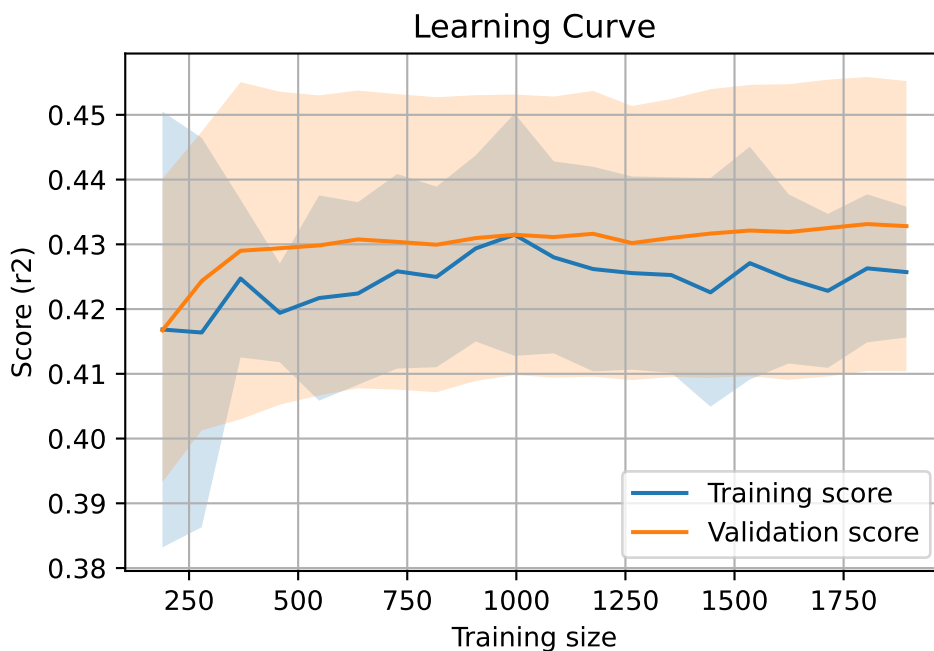
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 2025_Sep_1

`Len(group)` : 3867, `Len(training)` : 2707,

`Len(training_real)` : 1895, `Len(test_of_training)` : 812, `Len(test)` : 1160



2.3.13 Gridsearch — Group: 2025_Sep_2

Len(group) : 1129, Len(training) : 791, Len(test) : 338

Distribution y_train vs y_test: y_train: mean=1.926, std=0.742, min=0.959, max=6.804
y_test: mean=1.945, std=0.805, min=1.070, max=7.822

Fold 0 — y_test: mean=1.85, std=0.64, y_train: mean=1.96, std=0.78

Fold 1 — y_test: mean=1.95, std=0.82, y_train: mean=1.91, std=0.71

Fold 2 — y_test: mean=1.92, std=0.77, y_train: mean=1.93, std=0.73

Fold 3 — y_test: mean=1.88, std=0.76, y_train: mean=1.94, std=0.73

Fold 4 — y_test: mean=1.89, std=0.68, y_train: mean=1.94, std=0.77

train scores CV: [0.37483985 0.38789261 0.38559975 0.37728724 0.3613433]

test scores CV: [0.33651168 0.33696129 0.33543488 0.352859 0.39298824]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3774	0.351	0.022	0.3741	0.3745	0.0235	1.067	1.0753

Coefficients:

coef_r_CO2 = 0.0043

coef_r_NH3 = -0.0467

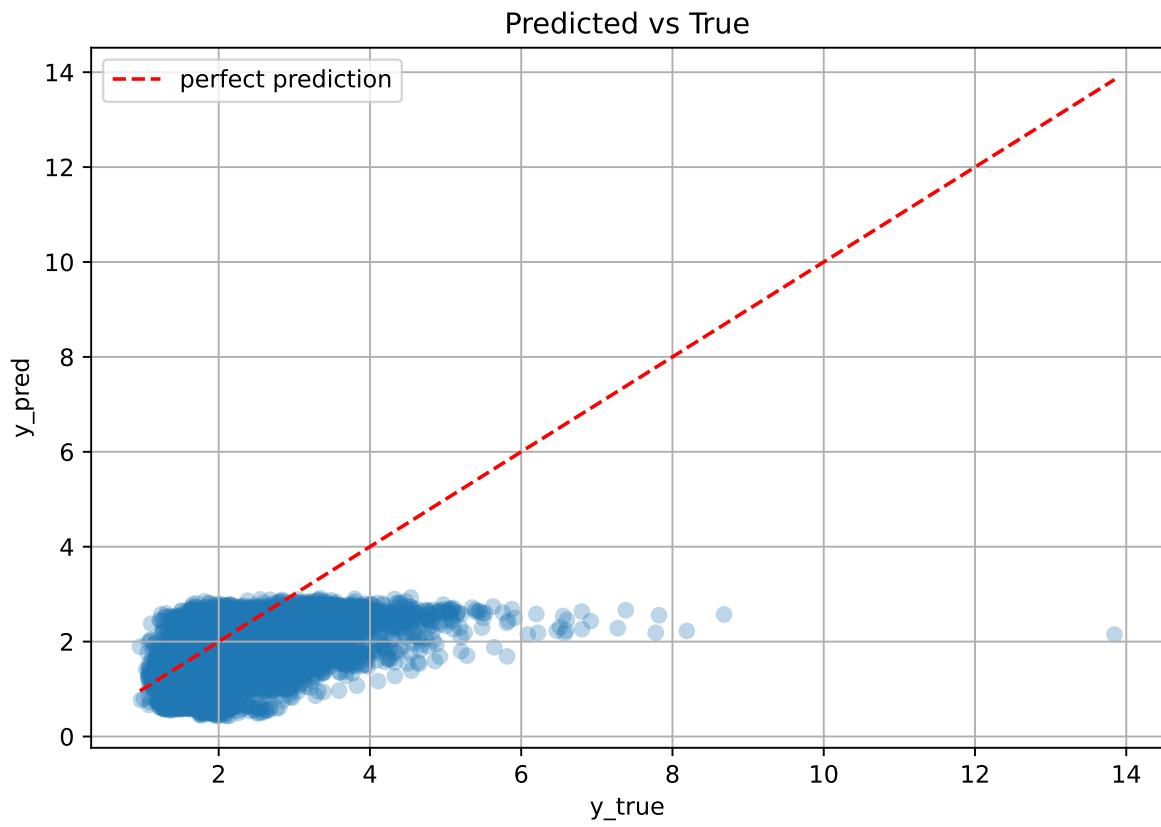
coef_r_THI = 1.3504

coef_r_temperature = -1.8574

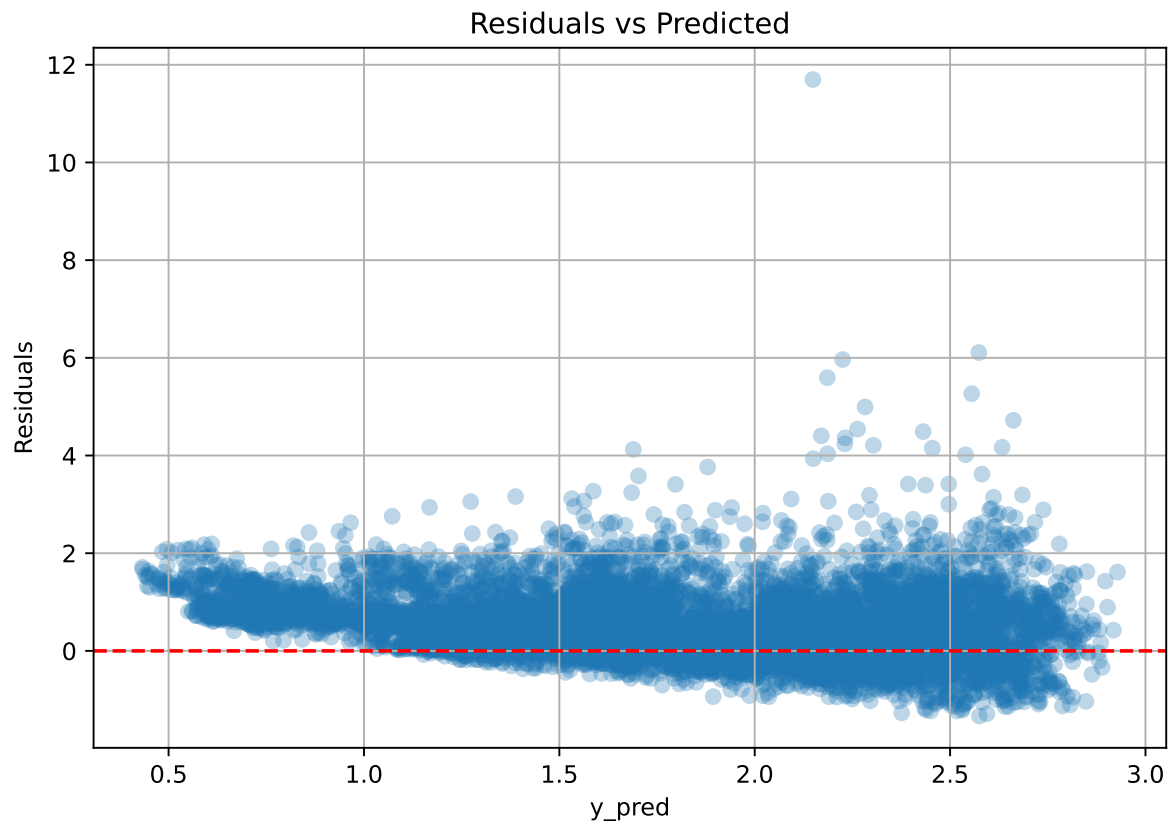
coef_r_umidity = -0.9884

intercept = 1.9256

2.3.14 Scatter



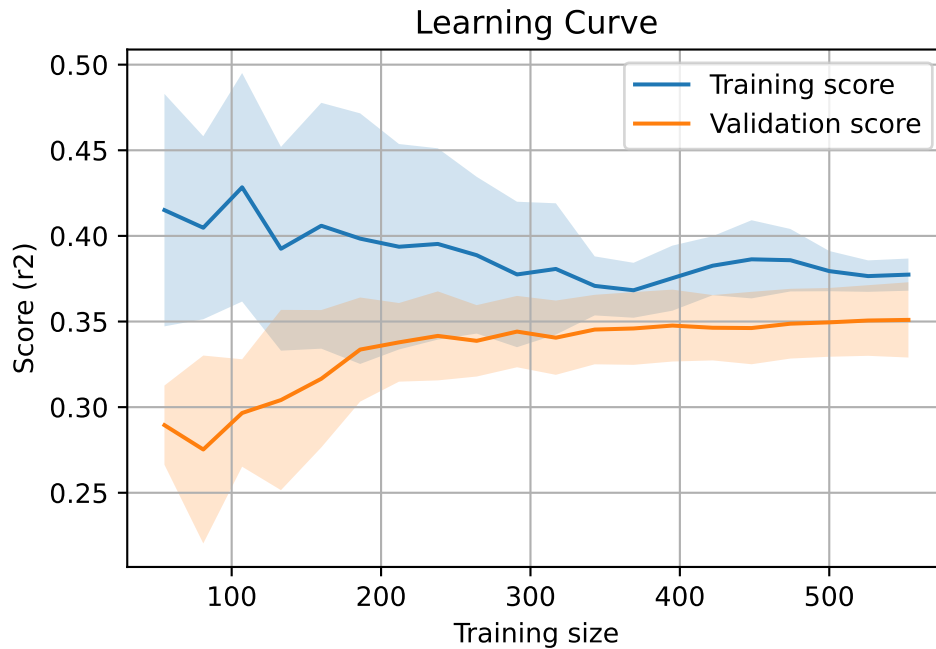
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 2025_Sep_2

`Len(group)` : 1129, `Len(training)` : 791,

`Len(training_real)` : 554, `Len(test_of_training)` : 237, `Len(test)` : 338



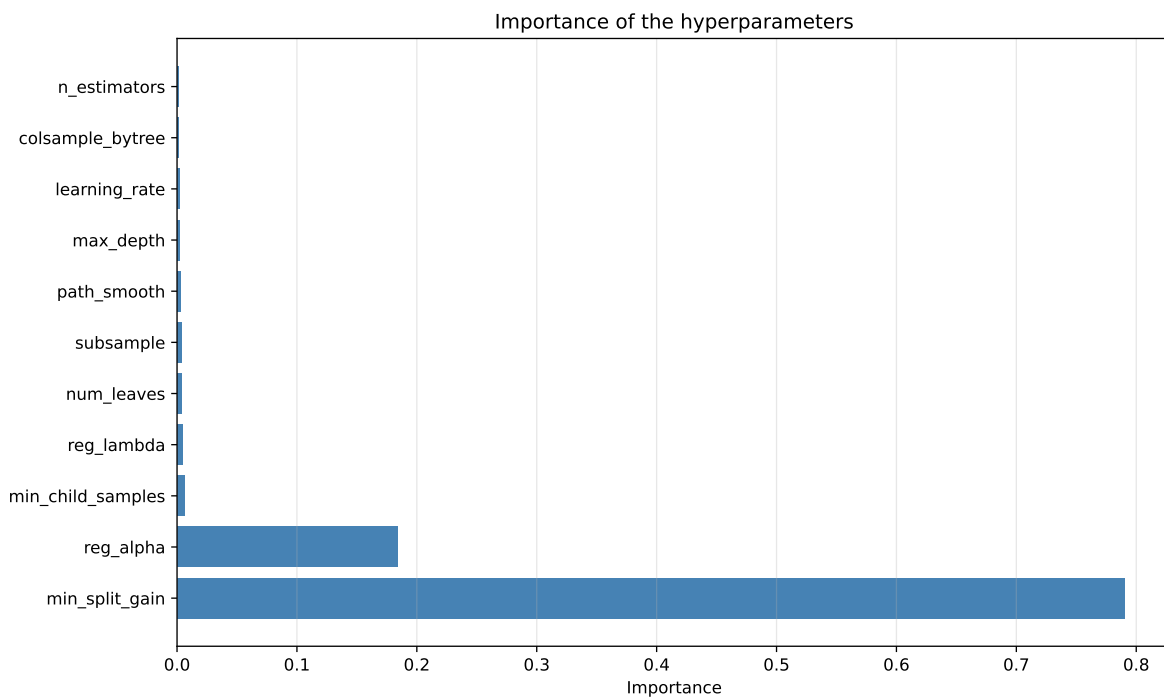
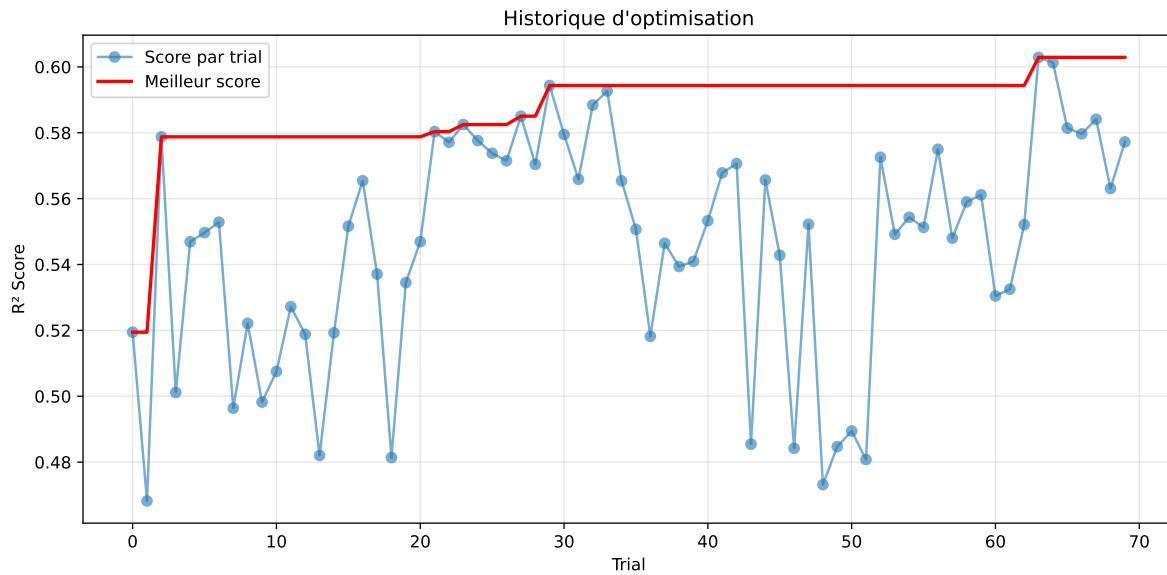
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 2025_Jun_1

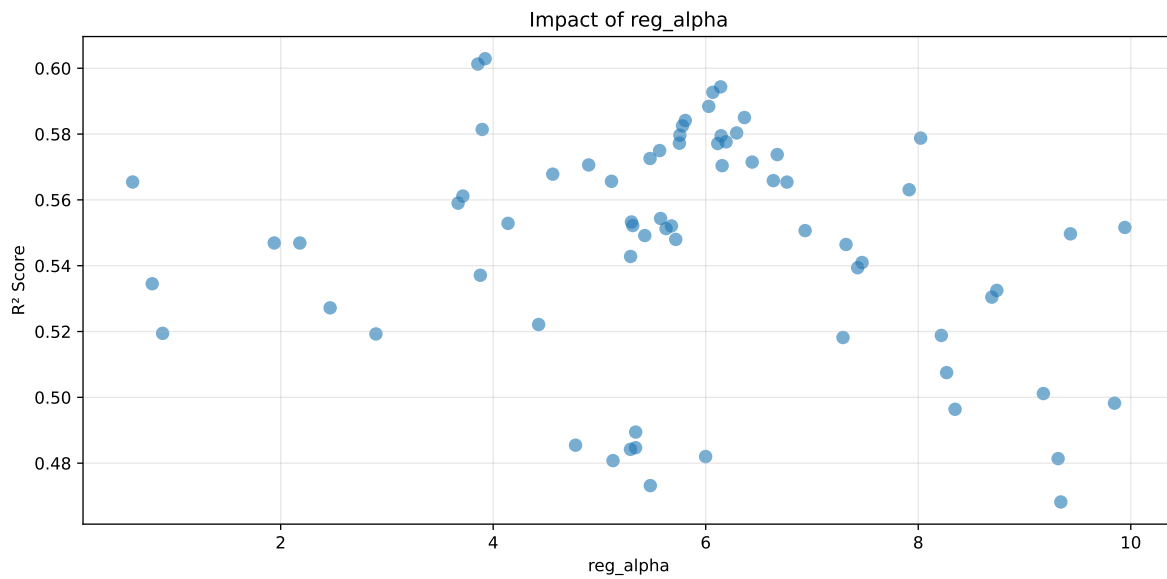
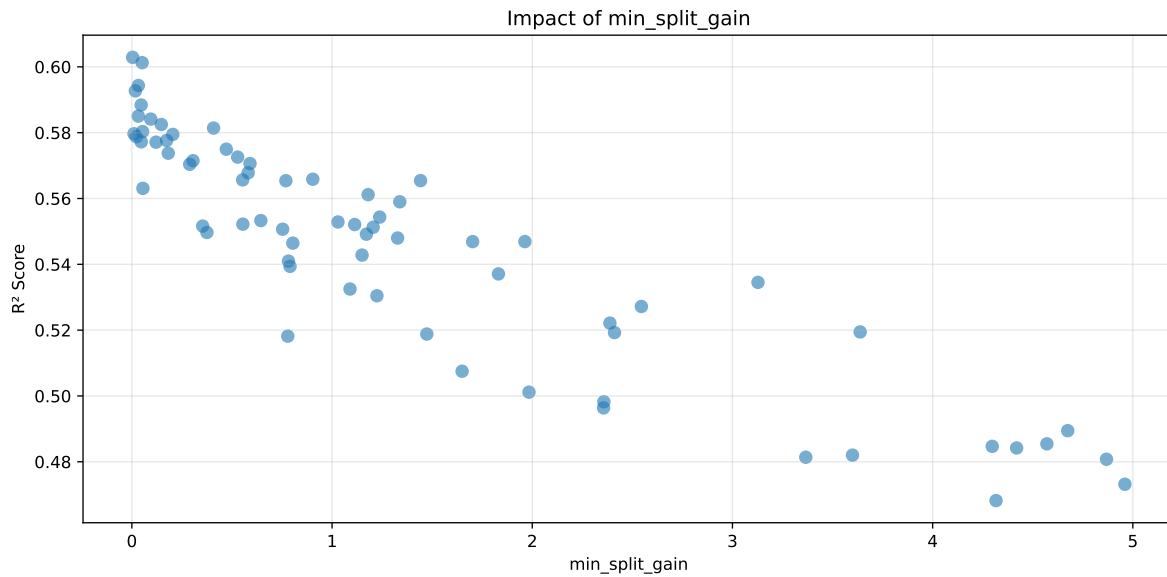
Len(group) : 1585, Len(training) : 1110, Len(test) : 475

Distribution y_train vs y_test: y_train: mean=2.772, std=0.805, min=1.432, max=13.845
y_test: mean=2.807, std=0.764, min=1.373, max=8.682

===== TOP Importance FEATURES =====
feature importance r_NH3 182 r_umidity 168 r_CO2 130 r_temperature 126 r_THI 101

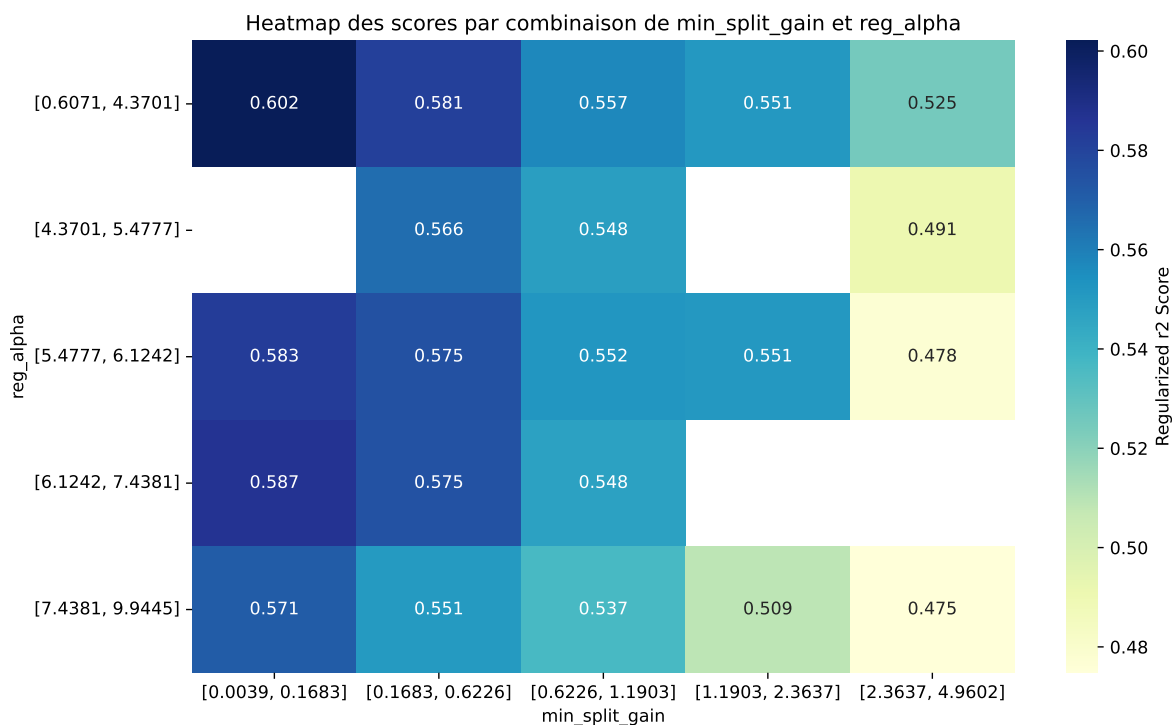


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of `observed=False` is deprecated and will change to `observed=True` in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6017	0.5841	0.5786	0.0349	0.5841	0.5994	0.0153	1.0262	1.0301
2	0.5876	0.5825	0.573	0.0335	0.5825	0.5967	0.0142	1.0243	1.0088
3	0.5697	0.5814	0.566	0.0335	0.5814	0.5894	0.008	1.0138	0.9798
4	0.5957	0.5803	0.5769	0.0385	0.5803	0.6075	0.0272	1.0468	1.0264
5	0.5814	0.5795	0.5771	0.0335	0.5795	0.6043	0.0248	1.0428	1.0034
6	0.5812	0.5788	0.5606	0.033	0.5788	0.5929	0.0141	1.0244	1.0043
7	0.5741	0.5776	0.5674	0.0325	0.5776	0.5915	0.0139	1.024	0.9939
8	0.589	0.5772	0.5717	0.0326	0.5772	0.6029	0.0257	1.0445	1.0203
9	0.5736	0.5771	0.5598	0.0362	0.5771	0.5876	0.0105	1.0182	0.9939
10	0.5599	0.575	0.5579	0.0343	0.575	0.5817	0.0067	1.0117	0.9738
---	---	---	---	---	---	---	---	---	---
70	0.7344	0.6029	0.6318	0.0479	0.2084	0.6491	0.0462	1.0767	1.2181

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.08584049119294529, 'reg__num_leaves': 25, 'reg__subsample': 0.8814094256473903, 'reg__colsample_bytree': 0.5511373870512652, 'reg__min_child_samples': 12, 'reg__reg_alpha': 5.80782035992744, 'reg__reg_lambda': 20.146270221938646, 'reg__min_split_gain': 0.09442800560097803, 'reg__path_smooth': 0.048893719885389805}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.08701520958034462, 'reg__num_leaves': 47, 'reg__subsample': 0.8970526425412897, 'reg__colsample_bytree': 0.5774460873008991, 'reg__min_child_samples': 10, 'reg__reg_alpha': 5.781479423694358, 'reg__reg_lambda': 19.185240637729034, 'reg__min_split_gain': 0.14712881622099405, 'reg__path_smooth': 0.7669972438094441}

Rank 3: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.06888779034719522, 'reg__num_leaves': 35, 'reg__subsample': 0.8771301955453843, 'reg__colsample_bytree': 0.5431857149889205, 'reg__min_child_samples': 12, 'reg__reg_alpha': 3.8964868554453025, 'reg__reg_lambda': 26.889844993627353, 'reg__min_split_gain': 0.4081214767922336, 'reg__path_smooth': 0.08948174288634173}

Rank 4: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.09480682148721586, 'reg__num_leaves': 15, 'reg__subsample': 0.8753483962334117, 'reg__colsample_bytree': 0.7565686869323973, 'reg__min_child_samples': 26, 'reg__reg_alpha': 6.290936007078809, 'reg__reg_lambda': 19.13242607292908, 'reg__min_split_gain': 0.053740856542928914, 'reg__path_smooth': 0.15738079454066778}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.09536514462751819, 'reg__num_leaves': 15, 'reg__subsample': 0.7986802745608147, 'reg__colsample_bytree': 0.7282382828307349, 'reg__min_child_samples': 13, 'reg__reg_alpha': 6.144586141734051, 'reg__reg_lambda': 19.59526065952423, 'reg__min_split_gain': 0.20442206502339555, 'reg__path_smooth': 0.8663992380370644}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.058002704939827815, 'reg__num_leaves': 19, 'reg__subsample': 0.8346910737238012, 'reg__colsample_bytree': 0.6655550083001026, 'reg__min_child_samples': 25, 'reg__reg_alpha': 8.0227401127374, 'reg__reg_lambda': 10.16111597251133, 'reg__min_split_gain': 0.022262080995602007, 'reg__path_smooth': 1.921376418098784}

Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.09388616411162992, 'reg__num_leaves': 46, 'reg__subsample': 0.89668413366439, 'reg__colsample_bytree': 0.5811884987371231, 'reg__min_child_samples': 11, 'reg__reg_alpha': 6.191520134615139, 'reg__reg_lambda': 19.097202133677897, 'reg__min_split_gain': 0.17362710214731808, 'reg__path_smooth': 2.4269850075031636}

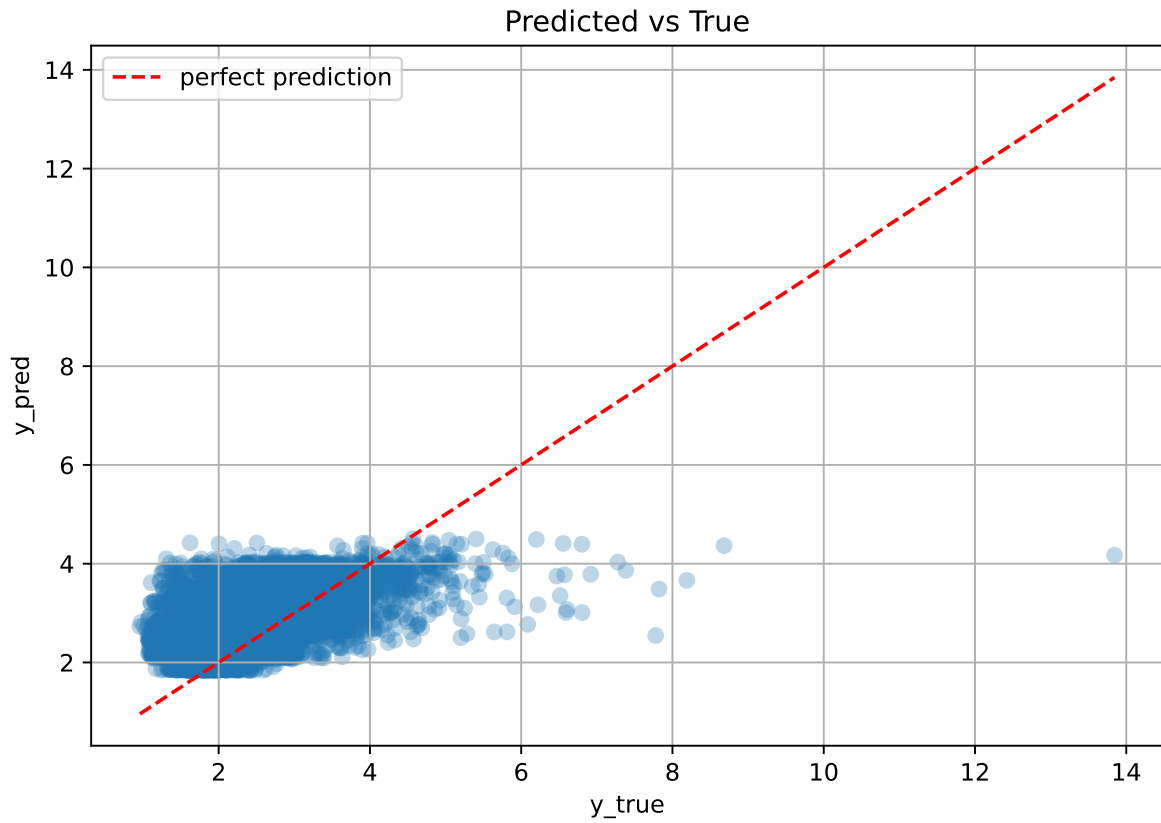
Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.08786705999980836, 'reg__num_leaves': 23, 'reg__subsample': 0.8720142708862078, 'reg__colsample_bytree': 0.7793104291323125, 'reg__min_child_samples': 36, 'reg__reg_alpha': 5.75241983179198, 'reg__reg_lambda': 27.260060586546693, 'reg__min_split_gain': 0.04683816675113406, 'reg__path_smooth': 0.017693730754995363}

Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 8, 'reg__learning_rate': 0.09356152350925323, 'reg__num_leaves': 24, 'reg__subsample': 0.7621020700359064,

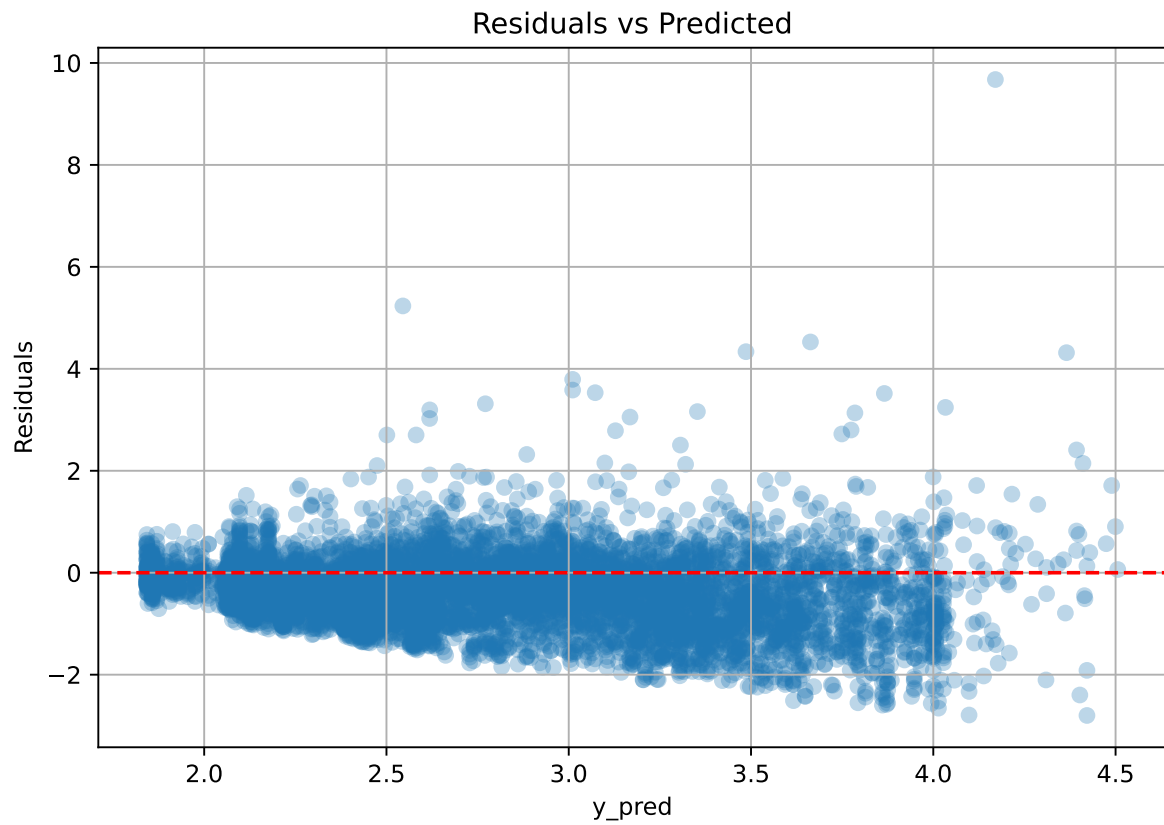
'reg__colsample_bytree': 0.5876328558954999, 'reg__min_child_samples': 25, 'reg__reg_alpha': 6.113245730956365, 'reg__reg_lambda': 18.371204478528533, 'reg__min_split_gain': 0.1208319821582877, 'reg__path_smooth': 4.579211485423168}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.06402209623707969, 'reg__num_leaves': 51, 'reg__subsample': 0.8486295352995585, 'reg__colsample_bytree': 0.5479692701643479, 'reg__min_child_samples': 12, 'reg__reg_alpha': 5.566645320029277, 'reg__reg_lambda': 8.355627712788323, 'reg__min_split_gain': 0.4718362440444914, 'reg__path_smooth': 1.9357827479661998}

2.4.2 Scatter



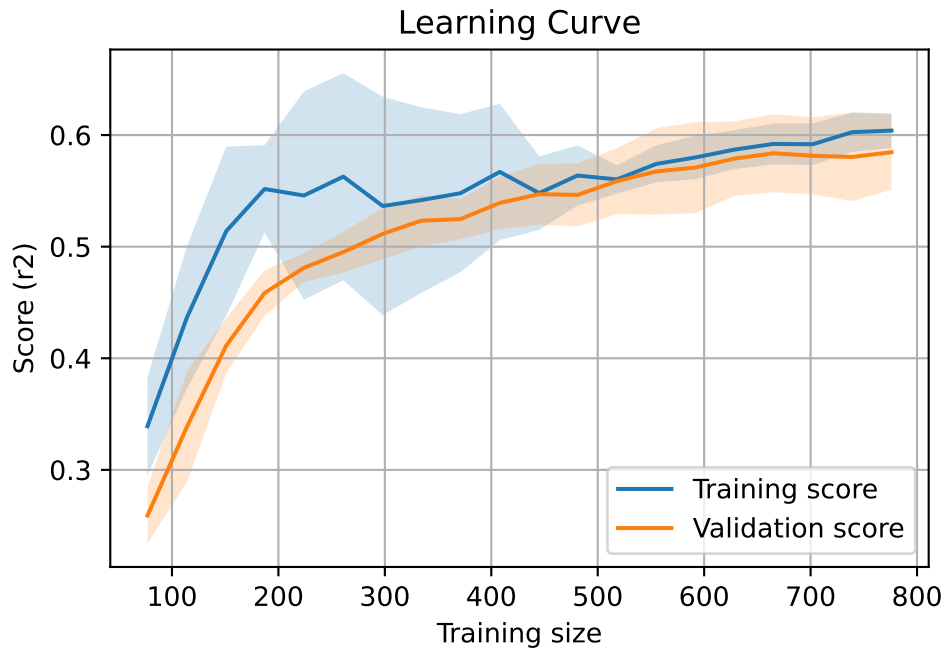
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 2025_Jun_1

`Len(group)` : 1585, `Len(training)` : 1110,

`Len(training_real)` : 777, `Len(test_of_training)` : 333, `Len(test)` : 475

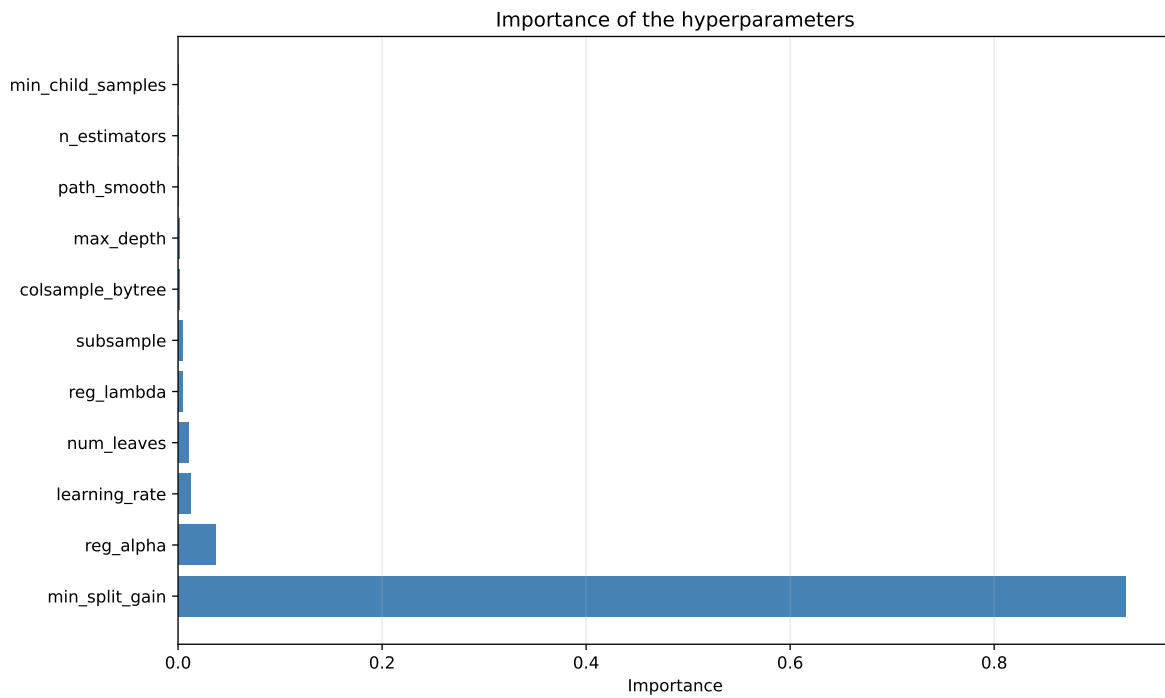
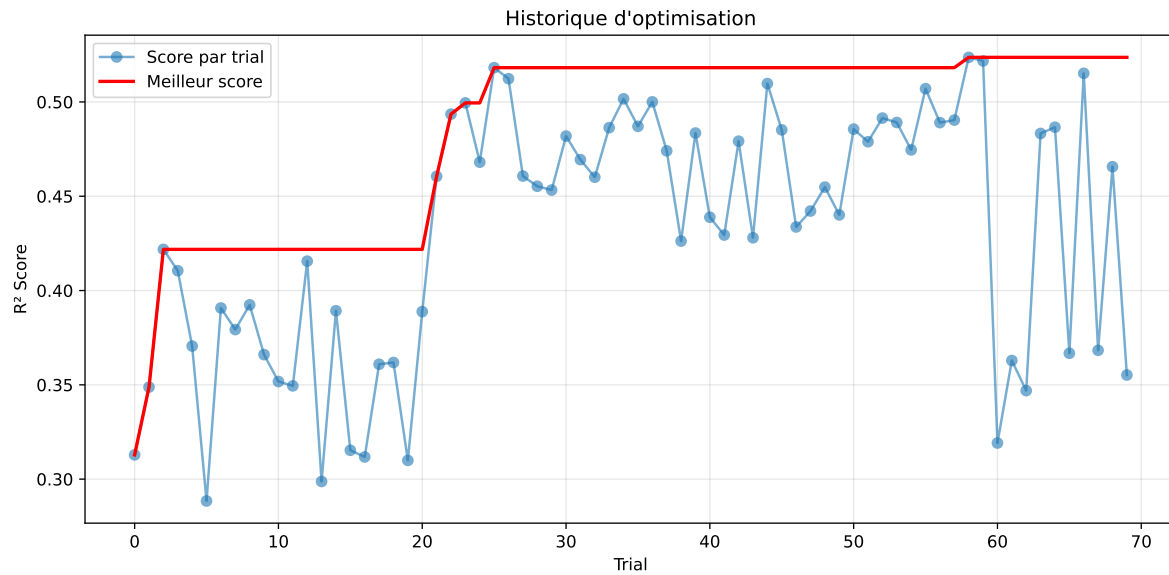


2.4.5 Gridsearch — Group: 2025_Jun_2

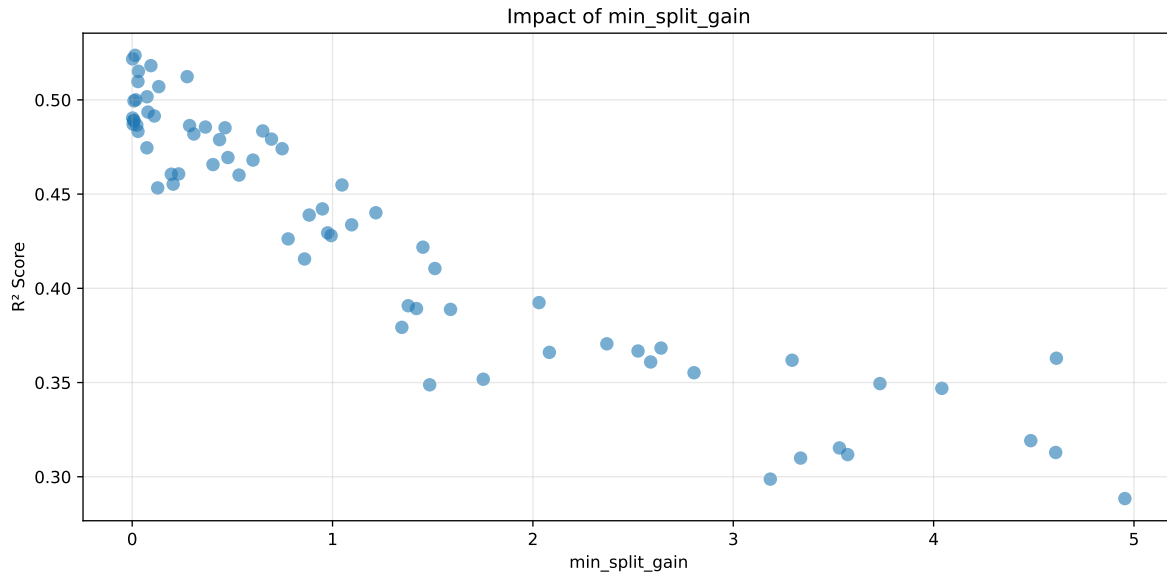
Len(group) : 1587, Len(training) : 1111, Len(test) : 476

Distribution y_train vs y_test: y_train: mean=2.509, std=0.675, min=1.253, max=6.222
y_test: mean=2.528, std=0.697, min=1.305, max=7.383

===== TOP Importance FEATURES =====
feature importance r_umidity 136 r_temperature 104 r_THI 101 r_CO2 69 r_NH3 60



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3997	0.3618	0.3545	0.0191	0.2482	0.3973	0.0355	1.0981	1.1047
2	0.4087	0.3683	0.3619	0.0171	0.2472	0.4071	0.0388	1.1053	1.1096
3	0.3837	0.3494	0.3391	0.0196	0.2466	0.3819	0.0324	1.0928	1.0981
4	0.4417	0.3924	0.3911	0.0115	0.2447	0.4335	0.0411	1.1048	1.1255
5	0.4079	0.3667	0.3623	0.0157	0.2431	0.4136	0.0469	1.1278	1.1124
6	0.4134	0.3706	0.3619	0.0143	0.242	0.3924	0.0218	1.0588	1.1157
7	0.4253	0.3793	0.3755	0.02	0.2414	0.4092	0.0299	1.0787	1.1212
8	0.4011	0.3609	0.356	0.0187	0.2404	0.3935	0.0325	1.0902	1.1113
9	0.4042	0.3629	0.3598	0.0145	0.2391	0.3959	0.033	1.091	1.1137
10	0.4084	0.366	0.3582	0.0168	0.2389	0.3894	0.0234	1.0638	1.1157
---	---	---	---	---	---	---	---	---	---
70	0.745	0.5097	0.5497	0.0173	-0.1959	0.5651	0.0553	1.1085	1.4614

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.017522696213193464, 'reg__num_leaves': 38, 'reg__subsample': 0.8437597604073453, 'reg__colsample_bytree': 0.7085947481243101, 'reg__min_child_samples': 34, 'reg__reg_alpha': 0.9008156831744707, 'reg__reg_lambda': 27.489394129432632, 'reg__min_split_gain': 3.2936275092396317, 'reg__path_smooth': 1.067606198388265}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.05822310288198462, 'reg__num_leaves': 57, 'reg__subsample': 0.8389008245286573,

'reg__colsample_bytree': 0.7807271848337755, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.931208816737403, 'reg__reg_lambda': 30.08392997522958, 'reg__min_split_gain': 2.639506097280684, 'reg__path_smooth': 1.2757400255090428}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.03509089237764475, 'reg__num_leaves': 15, 'reg__subsample': 0.806486888410299, 'reg__colsample_bytree': 0.8693813177443231, 'reg__min_child_samples': 24, 'reg__reg_alpha': 0.3526992859559763, 'reg__reg_lambda': 43.01137889322218, 'reg__min_split_gain': 3.7320492567766017, 'reg__path_smooth': 1.683103150471359}

Rank 4: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.06351953044057482, 'reg__num_leaves': 24, 'reg__subsample': 0.7852442881457262, 'reg__colsample_bytree': 0.7873678727867088, 'reg__min_child_samples': 11, 'reg__reg_alpha': 2.908659780042492, 'reg__reg_lambda': 18.58453228160964, 'reg__min_split_gain': 2.030528210946001, 'reg__path_smooth': 4.515096664724212}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.060571540230771534, 'reg__num_leaves': 36, 'reg__subsample': 0.8349005075322007, 'reg__colsample_bytree': 0.7746052453375063, 'reg__min_child_samples': 10, 'reg__reg_alpha': 5.326869348178365, 'reg__reg_lambda': 6.28198374538578, 'reg__min_split_gain': 2.52466476124509, 'reg__path_smooth': 0.35563797647695167}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.027302220213453448, 'reg__num_leaves': 26, 'reg__subsample': 0.5503903176195614, 'reg__colsample_bytree': 0.5688231308153585, 'reg__min_child_samples': 41, 'reg__reg_alpha': 2.652576613430254, 'reg__reg_lambda': 4.801132495376387, 'reg__min_split_gain': 2.3690099120301182, 'reg__path_smooth': 4.636575763791564}

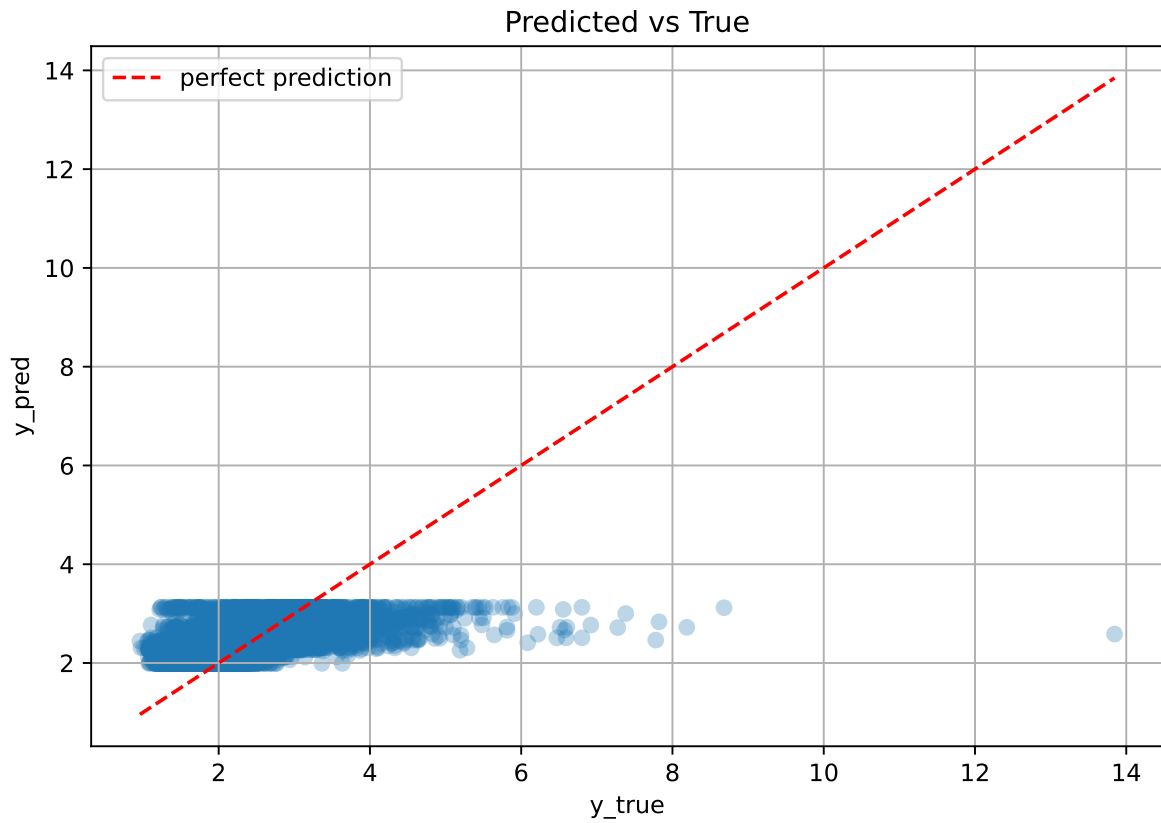
Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 3, 'reg__learning_rate': 0.07268384733657061, 'reg__num_leaves': 19, 'reg__subsample': 0.5386238503327988, 'reg__colsample_bytree': 0.8041284292327651, 'reg__min_child_samples': 38, 'reg__reg_alpha': 5.196570153414553, 'reg__reg_lambda': 37.96977738616707, 'reg__min_split_gain': 1.3459652508779425, 'reg__path_smooth': 0.3257020010158168}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.016345367306145143, 'reg__num_leaves': 20, 'reg__subsample': 0.8990139190096935, 'reg__colsample_bytree': 0.7711422259120811, 'reg__min_child_samples': 11, 'reg__reg_alpha': 4.119799029039647, 'reg__reg_lambda': 26.53856343081985, 'reg__min_split_gain': 2.5878015951732367, 'reg__path_smooth': 4.080736772366851}

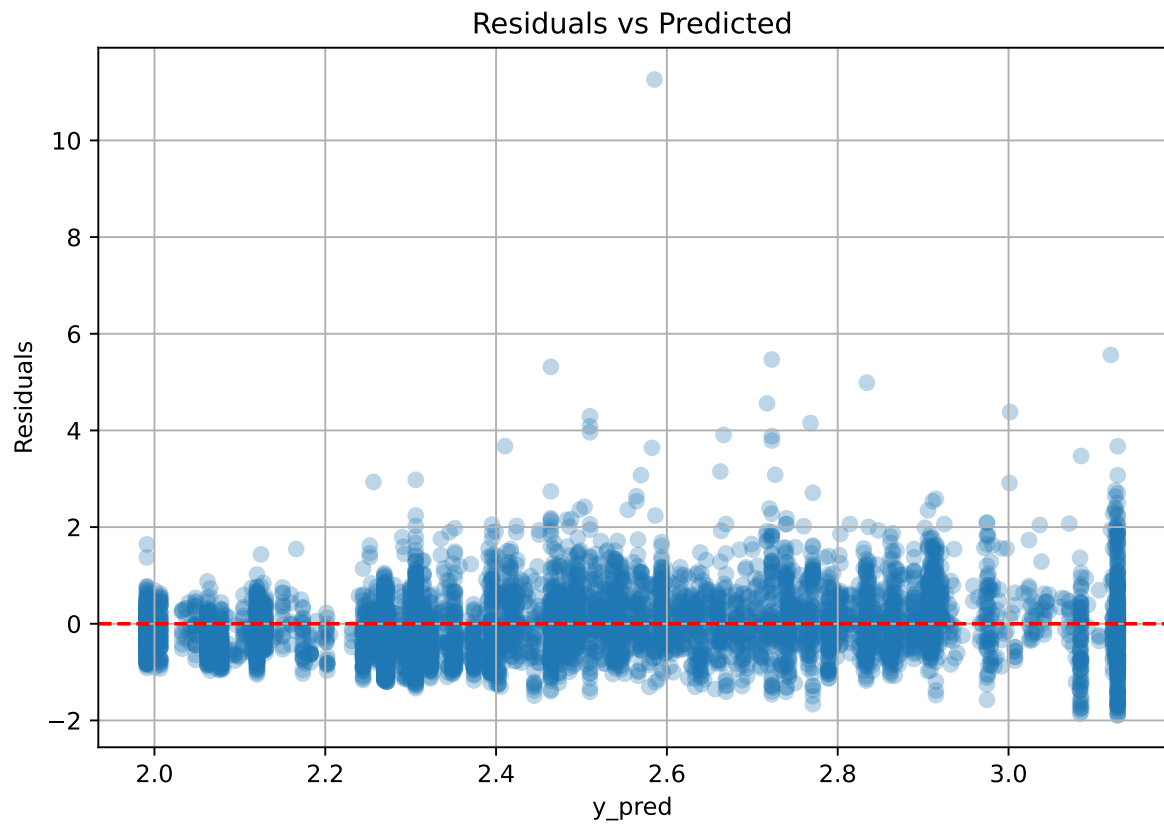
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.06529575683866086, 'reg__num_leaves': 42, 'reg__subsample': 0.7691490504783741, 'reg__colsample_bytree': 0.7670586074319894, 'reg__min_child_samples': 13, 'reg__reg_alpha': 0.010751307597650861, 'reg__reg_lambda': 6.572710640022928, 'reg__min_split_gain': 4.612798738285757, 'reg__path_smooth': 3.5537864035698536}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.014576002213404295, 'reg__num_leaves': 34, 'reg__subsample': 0.8385069124826903, 'reg__colsample_bytree': 0.5886378943319484, 'reg__min_child_samples': 50, 'reg__reg_alpha': 3.2343505883753663, 'reg__reg_lambda': 33.07469687613519, 'reg__min_split_gain': 2.082124495365847, 'reg__path_smooth': 0.6132800252346393}

2.4.6 Scatter



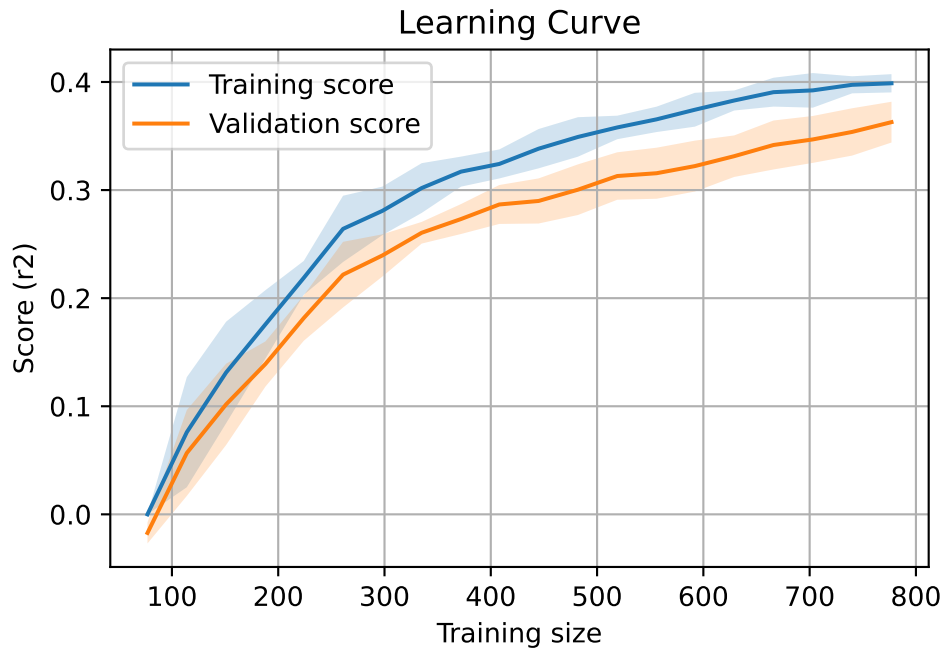
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2025_Jun_2

`Len(group)` : 1587, `Len(training)` : 1111,

`Len(training_real)` : 778, `Len(test_of_training)` : 333, `Len(test)` : 476

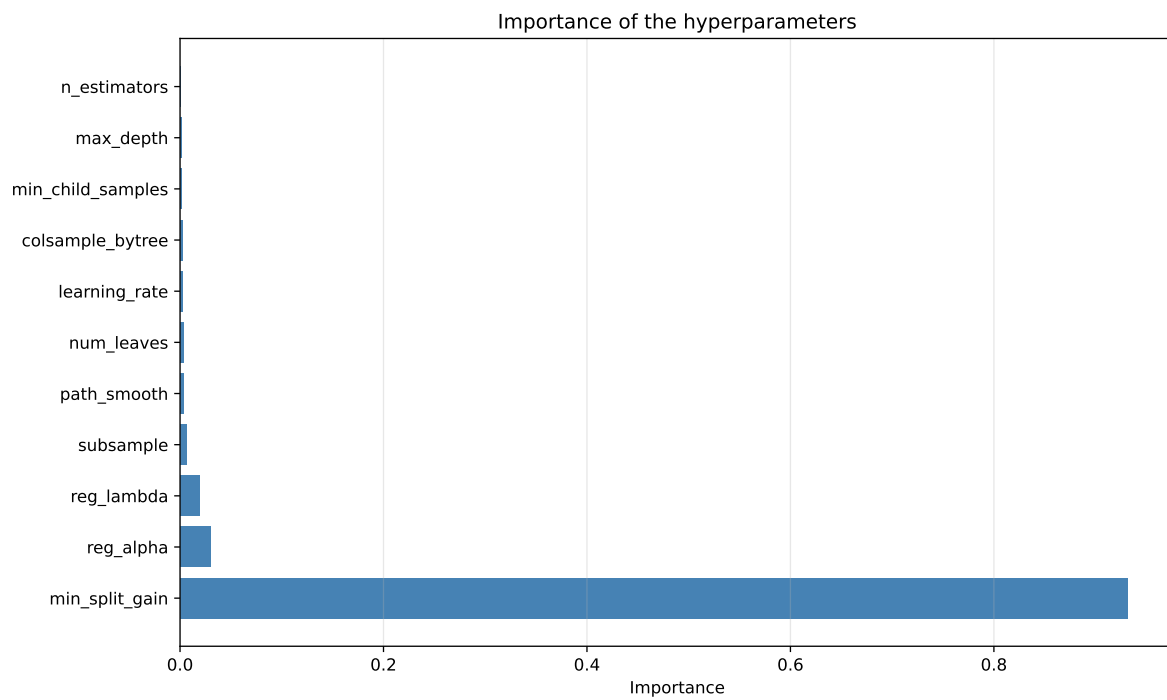
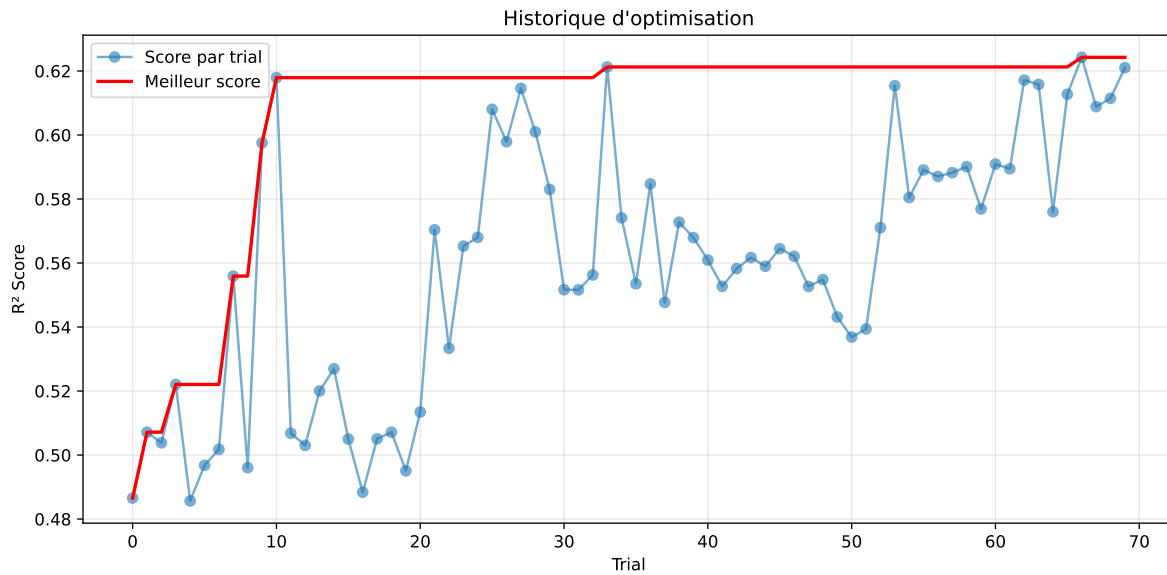


2.4.9 Gridsearch — Group: 2025_Sep_1

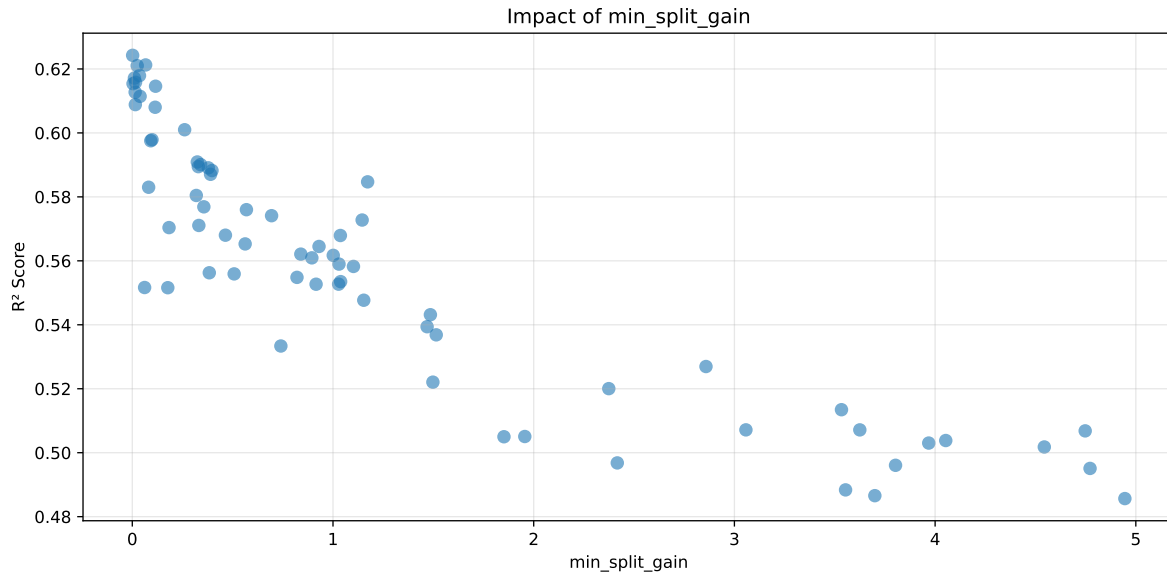
Len(group) : 3867, Len(training) : 2707, Len(test) : 1160

Distribution y_train vs y_test: y_train: mean=2.231, std=0.761, min=0.973, max=8.190
y_test: mean=2.216, std=0.723, min=1.110, max=6.920

===== TOP Importance FEATURES =====
feature importance r_NH3 595 r_umidity 555 r_CO2 431 r_THI 283 r_temperature 250



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.583	0.5563	0.5593	0.0307	0.5563	0.5784	0.0221	1.0398	1.0481
2	0.5812	0.5548	0.5495	0.0278	0.5548	0.5662	0.0114	1.0205	1.0475
3	0.5801	0.5535	0.5556	0.0296	0.5535	0.573	0.0195	1.0352	1.048
4	0.5798	0.5527	0.5529	0.0298	0.5527	0.574	0.0213	1.0386	1.049
5	0.578	0.5527	0.5494	0.028	0.5527	0.566	0.0133	1.024	1.0458
6	0.5701	0.5477	0.5491	0.0284	0.5477	0.5698	0.0221	1.0404	1.041
7	0.5631	0.5431	0.5466	0.031	0.5431	0.5634	0.0202	1.0372	1.0368
8	0.5573	0.5394	0.5388	0.0281	0.5394	0.5575	0.0181	1.0336	1.0331
9	0.5535	0.5369	0.5354	0.0275	0.5369	0.5533	0.0165	1.0307	1.031
10	0.5478	0.5334	0.5382	0.0311	0.5334	0.5516	0.0182	1.0341	1.027
---	---	---	---	---	---	---	---	---	---
70	0.768	0.6243	0.6089	0.0318	0.1929	0.6349	0.0106	1.017	1.2303

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.01856802565436653, 'reg__num_leaves': 61, 'reg__subsample': 0.5015138902403888, 'reg__colsample_bytree': 0.8096429135085378, 'reg__min_child_samples': 24, 'reg__reg_alpha': 6.251330642231211, 'reg__reg_lambda': 33.139699593374424, 'reg__min_split_gain': 0.38366632071895357, 'reg__path_smooth': 4.971164688120135}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.04108257409321123, 'reg__num_leaves': 46, 'reg__subsample': 0.8268658581698553,

'reg__colsample_bytree': 0.5039359156897876, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.2189040771667548, 'reg__reg_lambda': 39.027164604662794, 'reg__min_split_gain': 0.8205634251208922, 'reg__path_smooth': 1.4519724460356436}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.01676811228471481, 'reg__num_leaves': 20, 'reg__subsample': 0.5268494232442984, 'reg__colsample_bytree': 0.8214123725827394, 'reg__min_child_samples': 10, 'reg__reg_alpha': 0.6628311035419849, 'reg__reg_lambda': 35.55656635763092, 'reg__min_split_gain': 1.0380082879355121, 'reg__path_smooth': 4.4787116838334455}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.028747283577944878, 'reg__num_leaves': 46, 'reg__subsample': 0.7612193463448248, 'reg__colsample_bytree': 0.8964536267579585, 'reg__min_child_samples': 38, 'reg__reg_alpha': 0.631262991519153, 'reg__reg_lambda': 38.397612503026444, 'reg__min_split_gain': 1.0279962852866018, 'reg__path_smooth': 1.8285665019630886}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.027625680918062563, 'reg__num_leaves': 45, 'reg__subsample': 0.7594118374369409, 'reg__colsample_bytree': 0.5016141929904365, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.1954319267668905, 'reg__reg_lambda': 38.38446012899553, 'reg__min_split_gain': 0.9157440122612117, 'reg__path_smooth': 2.1645987515883602}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.015854985928079237, 'reg__num_leaves': 15, 'reg__subsample': 0.5474924567803765, 'reg__colsample_bytree': 0.8975312260689082, 'reg__min_child_samples': 39, 'reg__reg_alpha': 0.4451202855387244, 'reg__reg_lambda': 36.71926215791472, 'reg__min_split_gain': 1.1532817940777147, 'reg__path_smooth': 4.447312928549401}

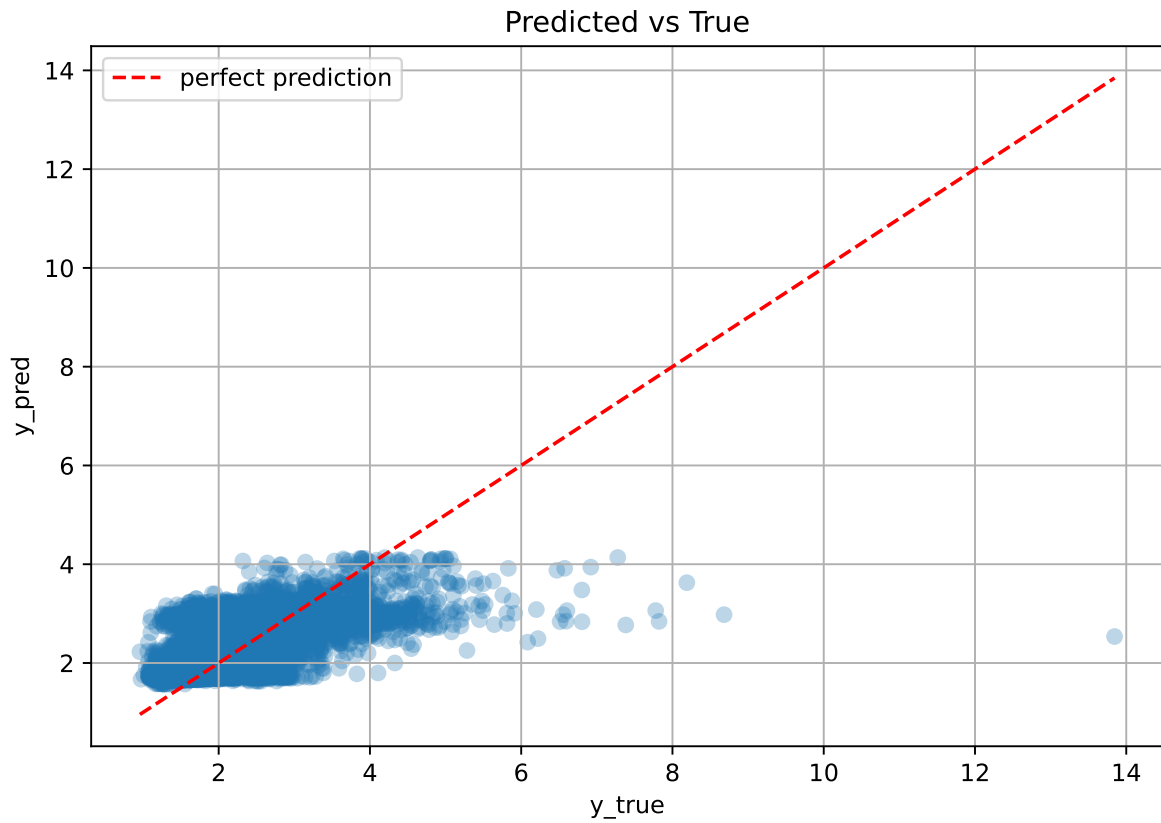
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.027110690934243738, 'reg__num_leaves': 44, 'reg__subsample': 0.5660135193160716, 'reg__colsample_bytree': 0.8492362318598387, 'reg__min_child_samples': 30, 'reg__reg_alpha': 1.0073190312476592, 'reg__reg_lambda': 38.86292327974222, 'reg__min_split_gain': 1.4852826491584703, 'reg__path_smooth': 2.042980268250399}

Rank 8: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.02550754716358517, 'reg__num_leaves': 28, 'reg__subsample': 0.5628065842722266, 'reg__colsample_bytree': 0.5001922725192963, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.2709401186885785, 'reg__reg_lambda': 28.524070522447257, 'reg__min_split_gain': 1.4688874961746898, 'reg__path_smooth': 2.5858453314369663}

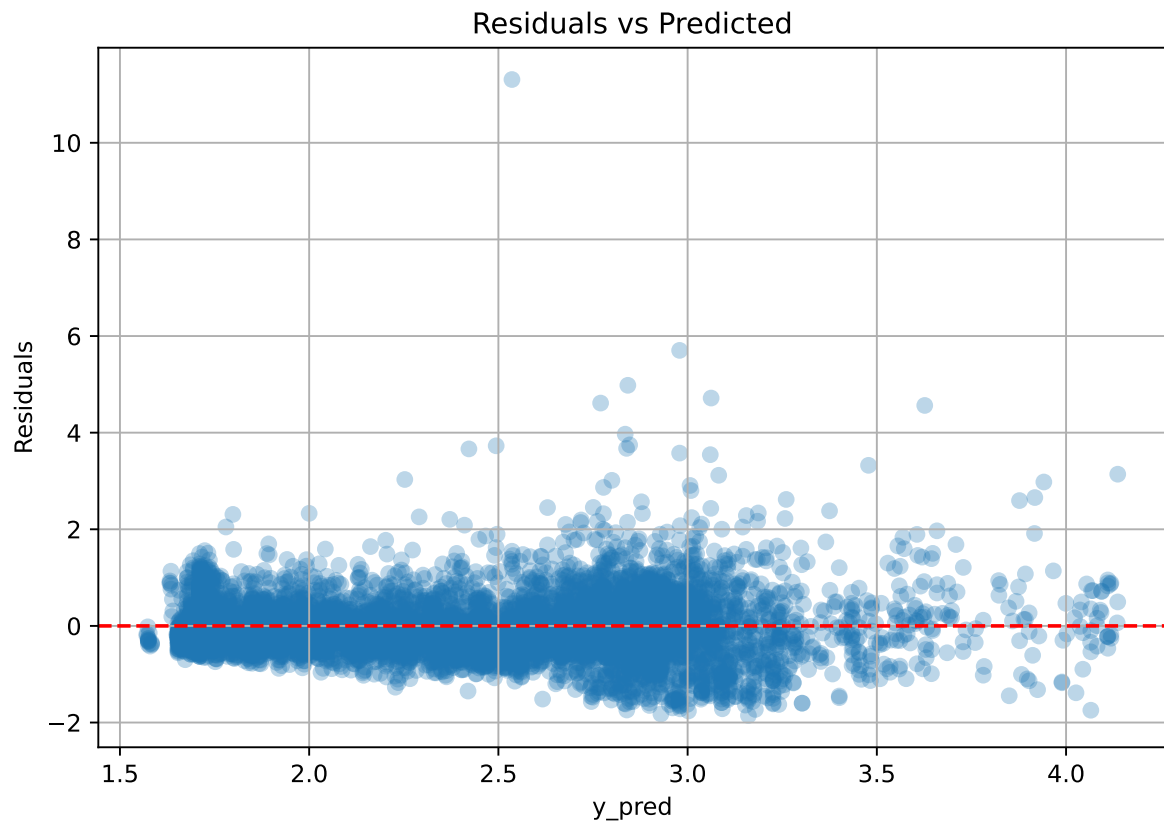
Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.03855866334678613, 'reg__num_leaves': 28, 'reg__subsample': 0.8283956197044589, 'reg__colsample_bytree': 0.5055588975588001, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.2574789007104332, 'reg__reg_lambda': 28.50734102520431, 'reg__min_split_gain': 1.514422431654524, 'reg__path_smooth': 1.293778701587427}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.02018720136428354, 'reg__num_leaves': 63, 'reg__subsample': 0.7476022618079006, 'reg__colsample_bytree': 0.812782536261819, 'reg__min_child_samples': 25, 'reg__reg_alpha': 6.438970105246623, 'reg__reg_lambda': 32.8431609022093, 'reg__min_split_gain': 0.7402689080067868, 'reg__path_smooth': 4.986109293247118}

2.4.10 Scatter



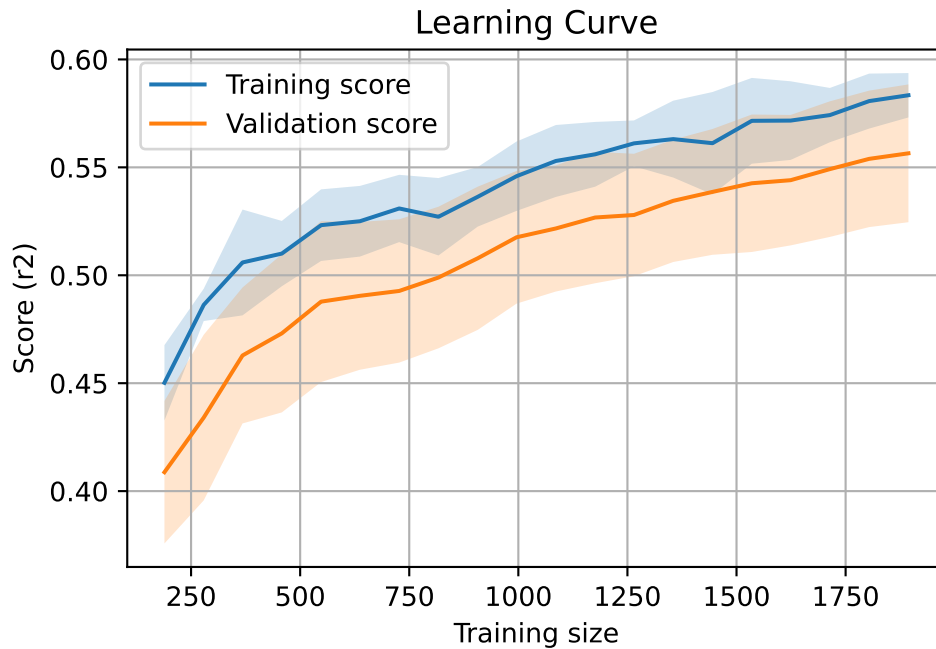
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 2025_Sep_1

`Len(group)` : 3867, `Len(training)` : 2707,

`Len(training_real)` : 1895, `Len(test_of_training)` : 812, `Len(test)` : 1160

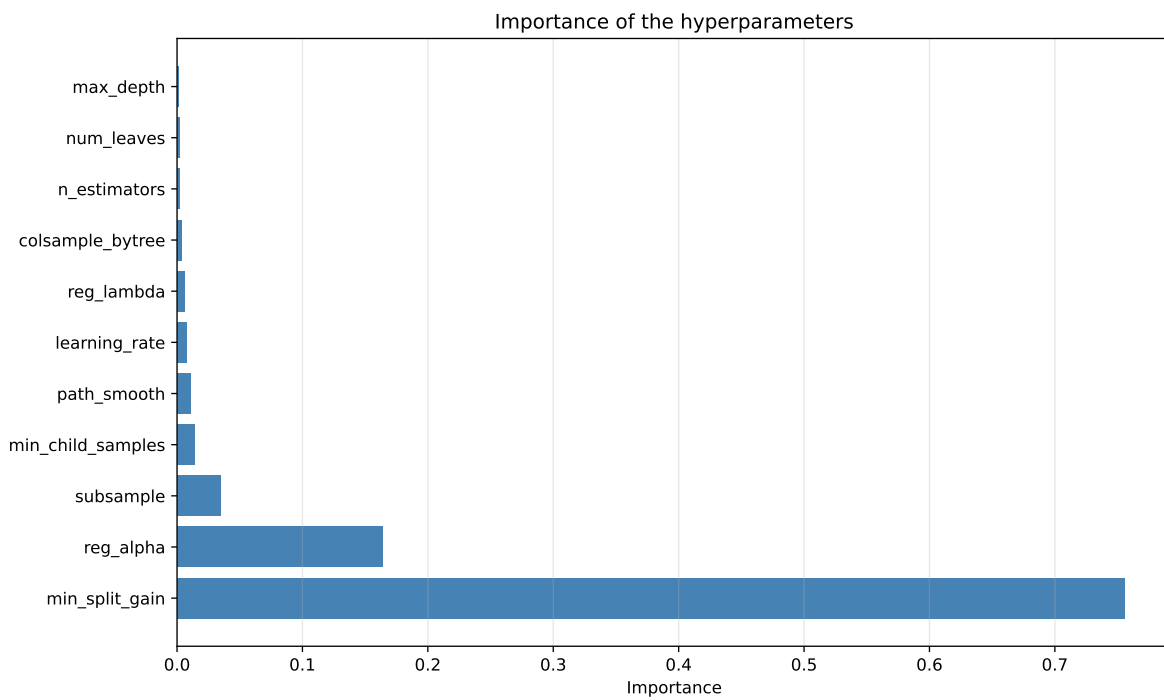
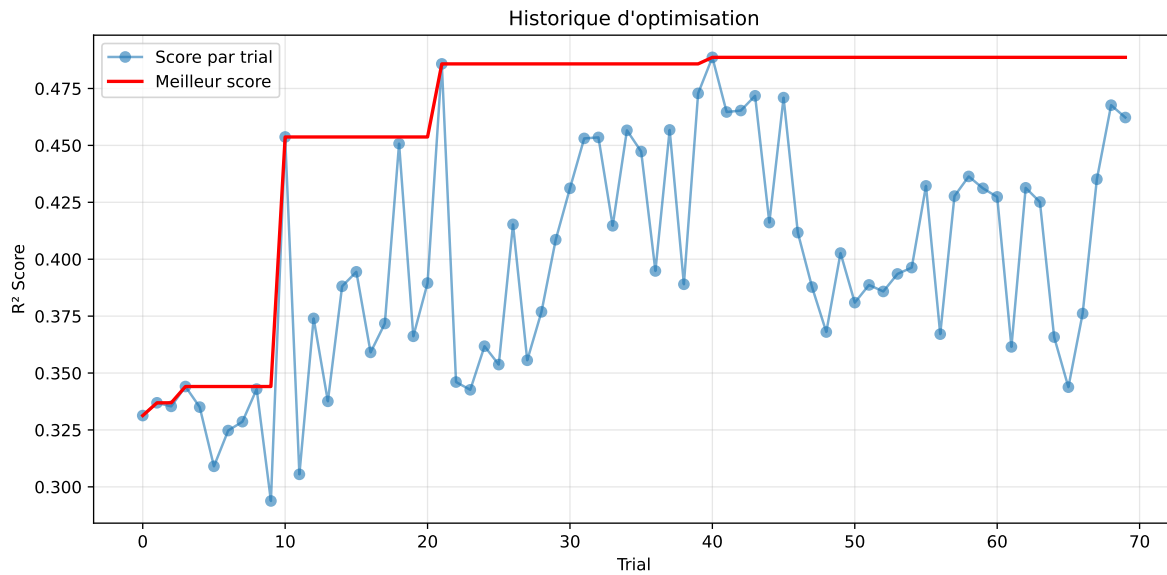


2.4.13 Gridsearch — Group: 2025_Sep_2

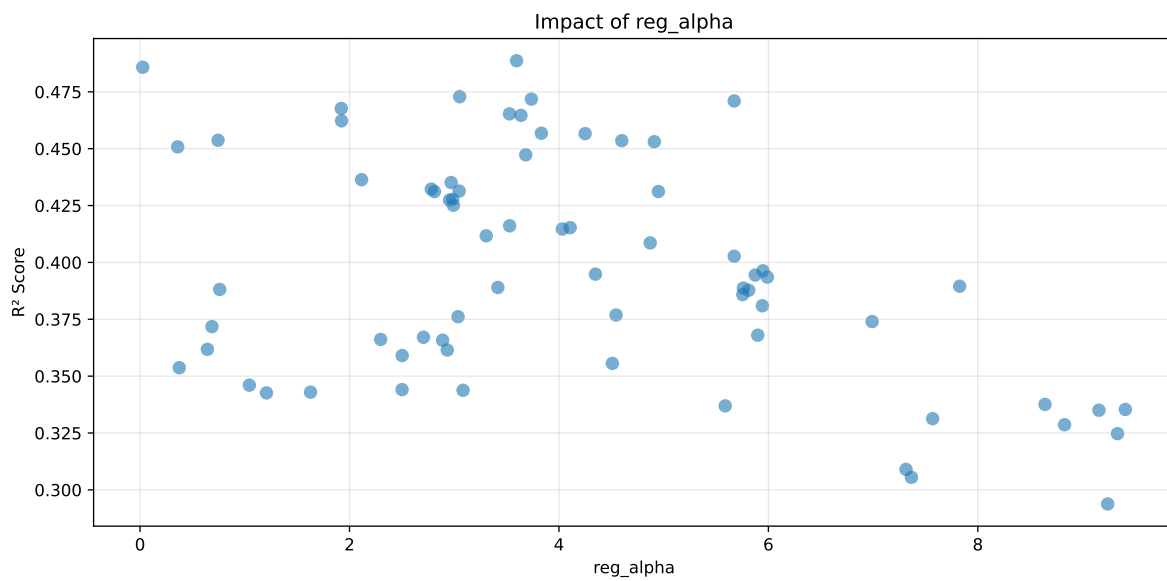
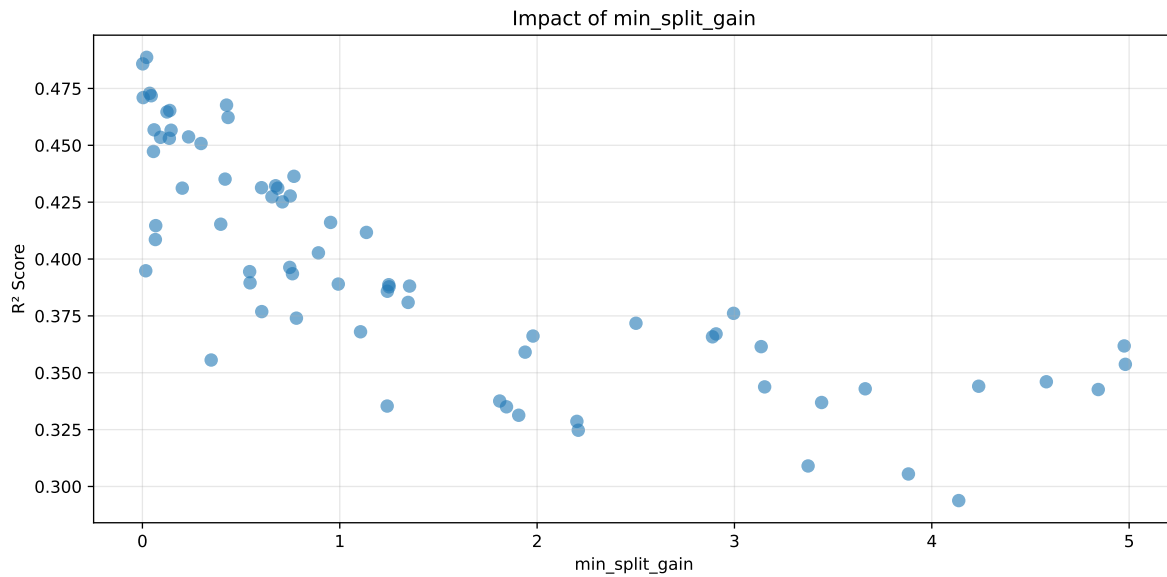
Len(group) : 1129, Len(training) : 791, Len(test) : 338

Distribution y_train vs y_test: y_train: mean=1.926, std=0.742, min=0.959, max=6.804
y_test: mean=1.945, std=0.805, min=1.070, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 83 r_THI 26 r_CO2 13 r_temperature 9 r_NH3 8

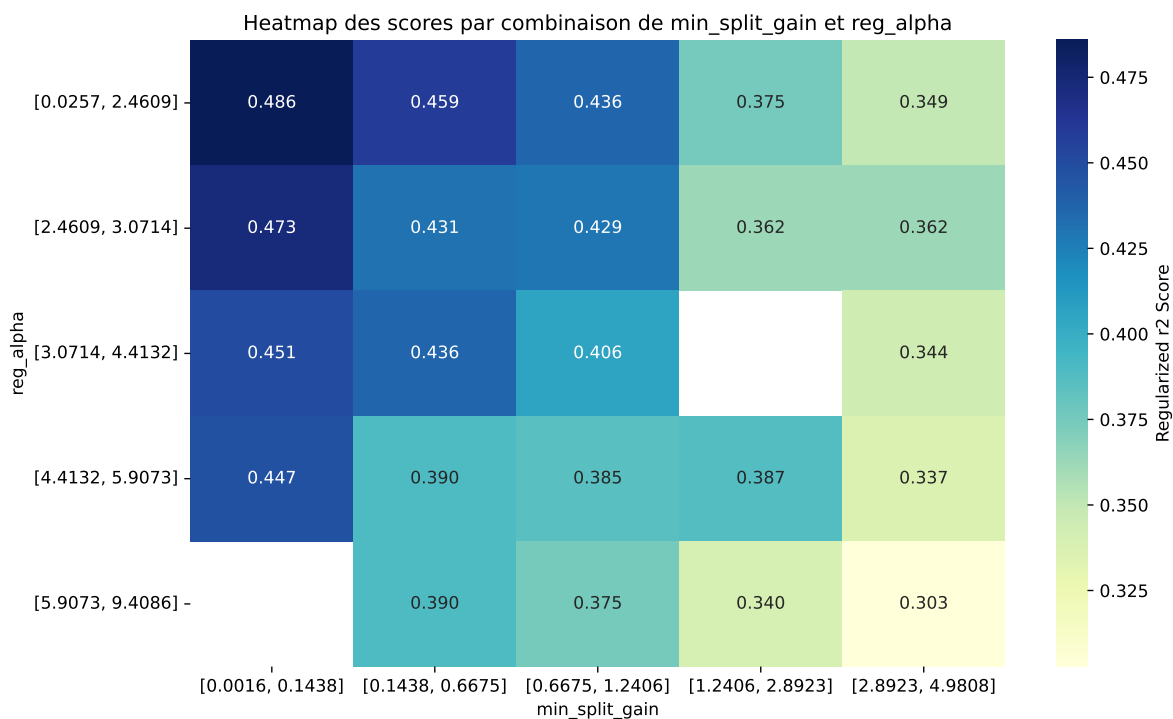


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.384	0.3441	0.3575	0.0322	0.2244	0.3867	0.0426	1.1238	1.1159
2	0.4091	0.3618	0.3816	0.0395	0.2198	0.4068	0.045	1.1243	1.1308
3	0.3858	0.3438	0.3588	0.0344	0.2179	0.3945	0.0507	1.1475	1.1221
4	0.3768	0.3369	0.3506	0.0318	0.2174	0.385	0.0481	1.1427	1.1183
5	0.4159	0.3658	0.3854	0.0364	0.2154	0.4121	0.0463	1.1266	1.1371
6	0.4186	0.3671	0.3899	0.0354	0.2125	0.4115	0.0444	1.121	1.1404
7	0.425	0.3718	0.395	0.0364	0.2119	0.4251	0.0533	1.1434	1.1433
8	0.4114	0.3615	0.3829	0.0349	0.2118	0.4102	0.0488	1.1349	1.1381
9	0.3691	0.3286	0.3441	0.0322	0.2071	0.3723	0.0437	1.133	1.1232
10	0.3813	0.3376	0.3553	0.0323	0.2064	0.377	0.0395	1.1169	1.1295
---	---	---	---	---	---	---	---	---	---
70	0.695	0.4858	0.5701	0.0351	-0.1418	0.573	0.0872	1.1795	1.4306

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.035063766290828344, 'reg__num_leaves': 19, 'reg__subsample': 0.6826154192234977, 'reg__colsample_bytree': 0.7795730118948808, 'reg__min_child_samples': 47, 'reg__reg_alpha': 2.5017385340888687, 'reg__reg_lambda': 14.757936939279642, 'reg__min_split_gain': 4.237571780472982, 'reg__path_smooth': 0.5884075166280739}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 7, 'reg__learning_rate': 0.017987492214260797, 'reg__num_leaves': 22, 'reg__subsample': 0.8787294464216007, 'reg__colsample_bytree': 0.895693125830986, 'reg__min_child_samples': 47, 'reg__reg_alpha': 0.6432811630244719, 'reg__reg_lambda': 3.119665713678117, 'reg__min_split_gain': 4.974934383632586, 'reg__path_smooth': 0.5284682976126781}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.039556349156033434, 'reg__num_leaves': 30, 'reg__subsample': 0.59460812658528, 'reg__colsample_bytree': 0.7481567360689894, 'reg__min_child_samples': 18, 'reg__reg_alpha': 3.0845234847041847, 'reg__reg_lambda': 33.64356724745295, 'reg__min_split_gain': 3.1529404337985882, 'reg__path_smooth': 2.6689464615221965}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.029889930563287464, 'reg__num_leaves': 54, 'reg__subsample': 0.6534672817571782, 'reg__colsample_bytree': 0.8041369723041082, 'reg__min_child_samples': 43, 'reg__reg_alpha': 5.5876227097989615, 'reg__reg_lambda': 15.074945461475025, 'reg__min_split_gain': 3.4416111680308363, 'reg__path_smooth': 2.220862226340854}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.03991651274853332, 'reg__num_leaves': 30, 'reg__subsample': 0.5413426085301165, 'reg__colsample_bytree': 0.7740126869069129, 'reg__min_child_samples': 18, 'reg__reg_alpha': 2.8885886924437765, 'reg__reg_lambda': 18.314654884964384, 'reg__min_split_gain': 2.888717157059496, 'reg__path_smooth': 2.4892103301577726}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.04001493447572174, 'reg__num_leaves': 36, 'reg__subsample': 0.5791441726389154, 'reg__colsample_bytree': 0.7475729117206837, 'reg__min_child_samples': 17, 'reg__reg_alpha': 2.70650161676901, 'reg__reg_lambda': 17.092530307750792, 'reg__min_split_gain': 2.906879904447369, 'reg__path_smooth': 1.1341154234936317}

Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.03403400128586836, 'reg__num_leaves': 15, 'reg__subsample': 0.602060187647743, 'reg__colsample_bytree': 0.7715015766288651, 'reg__min_child_samples': 15, 'reg__reg_alpha': 0.6866949043135528, 'reg__reg_lambda': 39.13398577210704, 'reg__min_split_gain': 2.5009197714761915, 'reg__path_smooth': 3.0042361761075713}

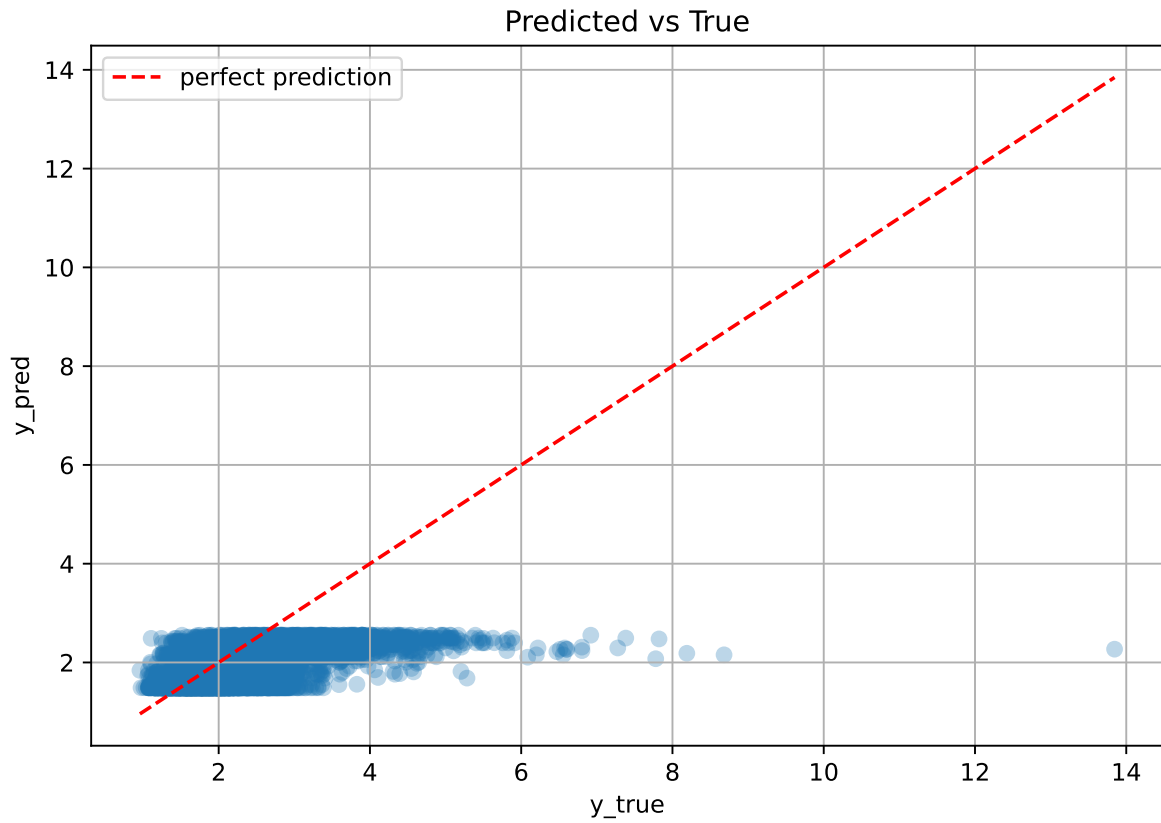
Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.038575729749444114, 'reg__num_leaves': 30, 'reg__subsample': 0.581776348612793, 'reg__colsample_bytree': 0.7474849867999682, 'reg__min_child_samples': 18, 'reg__reg_alpha': 2.933712215536626, 'reg__reg_lambda': 16.731662544480248, 'reg__min_split_gain': 3.1354243081223103, 'reg__path_smooth': 1.1069632359504449}

Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.06443853244712373, 'reg__num_leaves': 17, 'reg__subsample': 0.7614899522405403,

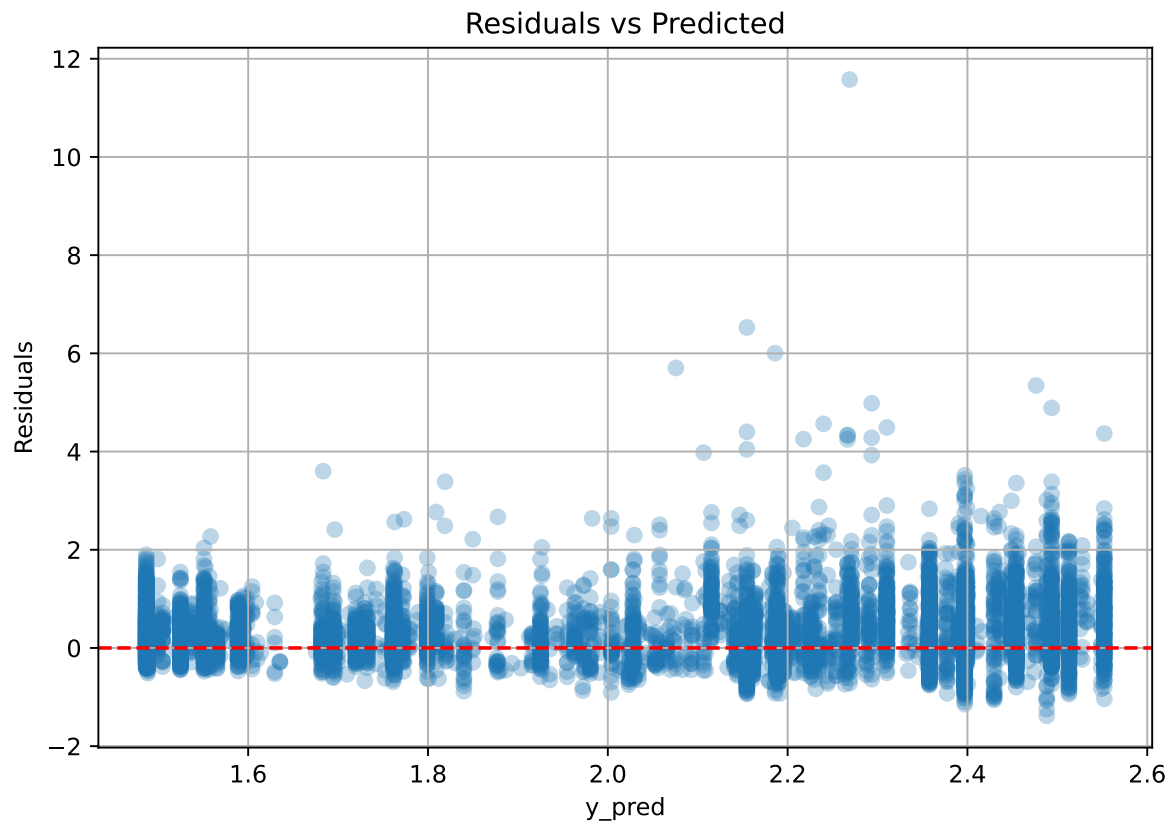
'reg__colsample_bytree': 0.7194105308204313, 'reg__min_child_samples': 27, 'reg__reg_alpha': 8.828081569188445, 'reg__reg_lambda': 43.69584486401831, 'reg__min_split_gain': 2.201551669028024, 'reg__path_smooth': 2.6512478126283394}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.09481141313347542, 'reg__num_leaves': 34, 'reg__subsample': 0.6495130965807104, 'reg__colsample_bytree': 0.7828787144804918, 'reg__min_child_samples': 22, 'reg__reg_alpha': 8.64148733637723, 'reg__reg_lambda': 43.036902477137104, 'reg__min_split_gain': 1.810452901616848, 'reg__path_smooth': 4.988297226523581}

2.4.14 Scatter



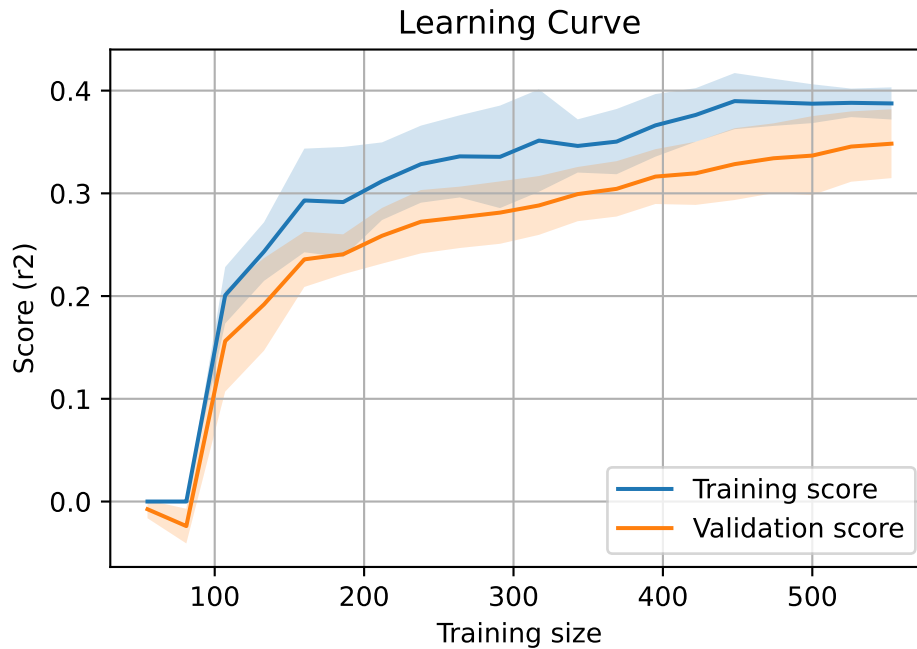
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 2025_Sep_2

`Len(group)` : 1129, `Len(training)` : 791,

`Len(training_real)` : 554, `Len(test_of_training)` : 237, `Len(test)` : 338



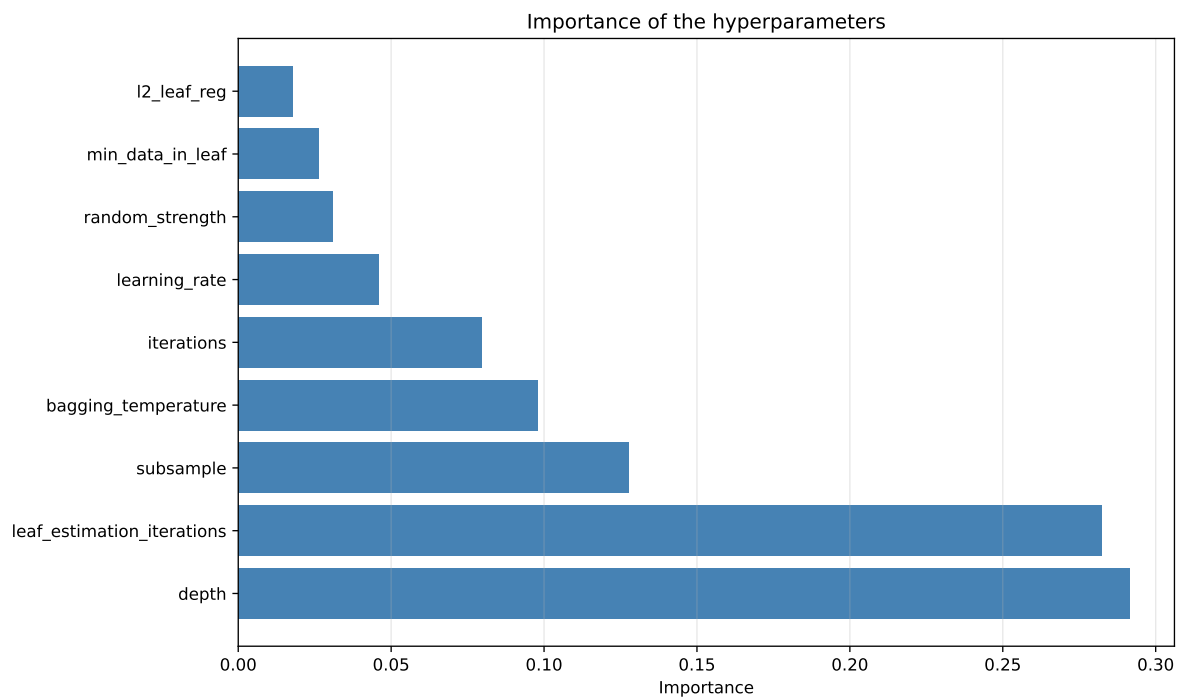
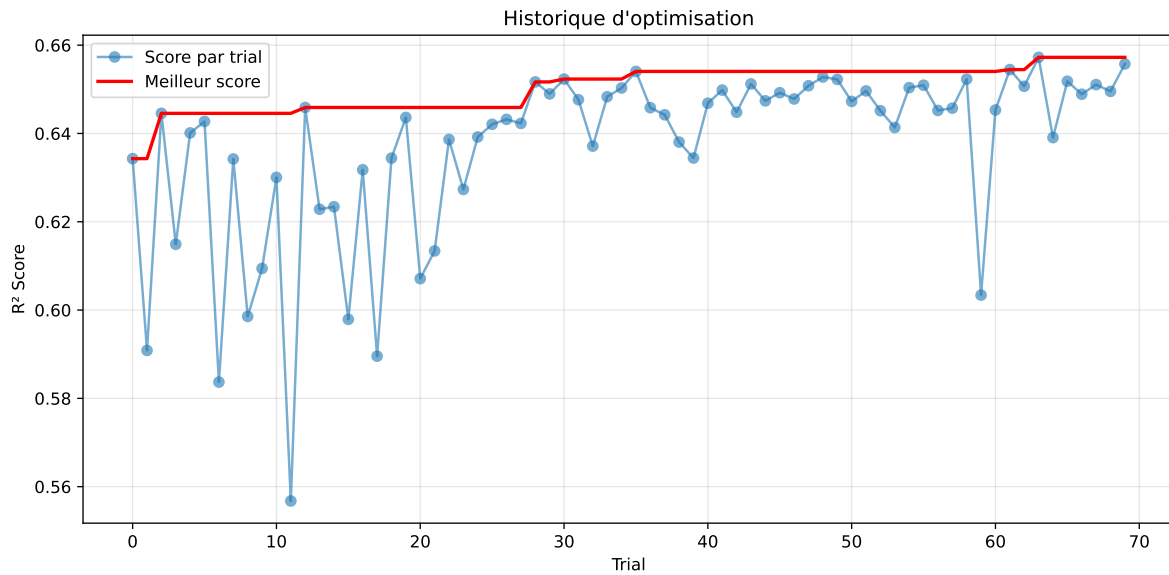
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 2025_Jun_1

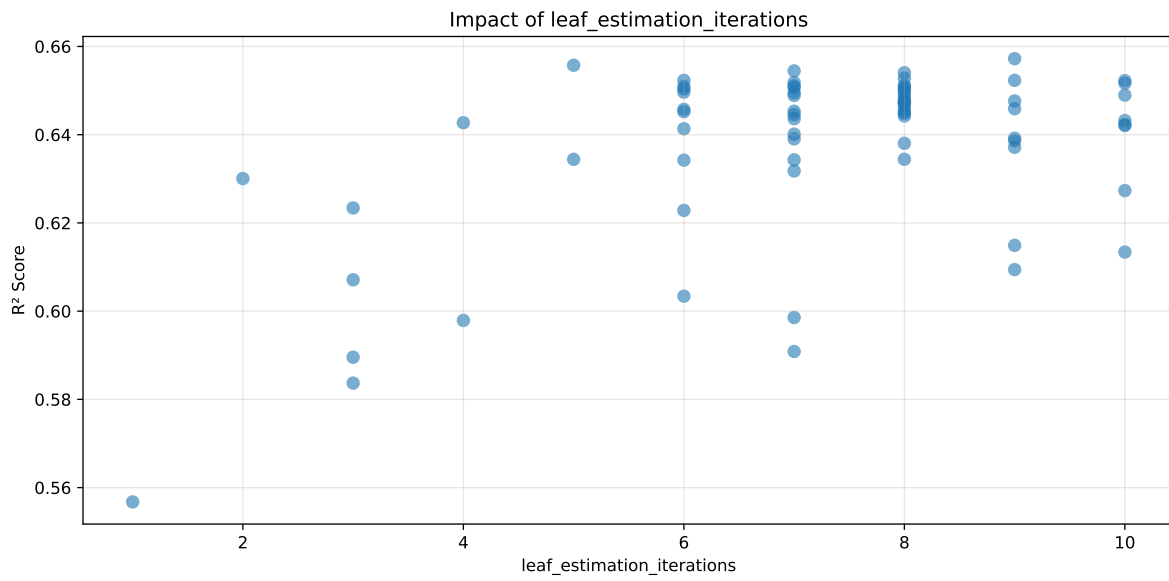
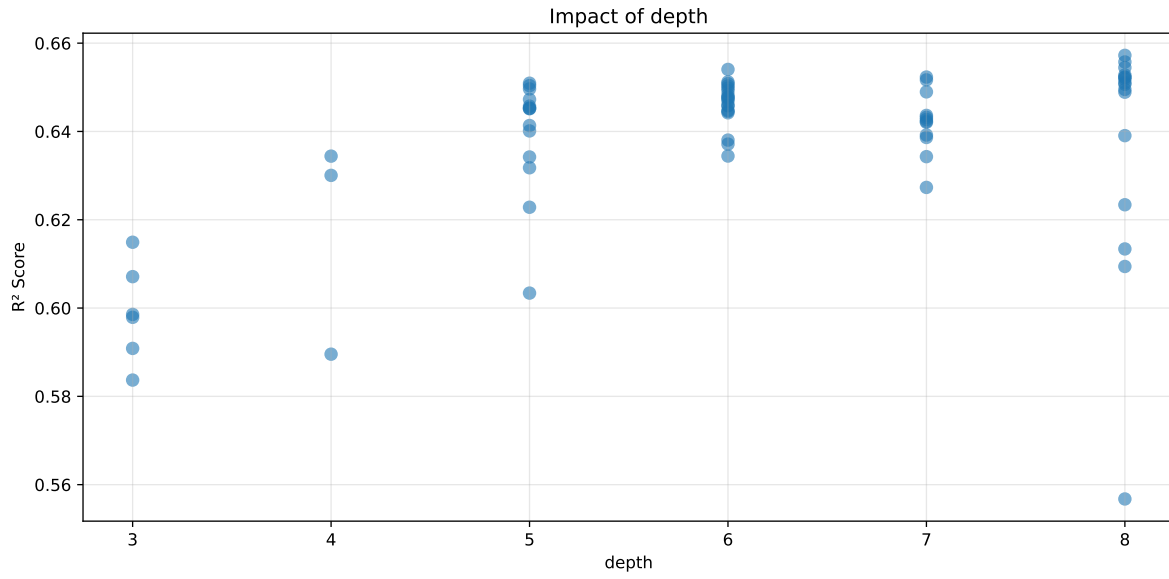
Len(group) : 1585, Len(training) : 1110, Len(test) : 475

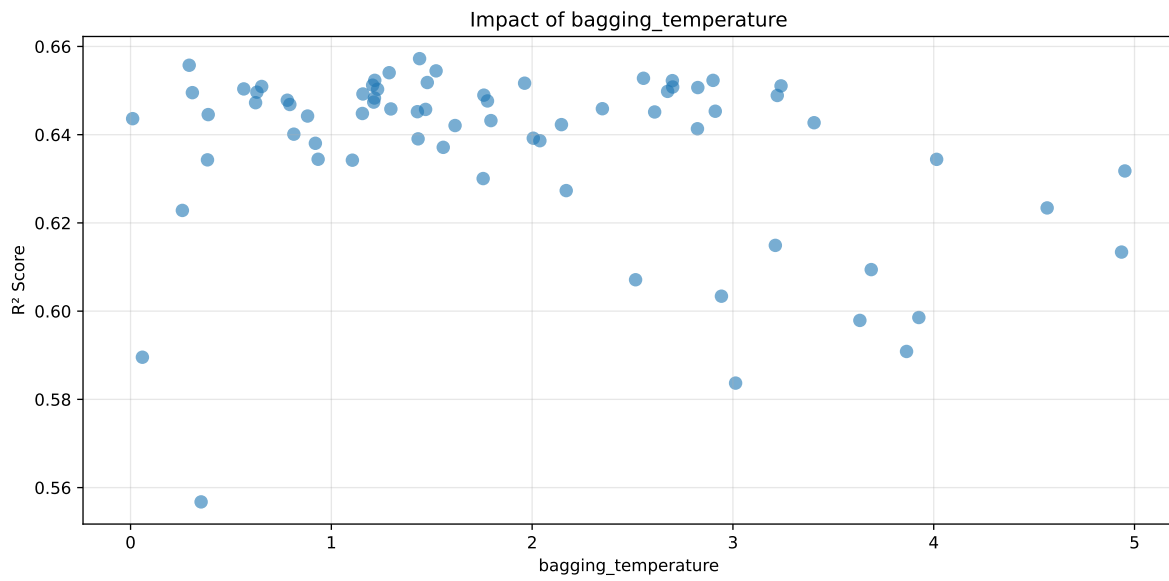
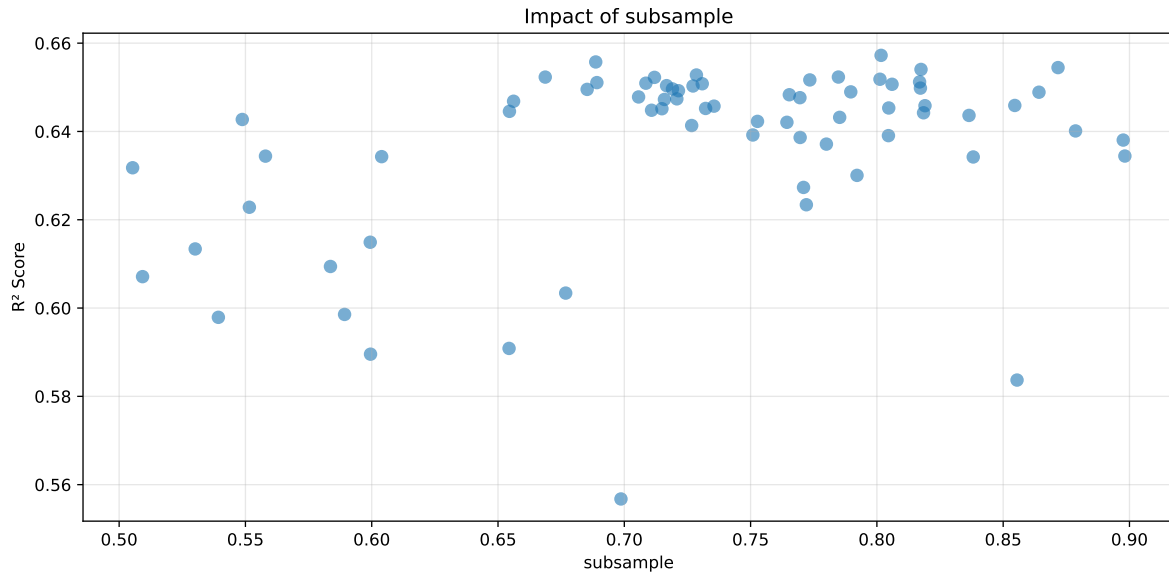
Distribution y_train vs y_test: y_train: mean=2.772, std=0.805, min=1.432, max=13.845
y_test: mean=2.807, std=0.764, min=1.373, max=8.682

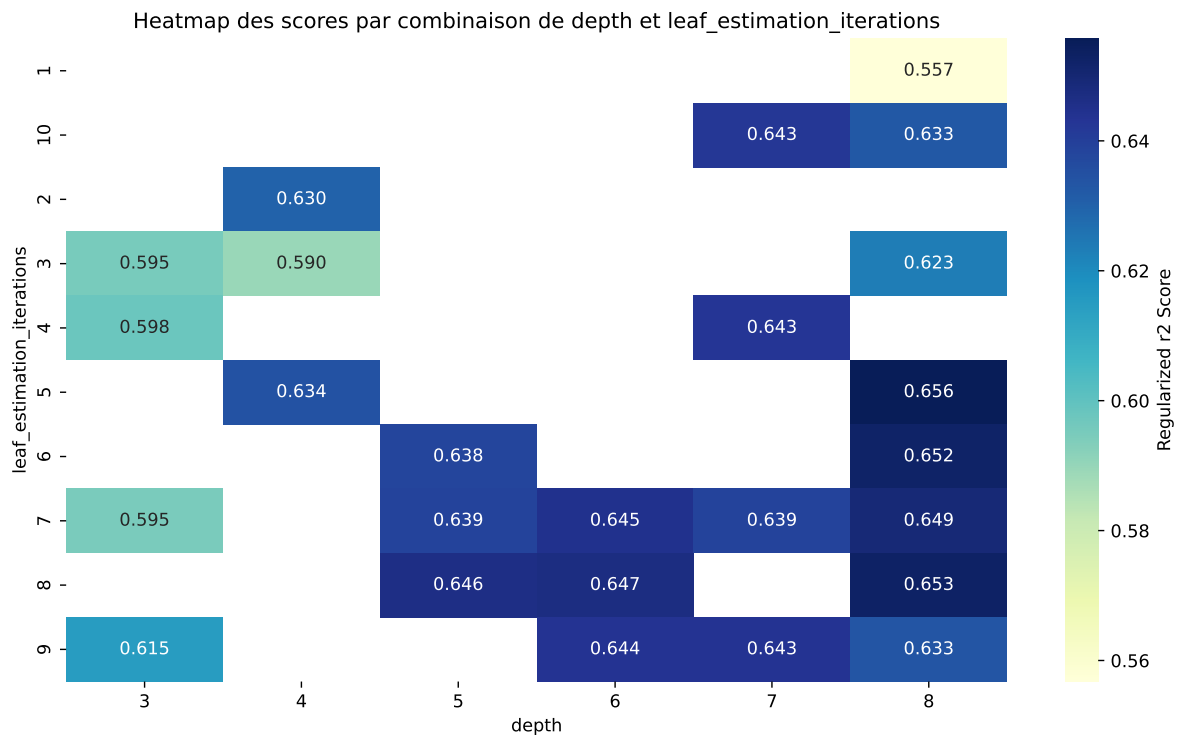
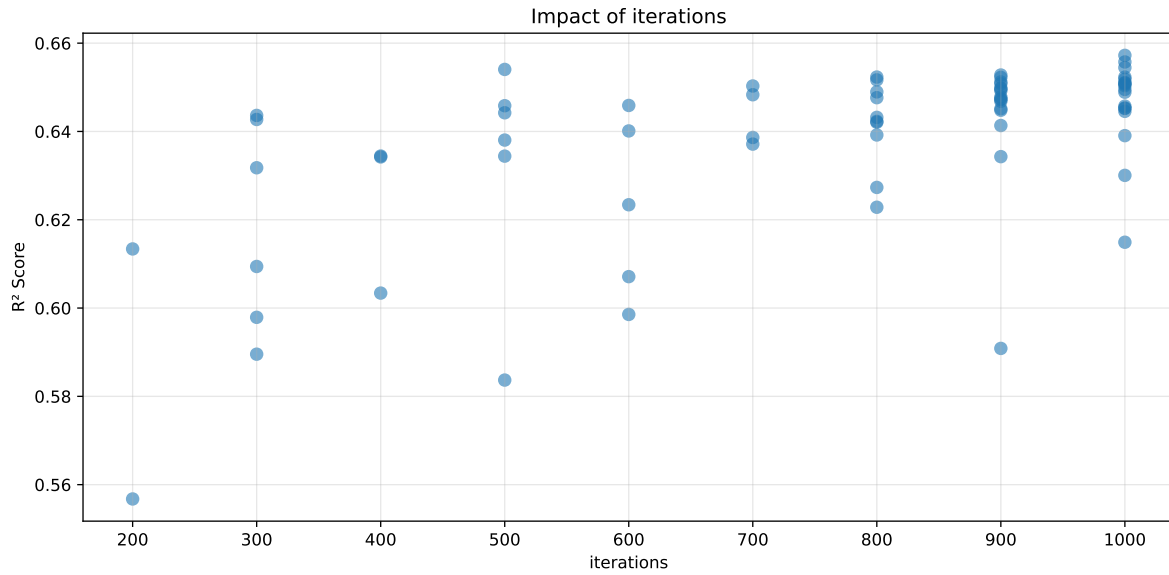
===== TOP Importance FEATURES =====
feature importance r_NH3 31.408515 r_CO2 18.847899 r_umidity 18.059839 r_THI 15.879732
r_temperature 15.804015



Paramètres avec >5% d'importance : ['depth', 'leaf_estimation_iterations', 'subsample', 'bagging_temperature', 'iterations']







rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5222	0.5568	0.5323	0.0256	0.5568	0.5582	0.0014	1.0025	0.938
2	0.6292	0.5979	0.6133	0.0424	0.5039	0.631	0.0331	1.0553	1.0524
3	0.6391	0.6034	0.6099	0.0437	0.4962	0.632	0.0287	1.0475	1.0592
4	0.6351	0.5896	0.5849	0.0385	0.453	0.5953	0.0058	1.0098	1.0772
5	0.686	0.6094	0.6028	0.0321	0.3797	0.62	0.0106	1.0173	1.1257
6	0.7735	0.6413	0.6711	0.037	0.2449	0.6918	0.0504	1.0787	1.206
7	0.7965	0.6457	0.6787	0.0365	0.1935	0.6954	0.0497	1.077	1.2335
8	0.7874	0.6371	0.6733	0.0348	0.1863	0.6896	0.0525	1.0823	1.2358
9	0.803	0.6453	0.6836	0.0359	0.1723	0.6988	0.0535	1.0829	1.2443
10	0.8	0.6423	0.6741	0.0389	0.169	0.6901	0.0479	1.0745	1.2456
---	---	---	---	---	---	---	---	---	---
70	0.9948	0.6273	0.6954	0.0532	-0.4751	0.6987	0.0714	1.1138	1.5858

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.017774436319927308, 'reg__l2_leaf_reg': 22.158563538307938, 'reg__bagging_temperature': 0.351330516211123, 'reg__random_strength': 2.428805992847129, 'reg__subsample': 0.6986769801651771, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.045046134087311805, 'reg__l2_leaf_reg': 47.64295393890445, 'reg__bagging_temperature': 3.632165103442348, 'reg__random_strength': 2.754247371702887, 'reg__subsample': 0.5392867083656018, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 4}

Rank 3: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.014087373883322978, 'reg__l2_leaf_reg': 45.22456789835694, 'reg__bagging_temperature': 2.942749500523343, 'reg__random_strength': 0.9562321108048759, 'reg__subsample': 0.6767954895633946, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 6}

Rank 4: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.023076057721330184, 'reg__l2_leaf_reg': 4.369153238578607, 'reg__bagging_temperature': 0.05895189208198093, 'reg__random_strength': 3.91398673140916, 'reg__subsample': 0.5995286480982027, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 3}

Rank 5: {'reg__iterations': 300, 'reg__depth': 8, 'reg__learning_rate': 0.011685960764958068, 'reg__l2_leaf_reg': 12.611781911806302, 'reg__bagging_temperature': 3.688750195908792, 'reg__random_strength': 4.138453137113104, 'reg__subsample': 0.5836616894475558, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 9}

Rank 6: {'reg__iterations': 900, 'reg__depth': 5, 'reg__learning_rate': 0.014253498037399172, 'reg__l2_leaf_reg': 41.83205018853873, 'reg__bagging_temperature': 2.8233122703666385, 'reg__random_strength': 1.306274457045487, 'reg__subsample': 0.726669205337501, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 6}

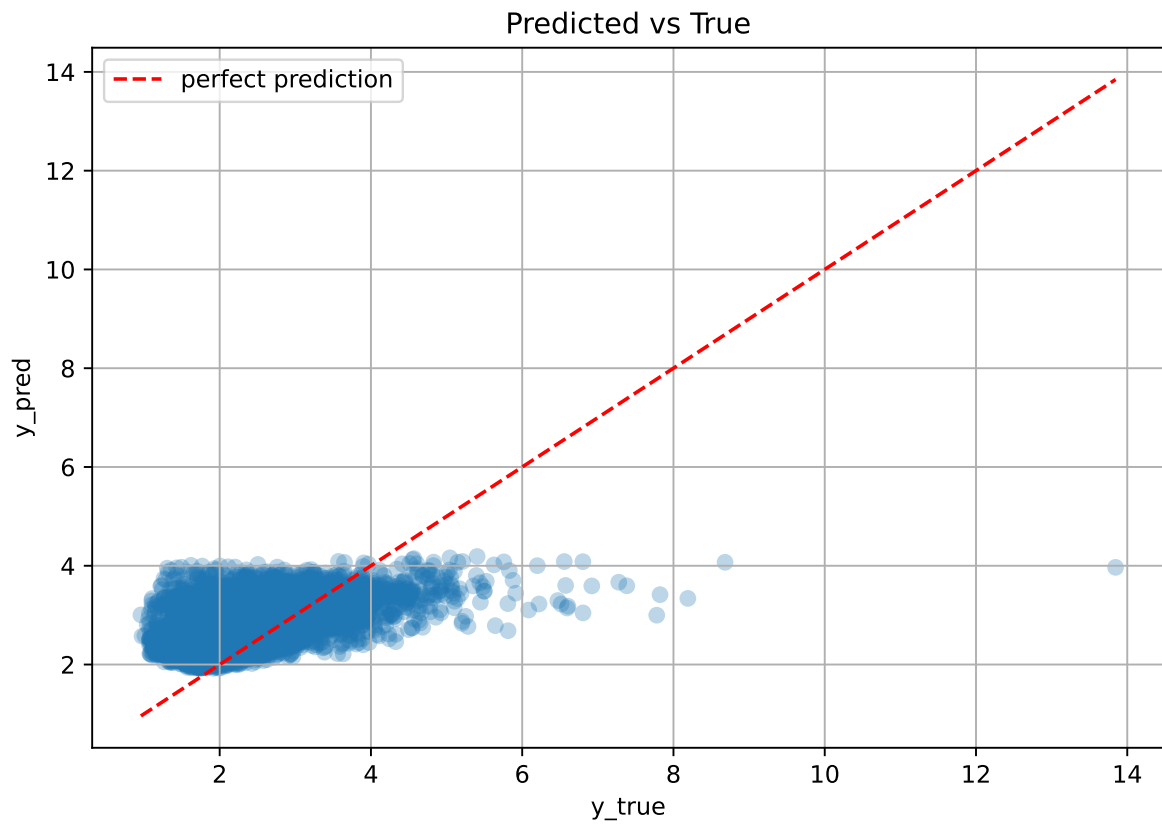
Rank 7: {'reg__iterations': 1000, 'reg__depth': 5, 'reg__learning_rate': 0.014734263828662258, 'reg__l2_leaf_reg': 45.05979828053295, 'reg__bagging_temperature': 1.469381664594419, 'reg__random_strength': 0.9914390912131772, 'reg__subsample': 0.7355167346335196, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 6}

Rank 8: {'reg__iterations': 700, 'reg__depth': 6, 'reg__learning_rate': 0.016346870276437612, 'reg__l2_leaf_reg': 37.65102575934116, 'reg__bagging_temperature': 1.5572495392434194, 'reg__random_strength': 4.816911388477891, 'reg__subsample': 0.7800120983641069, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 9}

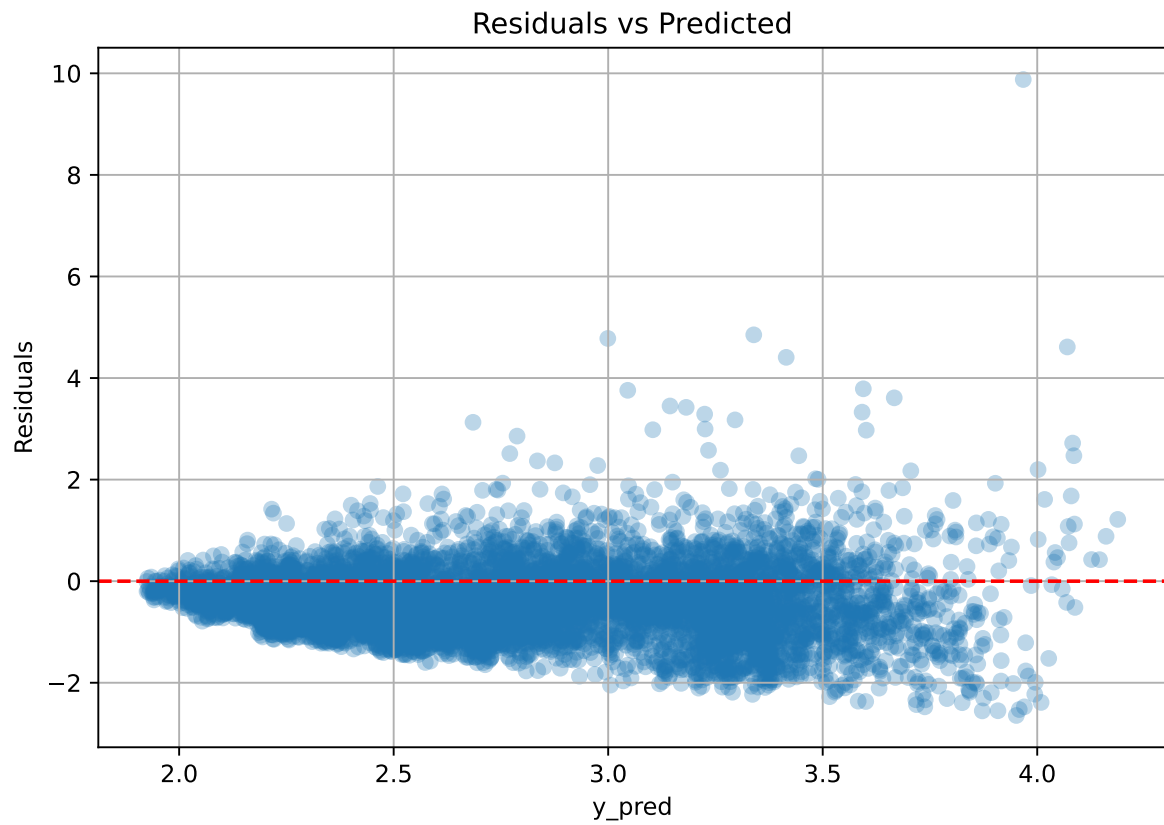
Rank 9: {'reg__iterations': 1000, 'reg__depth': 5, 'reg__learning_rate': 0.014473448486679882, 'reg__l2_leaf_reg': 45.31234594826323, 'reg__bagging_temperature': 2.9120796140860072, 'reg__random_strength': 0.9398027790825084, 'reg__subsample': 0.8046970019135661, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 7}

Rank 10: {'reg__iterations': 800, 'reg__depth': 7, 'reg__learning_rate': 0.010369274899254106, 'reg__l2_leaf_reg': 36.808584957865186, 'reg__bagging_temperature': 2.1459186487457442, 'reg__random_strength': 1.7810571651165774, 'reg__subsample': 0.752728726298446, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 10}

2.5.2 Scatter



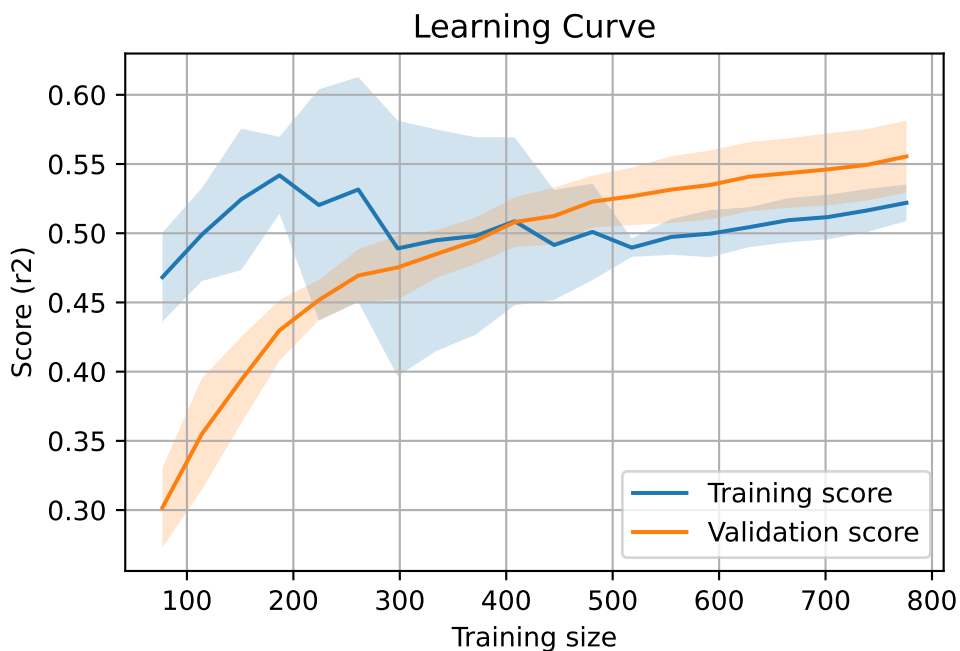
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 2025_Jun_1

`Len(group)` : 1585, `Len(training)` : 1110,

`Len(training_real)` : 777, `Len(test_of_training)` : 333, `Len(test)` : 475

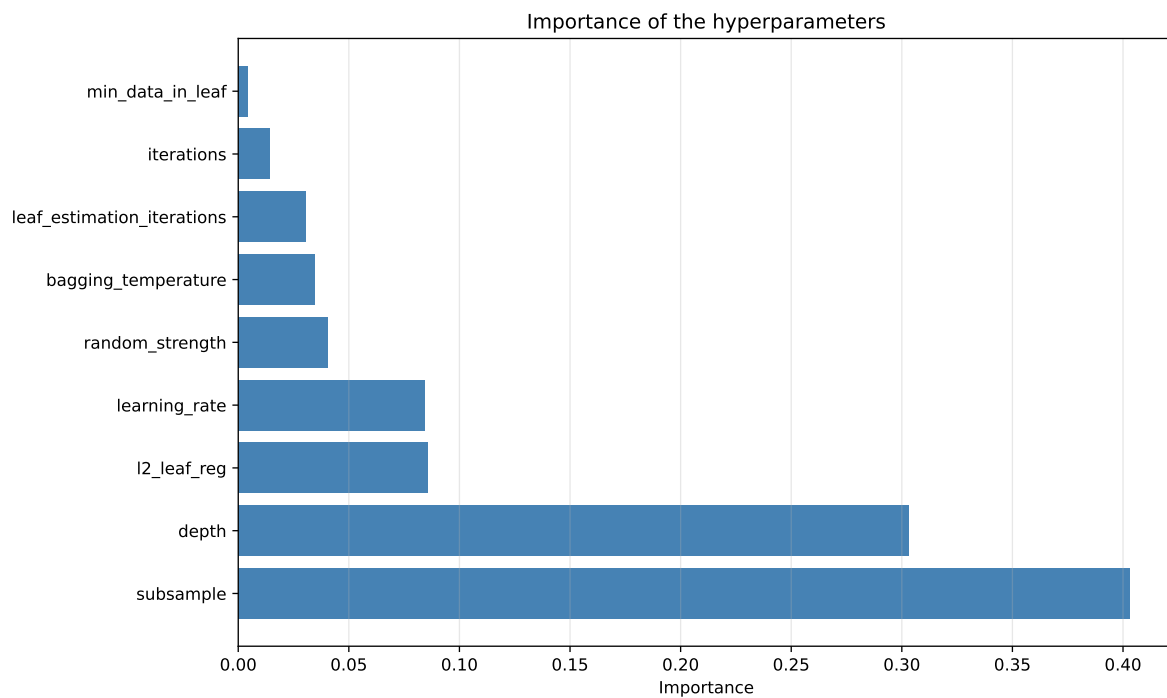
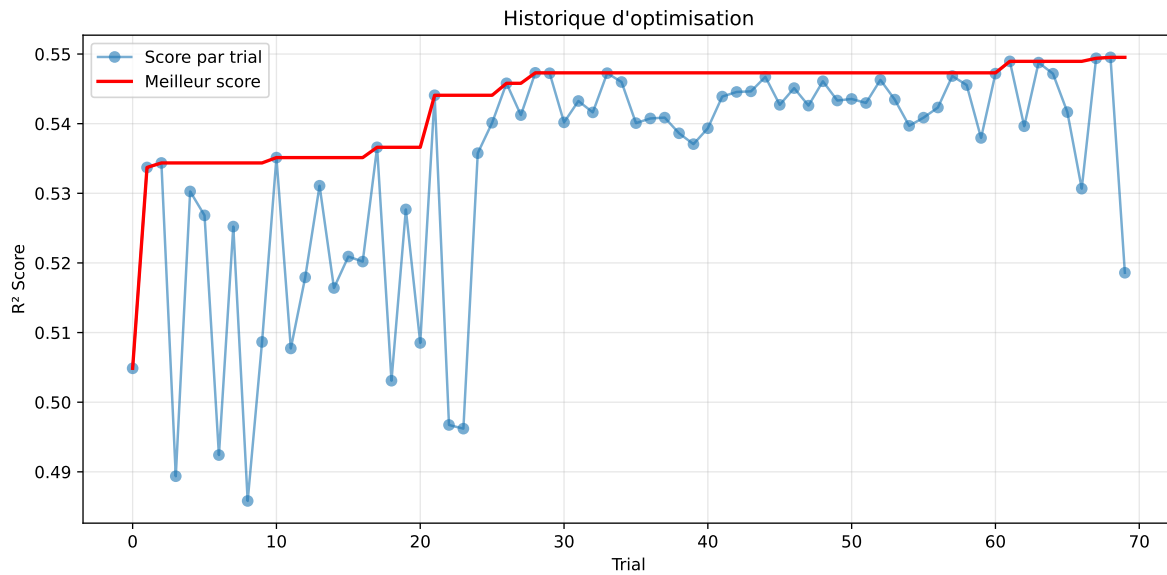


2.5.5 Gridsearch — Group: 2025_Jun_2

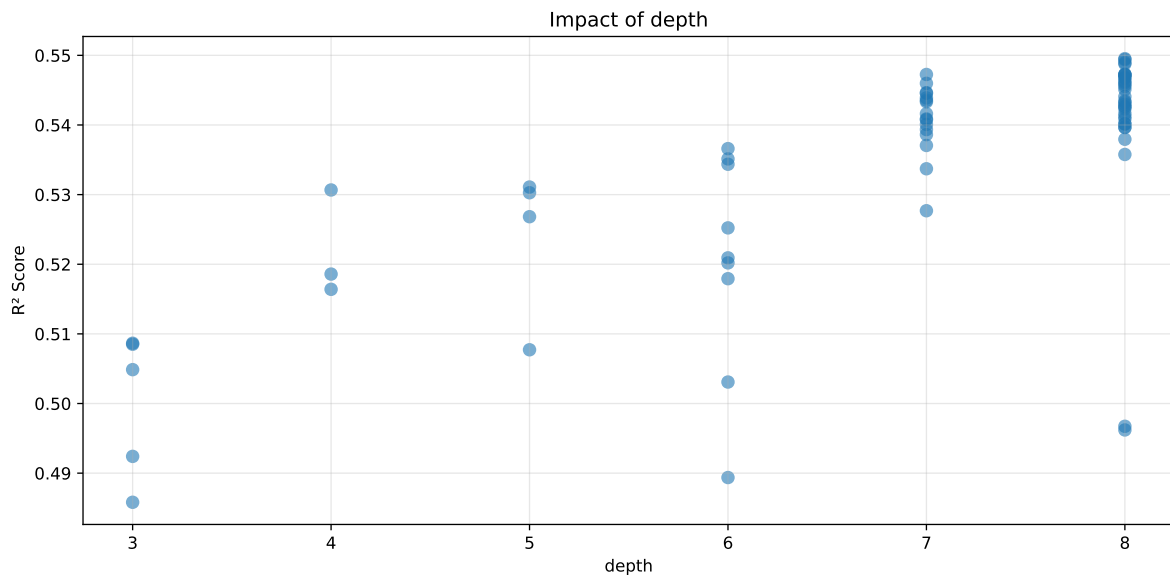
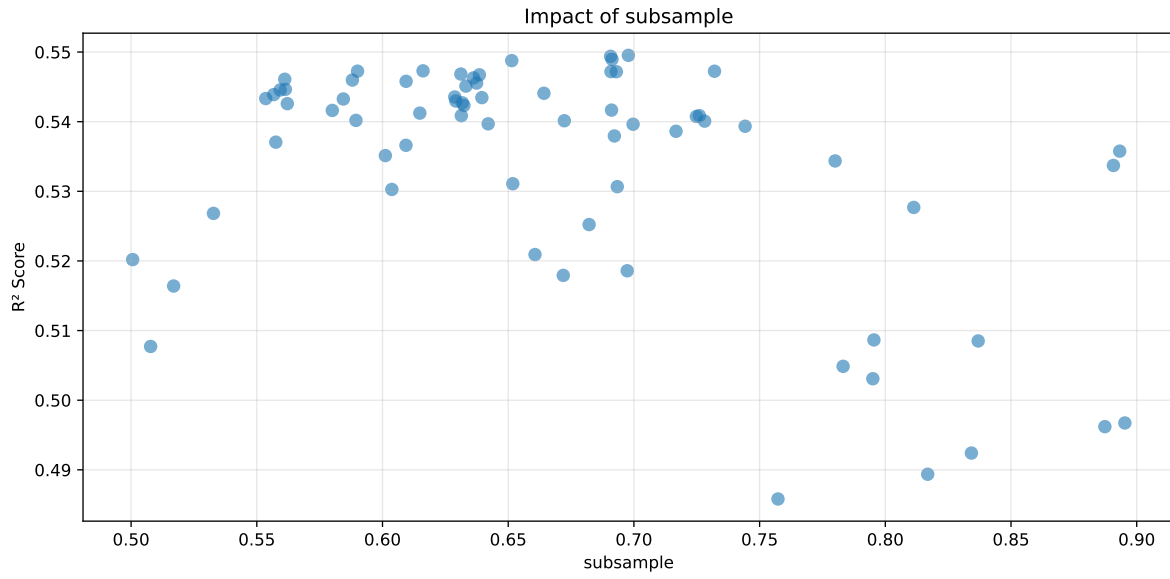
Len(group) : 1587, Len(training) : 1111, Len(test) : 476

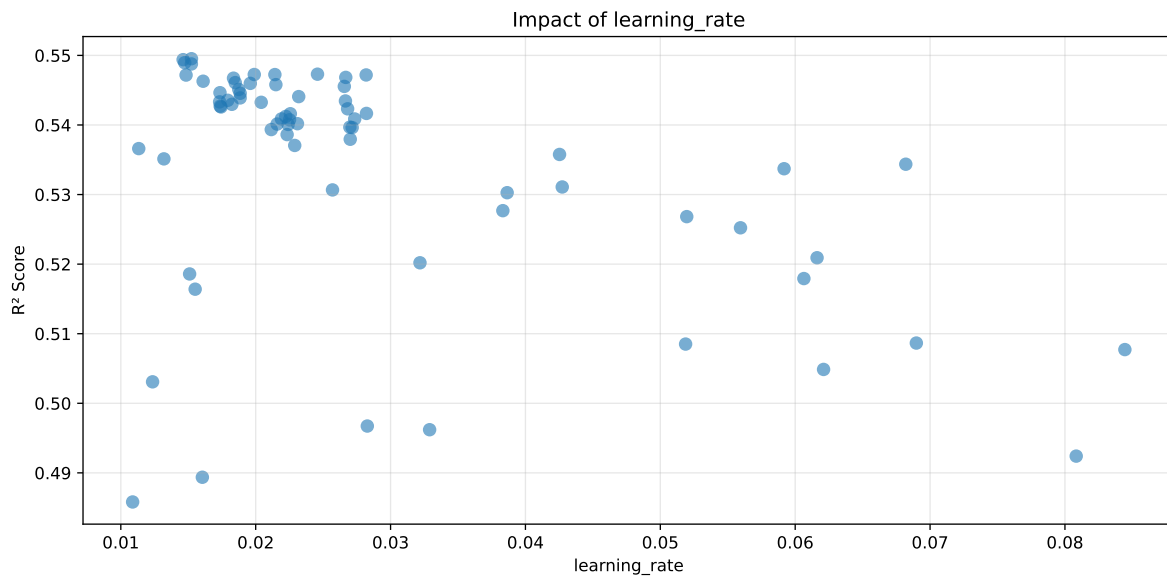
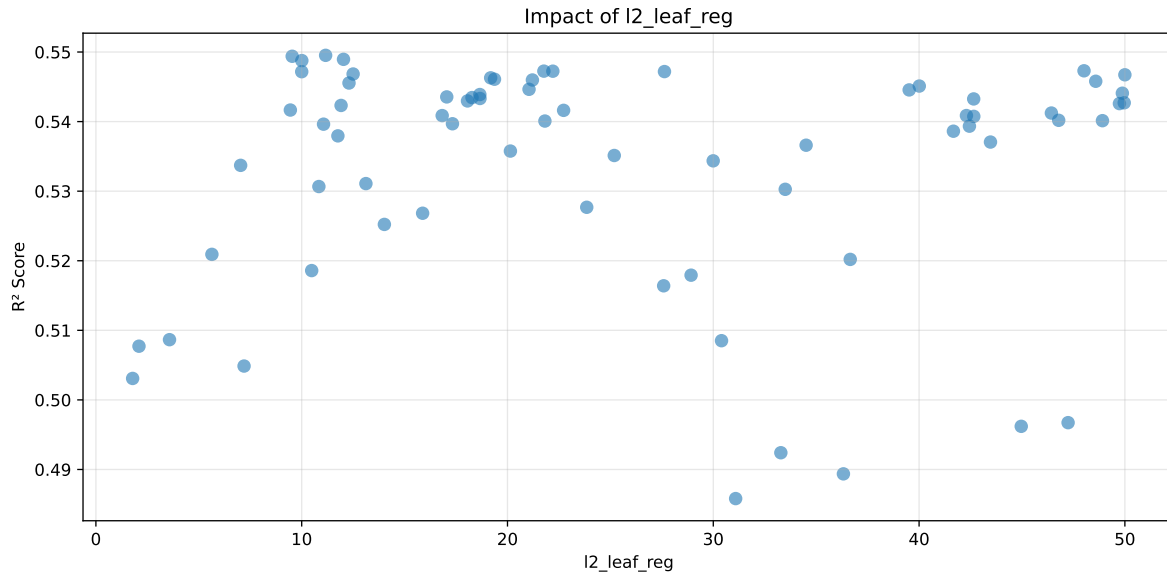
Distribution y_train vs y_test: y_train: mean=2.509, std=0.675, min=1.253, max=6.222
y_test: mean=2.528, std=0.697, min=1.305, max=7.383

===== TOP Importance FEATURES =====
feature importance r_CO2 25.450648 r_THI 24.106411 r_umidity 20.254912 r_temperature
16.005966 r_NH3 14.182063



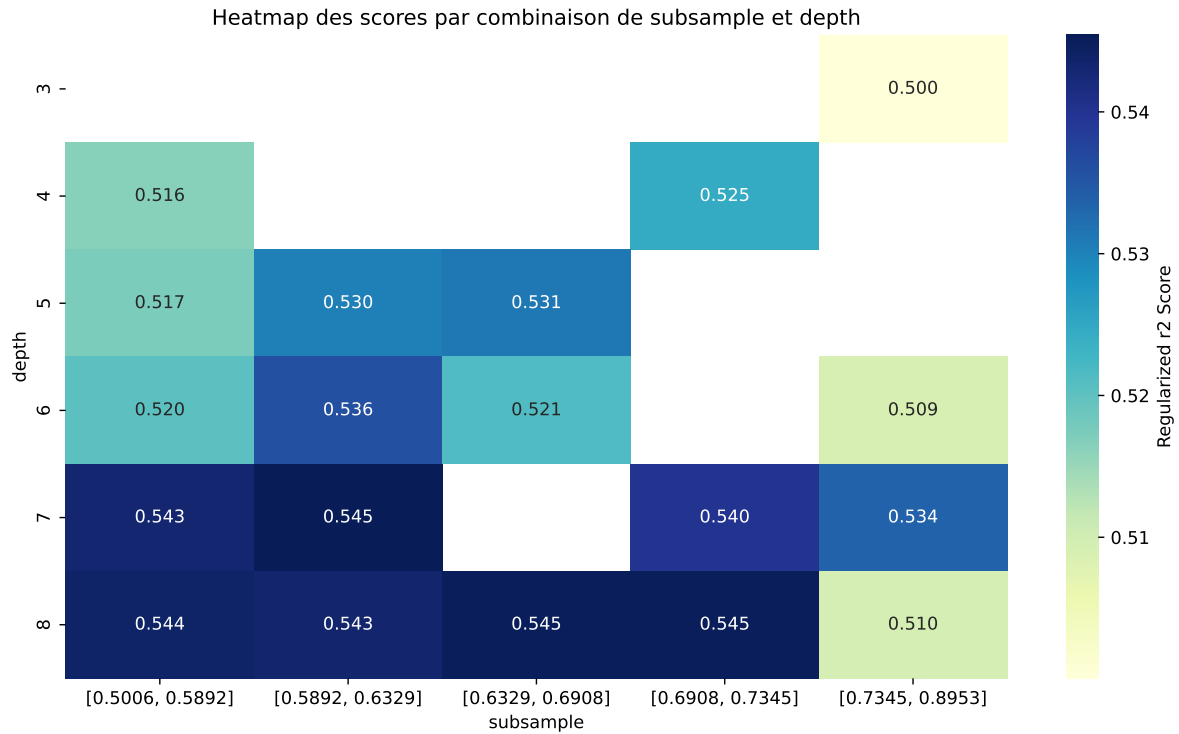
Paramètres avec >5% d'importance : ['subsample', 'depth', 'l2_leaf_reg', 'learning_rate']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6026	0.4962	0.5258	0.0143	0.1771	0.5505	0.0543	1.1095	1.2144
2	0.5947	0.4894	0.5206	0.012	0.1733	0.5407	0.0513	1.1048	1.2153
3	0.6164	0.5031	0.5342	0.0064	0.1632	0.5405	0.0375	1.0745	1.2252
4	0.6098	0.4967	0.5296	0.0148	0.1576	0.5587	0.062	1.1248	1.2276
5	0.651	0.5164	0.5565	0.0076	0.1126	0.5654	0.049	1.0949	1.2607
6	0.6544	0.5186	0.559	0.0115	0.111	0.5656	0.047	1.0907	1.262
7	0.6144	0.4858	0.5224	0.0083	0.1001	0.5289	0.043	1.0886	1.2646
8	0.6625	0.5202	0.5637	0.0123	0.0933	0.5719	0.0517	1.0995	1.2735
9	0.6453	0.5049	0.5441	0.0182	0.0834	0.5429	0.038	1.0753	1.2782
10	0.6989	0.5366	0.5848	0.0075	0.0498	0.5931	0.0565	1.1052	1.3024
---	---	---	---	---	---	---	---	---	---
70	0.9658	0.5077	0.5603	0.0135	-0.8667	0.5943	0.0866	1.1706	1.9023

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 8, 'reg__learning_rate': 0.03289514048596848, 'reg__l2_leaf_reg': 44.96412359276624, 'reg__bagging_temperature': 4.890903178060327, 'reg__random_strength': 3.3508981700841933, 'reg__subsample': 0.8873423351992932, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 1}

Rank 2: {'reg__iterations': 500, 'reg__depth': 6, 'reg__learning_rate': 0.016045270678954518, 'reg__l2_leaf_reg': 36.32305866835038, 'reg__bagging_temperature': 3.7986914112877286, 'reg__random_strength': 0.9970823648632543, 'reg__subsample': 0.8168609021507478, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 1}

Rank 3: {'reg__iterations': 400, 'reg__depth': 6, 'reg__learning_rate': 0.012355152233205963, 'reg__l2_leaf_reg': 1.784997198210827, 'reg__bagging_temperature': 1.9787862629485897, 'reg__random_strength': 4.490238522559953, 'reg__subsample': 0.7950450343681049, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 3}

Rank 4: {'reg__iterations': 400, 'reg__depth': 8, 'reg__learning_rate': 0.02827654548890208, 'reg__l2_leaf_reg': 47.24087219062136, 'reg__bagging_temperature': 4.940542739033052, 'reg__random_strength': 3.4640160356904888, 'reg__subsample': 0.8952908109511053, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 1}

Rank 5: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.015512151636470054, 'reg__l2_leaf_reg': 27.588363813459146, 'reg__bagging_temperature': 3.7306322783824193, 'reg__random_strength': 1.2994262879826302, 'reg__subsample': 0.5169161914285272, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 5}

Rank 6: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.015101117633065864, 'reg__l2_leaf_reg': 10.48511883660854, 'reg__bagging_temperature': 4.589697554503868, 'reg__random_strength': 1.9253650859186933, 'reg__subsample': 0.6973356991845987, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 2}

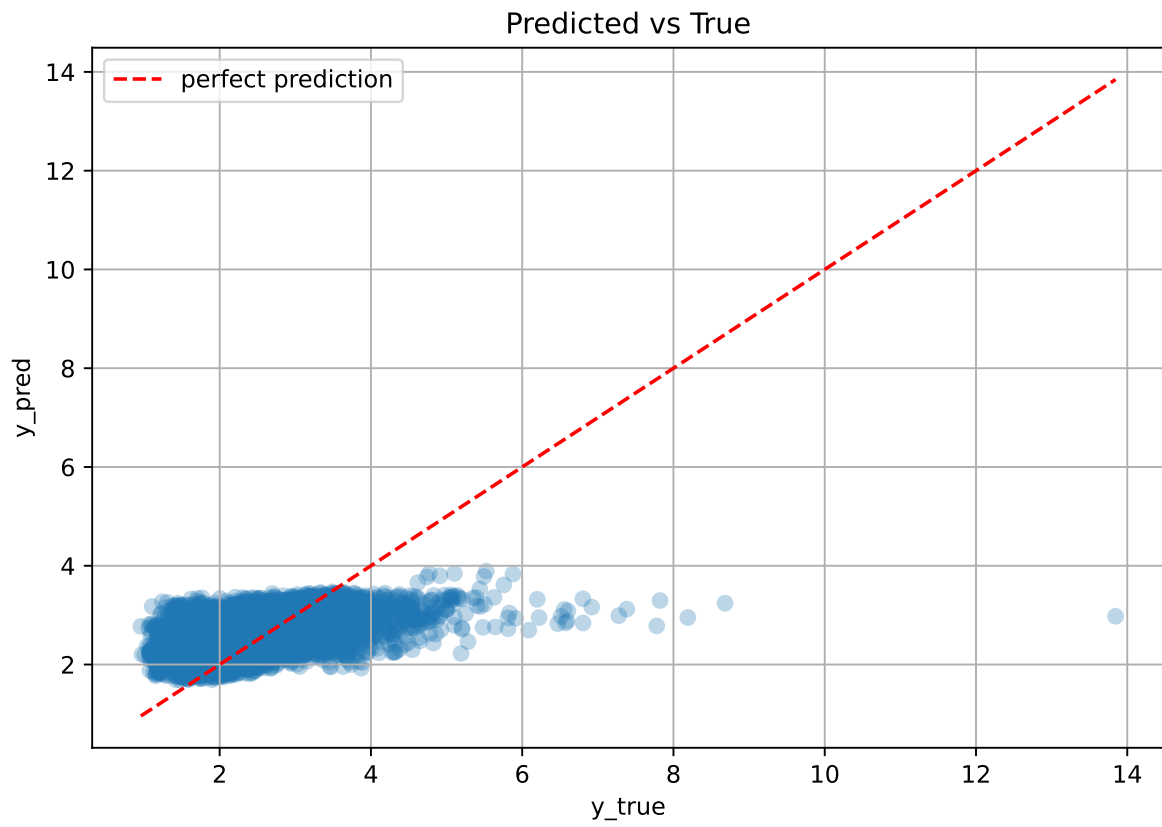
Rank 7: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.01087650456800175, 'reg__l2_leaf_reg': 31.086478795548153, 'reg__bagging_temperature': 1.6659838196650316, 'reg__random_strength': 0.24558662610470738, 'reg__subsample': 0.7573120550494352, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 4}

Rank 8: {'reg__iterations': 200, 'reg__depth': 6, 'reg__learning_rate': 0.03218075277361675, 'reg__l2_leaf_reg': 36.65523308569996, 'reg__bagging_temperature': 4.484248722612957, 'reg__random_strength': 1.4935539856186613, 'reg__subsample': 0.500609208259236, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 7}

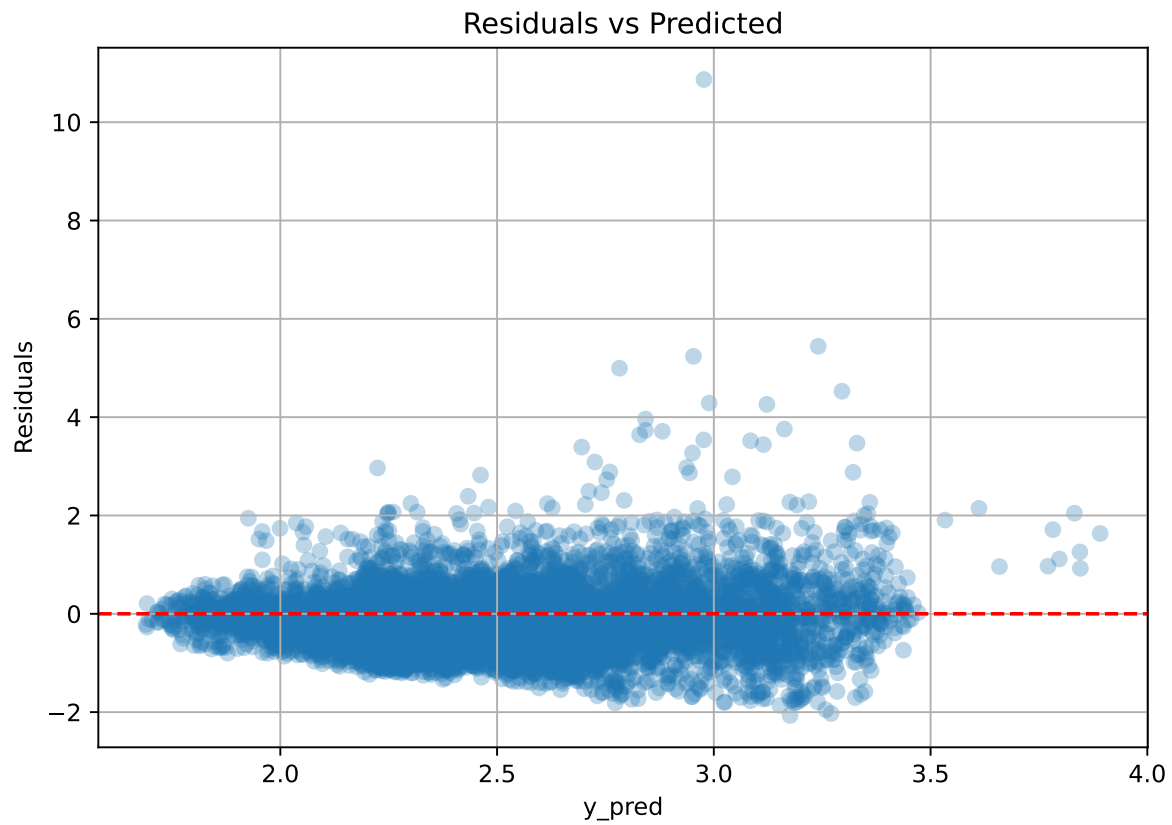
Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.06210391758444661, 'reg__l2_leaf_reg': 7.199366140373704, 'reg__bagging_temperature': 3.767160524654438, 'reg__random_strength': 2.2619721963368224, 'reg__subsample': 0.7832108471265952, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 3}

Rank 10: {'reg__iterations': 800, 'reg__depth': 6, 'reg__learning_rate': 0.011329761951327672, 'reg__l2_leaf_reg': 34.511023893106895, 'reg__bagging_temperature': 2.2634984890760146, 'reg__random_strength': 3.371223222565876, 'reg__subsample': 0.6093337505318145, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 8}

2.5.6 Scatter



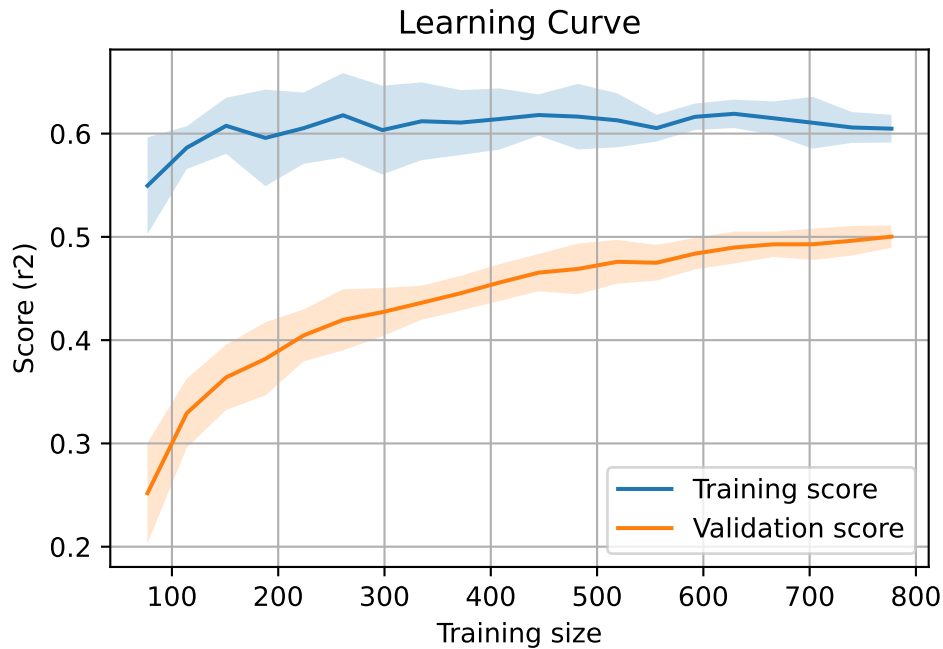
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 2025_Jun_2

`Len(group)` : 1587, `Len(training)` : 1111,

`Len(training_real)` : 778, `Len(test_of_training)` : 333, `Len(test)` : 476

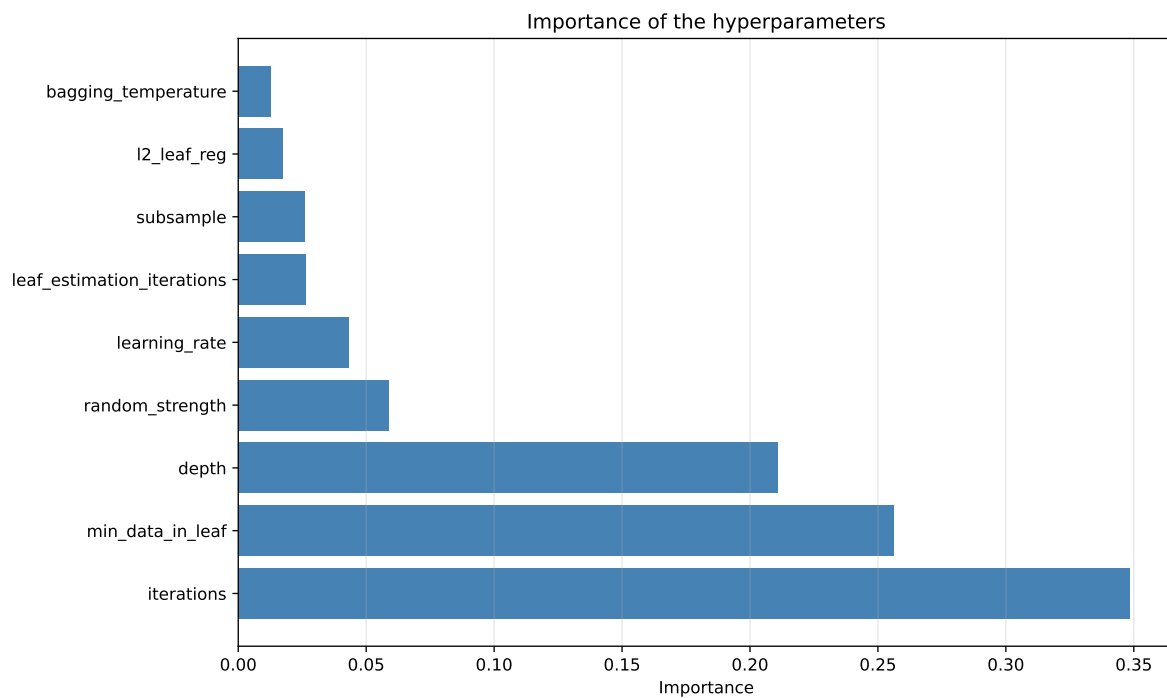
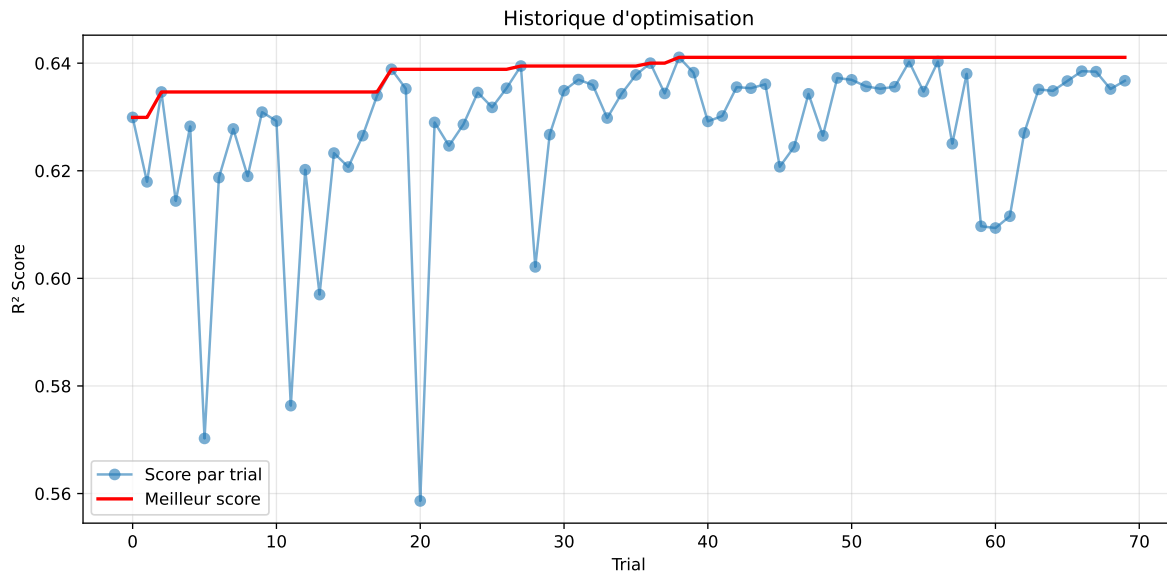


2.5.9 Gridsearch — Group: 2025_Sep_1

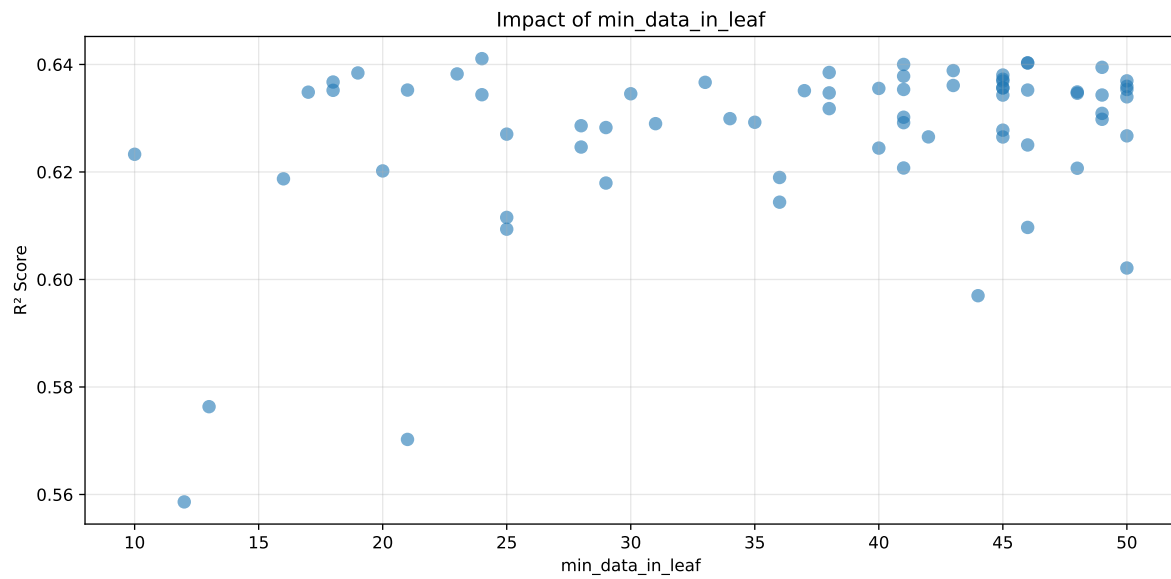
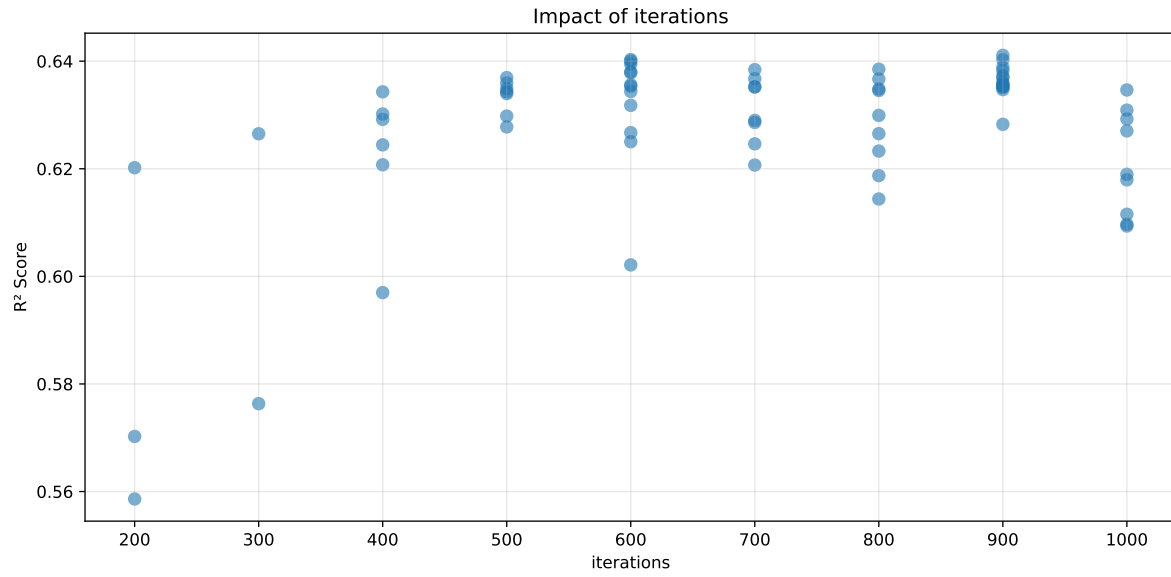
Len(group) : 3867, Len(training) : 2707, Len(test) : 1160

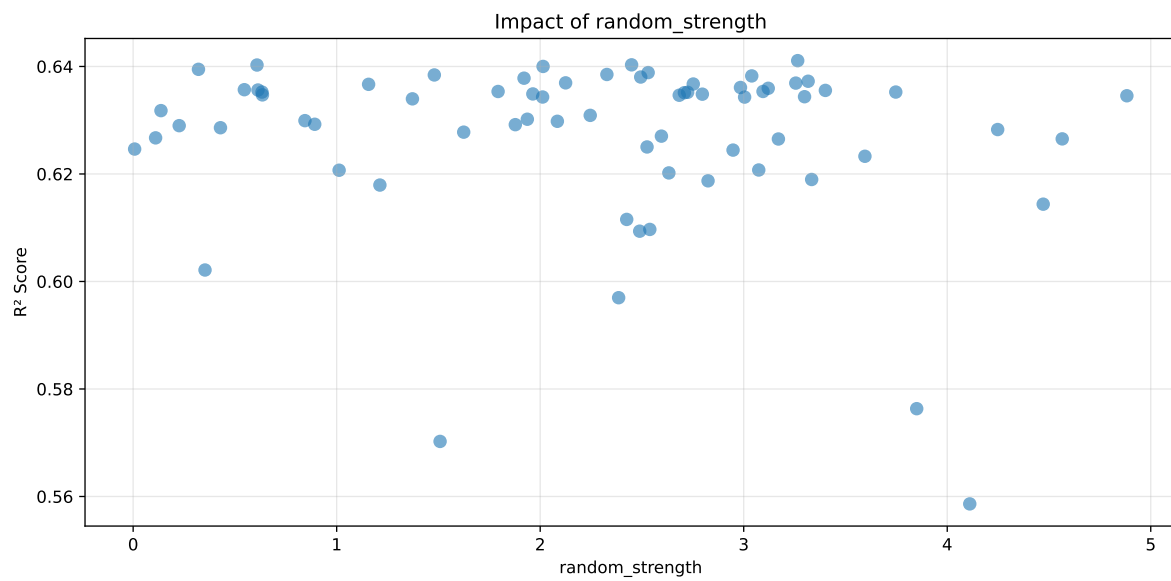
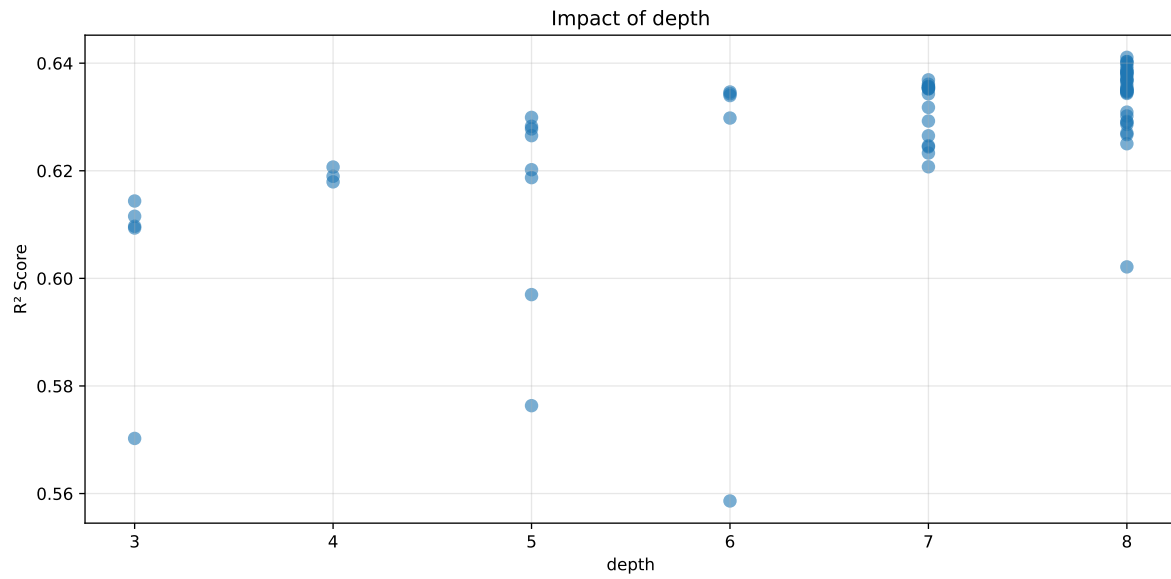
Distribution y_train vs y_test: y_train: mean=2.231, std=0.761, min=0.973, max=8.190
y_test: mean=2.216, std=0.723, min=1.110, max=6.920

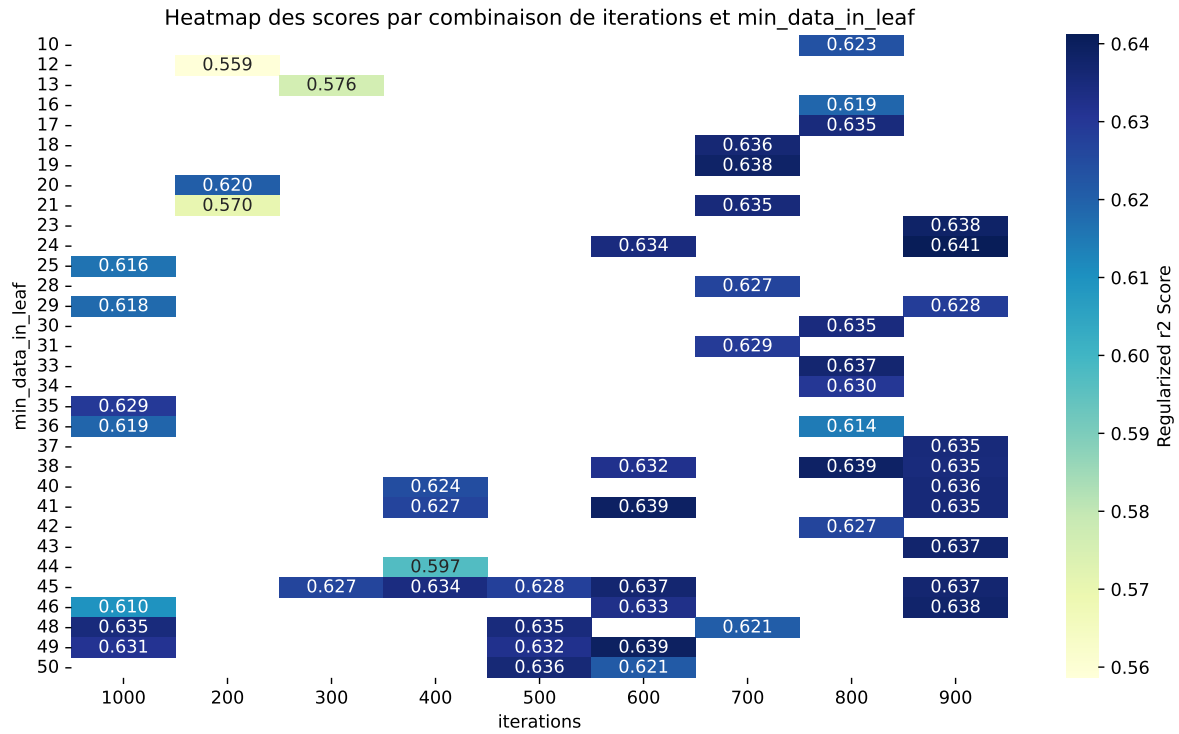
===== TOP Importance FEATURES =====
feature importance r_NH3 36.486955 r_umidity 34.960856 r_CO2 12.771226 r_temperature
8.791788 r_THI 6.989175



Paramètres avec >5% d'importance : ['iterations', 'min_data_in_leaf', 'depth', 'random_strength']







rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5912	0.5763	0.5877	0.0283	0.5763	0.5964	0.02	1.0347	1.0259
2	0.5818	0.5702	0.5817	0.0262	0.5702	0.5876	0.0173	1.0304	1.0202
3	0.5735	0.5586	0.5693	0.0283	0.5586	0.5796	0.021	1.0376	1.0266
4	0.6295	0.597	0.6039	0.0287	0.4995	0.6129	0.016	1.0267	1.0545
5	0.6676	0.6021	0.6021	0.0304	0.4058	0.6175	0.0154	1.0255	1.1087
6	0.6986	0.6233	0.6258	0.0287	0.3973	0.6358	0.0125	1.0201	1.1208
7	0.6963	0.6207	0.6224	0.029	0.3942	0.6319	0.0112	1.018	1.1217
8	0.7017	0.6202	0.6216	0.0264	0.3757	0.6341	0.0139	1.0224	1.1314
9	0.7156	0.6244	0.6282	0.027	0.3509	0.6357	0.0113	1.0181	1.146
10	0.7079	0.6144	0.6199	0.0214	0.3338	0.6333	0.0189	1.0307	1.1522
---	---	---	---	---	---	---	---	---	---
70	0.9693	0.625	0.6254	0.0159	-0.4077	0.6549	0.0299	1.0478	1.5508

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.01813532804537802, 'reg__l2_leaf_reg': 33.43622397070643, 'reg__bagging_temperature': 2.274365860024918, 'reg__random_strength': 3.850018089520475, 'reg__subsample': 0.8638059856656213, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 6}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.03379306661332952, 'reg__l2_leaf_reg': 5.9953337647479295, 'reg__bagging_temperature': 0.7630841816419914, 'reg__random_strength': 1.508095584907649, 'reg__subsample': 0.7011873552530451, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 3}

Rank 3: {'reg__iterations': 200, 'reg__depth': 6, 'reg__learning_rate': 0.018186212081820672, 'reg__l2_leaf_reg': 34.2545649501828, 'reg__bagging_temperature': 1.877534720921819, 'reg__random_strength': 4.1106923882877435, 'reg__subsample': 0.5845212402236073, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 4}

Rank 4: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.018194448037414646, 'reg__l2_leaf_reg': 30.675599424048116, 'reg__bagging_temperature': 2.7106974187955246, 'reg__random_strength': 2.385456513567811, 'reg__subsample': 0.6836514418003904, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 4}

Rank 5: {'reg__iterations': 600, 'reg__depth': 8, 'reg__learning_rate': 0.010051961931652595, 'reg__l2_leaf_reg': 21.313355856374848, 'reg__bagging_temperature': 3.593138839646124, 'reg__random_strength': 0.3531326849745122, 'reg__subsample': 0.8991519102601471, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 1}

Rank 6: {'reg__iterations': 800, 'reg__depth': 7, 'reg__learning_rate': 0.012378464839149849, 'reg__l2_leaf_reg': 39.81659624381865, 'reg__bagging_temperature': 1.9555320954226685, 'reg__random_strength': 3.5955679400962515, 'reg__subsample': 0.6476940866032238, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 7}

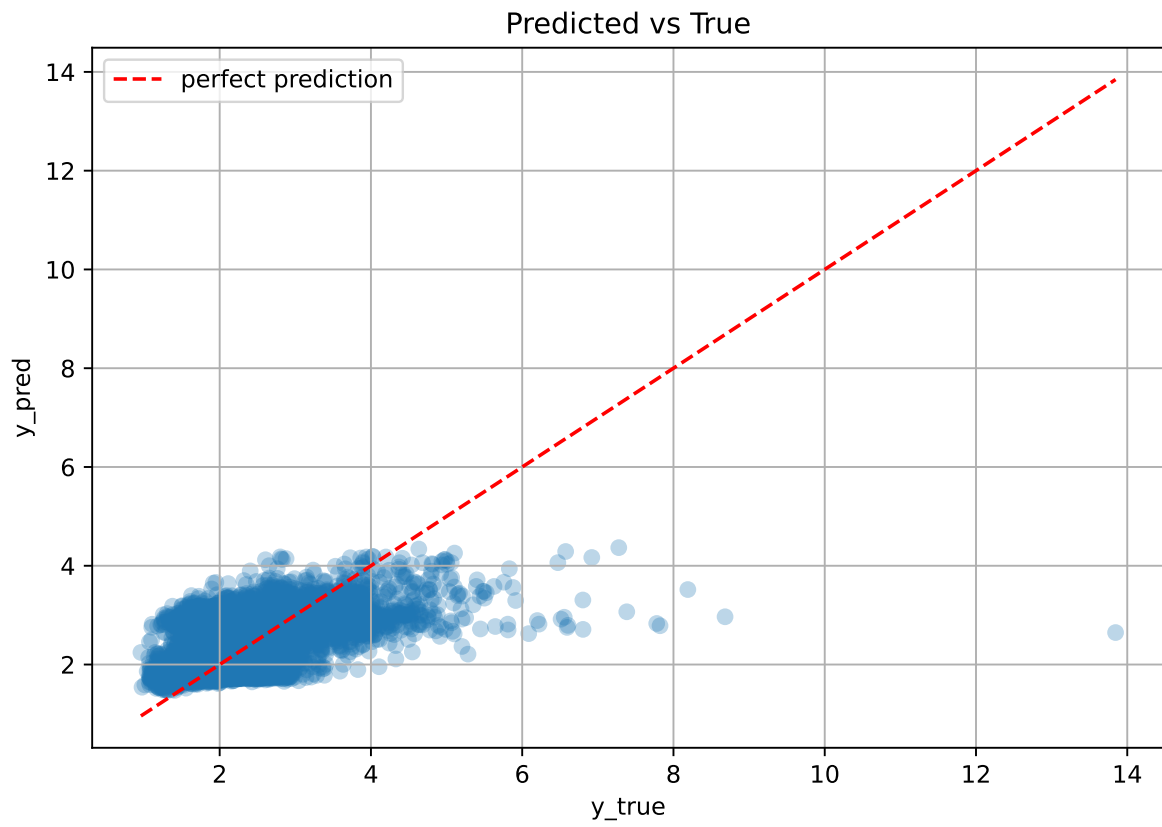
Rank 7: {'reg__iterations': 400, 'reg__depth': 7, 'reg__learning_rate': 0.021802595443471053, 'reg__l2_leaf_reg': 27.314592204680817, 'reg__bagging_temperature': 1.5488705088466355, 'reg__random_strength': 3.073871218366343, 'reg__subsample': 0.7241435345593703, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 8}

Rank 8: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.08918963150005478, 'reg__l2_leaf_reg': 47.44456179982939, 'reg__bagging_temperature': 3.15049136384686, 'reg__random_strength': 2.6321856899304468, 'reg__subsample': 0.5576117600828625, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 3}

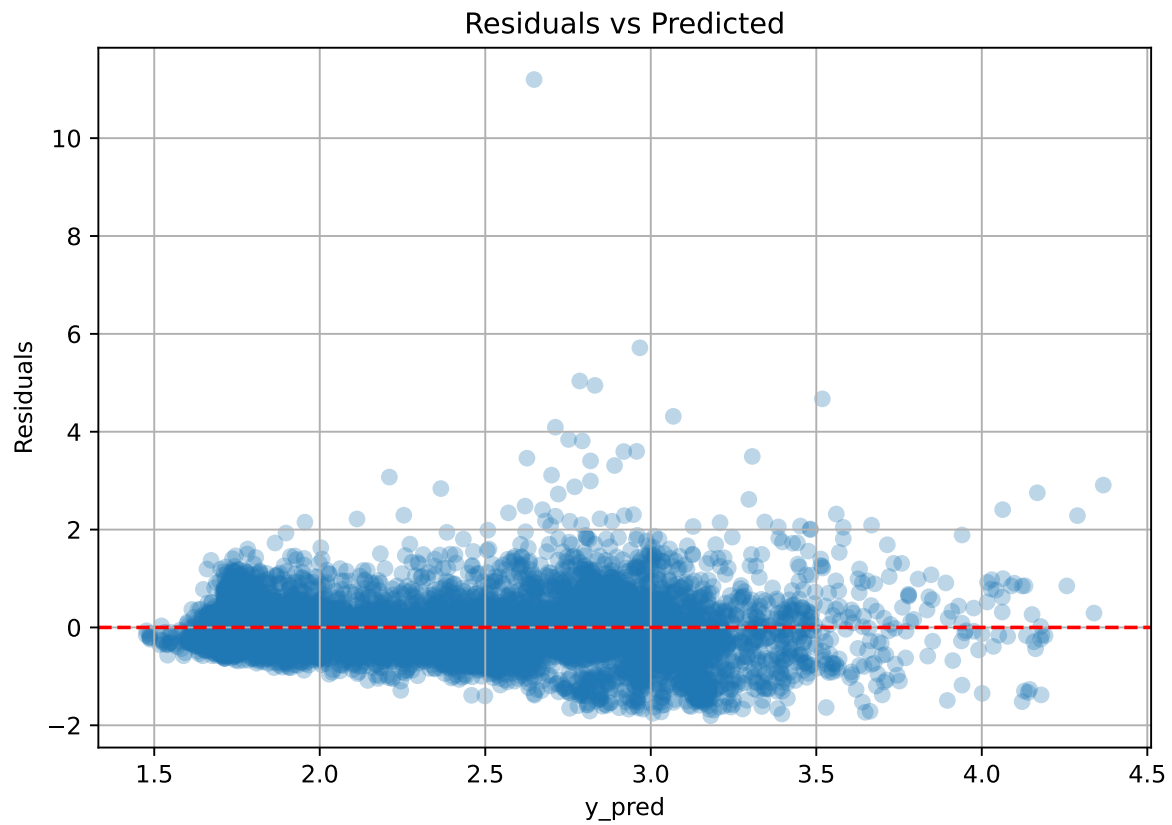
Rank 9: {'reg__iterations': 400, 'reg__depth': 7, 'reg__learning_rate': 0.021976523405787683, 'reg__l2_leaf_reg': 9.144478395014767, 'reg__bagging_temperature': 1.3960931246042154, 'reg__random_strength': 2.94759450222221, 'reg__subsample': 0.876555455353602, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 8}

Rank 10: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.04011919760580765, 'reg__l2_leaf_reg': 44.19272762382156, 'reg__bagging_temperature': 4.4987150921604595, 'reg__random_strength': 4.471558603956003, 'reg__subsample': 0.6794897720751019, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 5}

2.5.10 Scatter



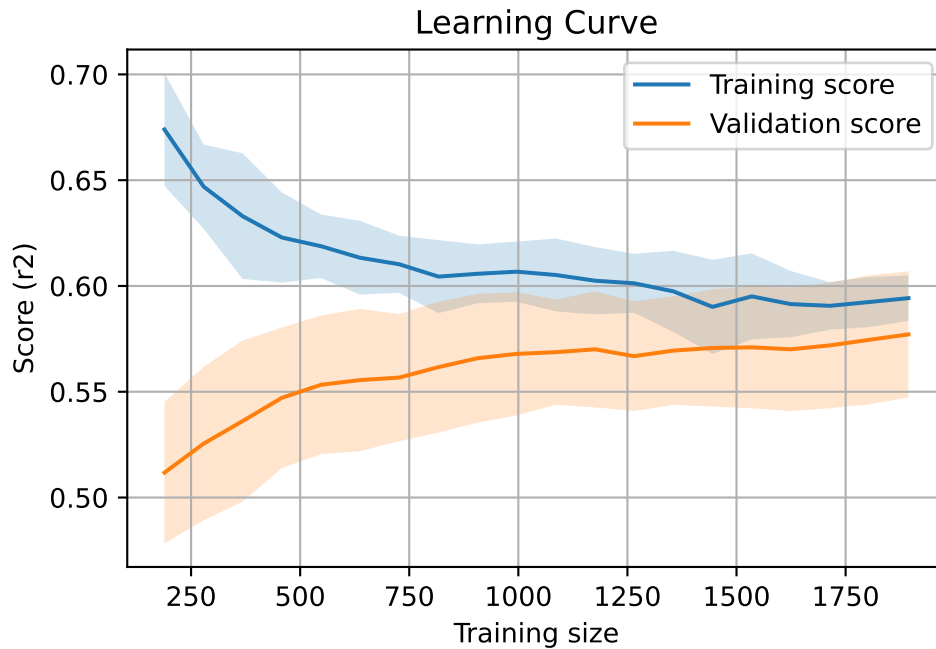
2.5.11 Scatter residuals



2.5.12 Learning curve — Group: 2025_Sep_1

`Len(group)` : 3867, `Len(training)` : 2707,

`Len(training_real)` : 1895, `Len(test_of_training)` : 812, `Len(test)` : 1160

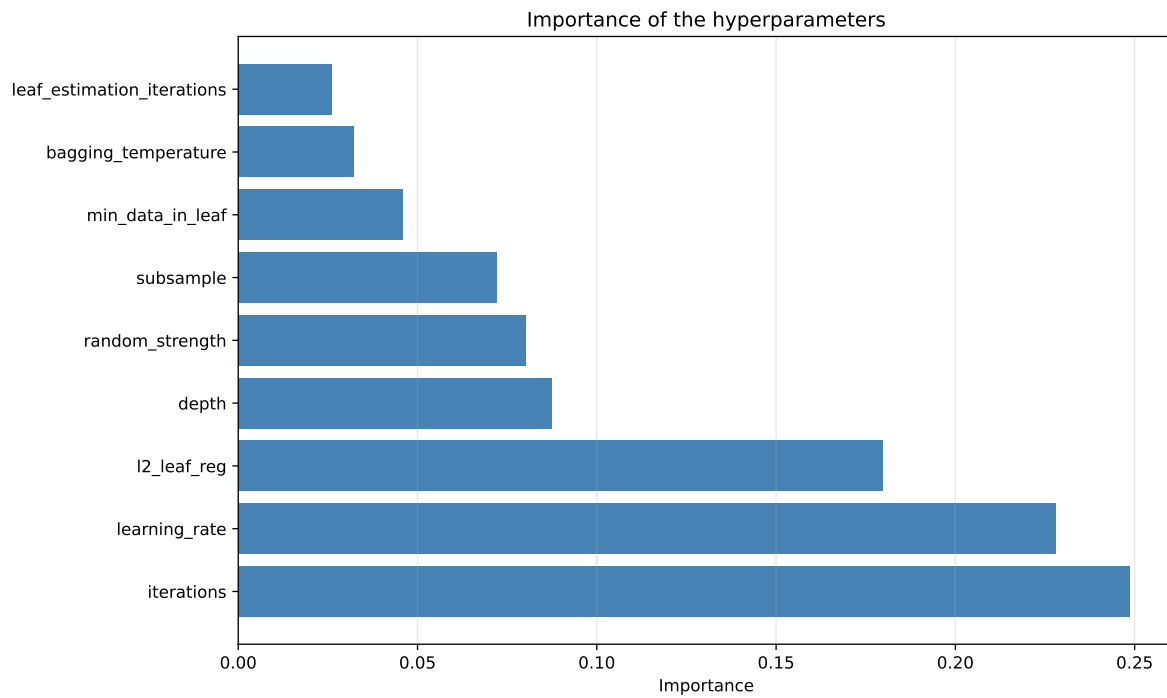
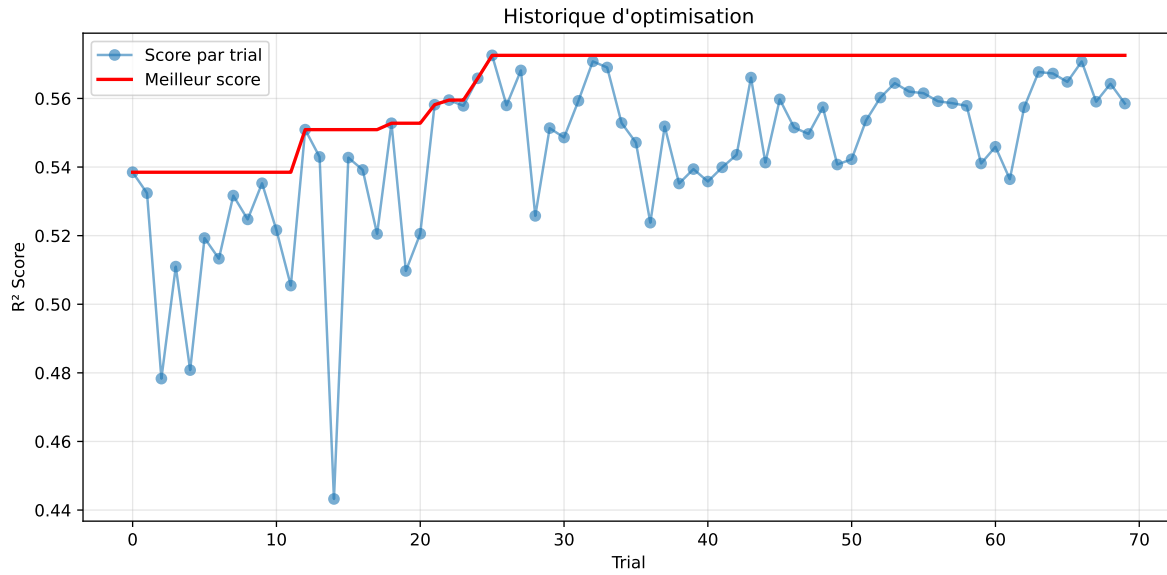


2.5.13 Gridsearch — Group: 2025_Sep_2

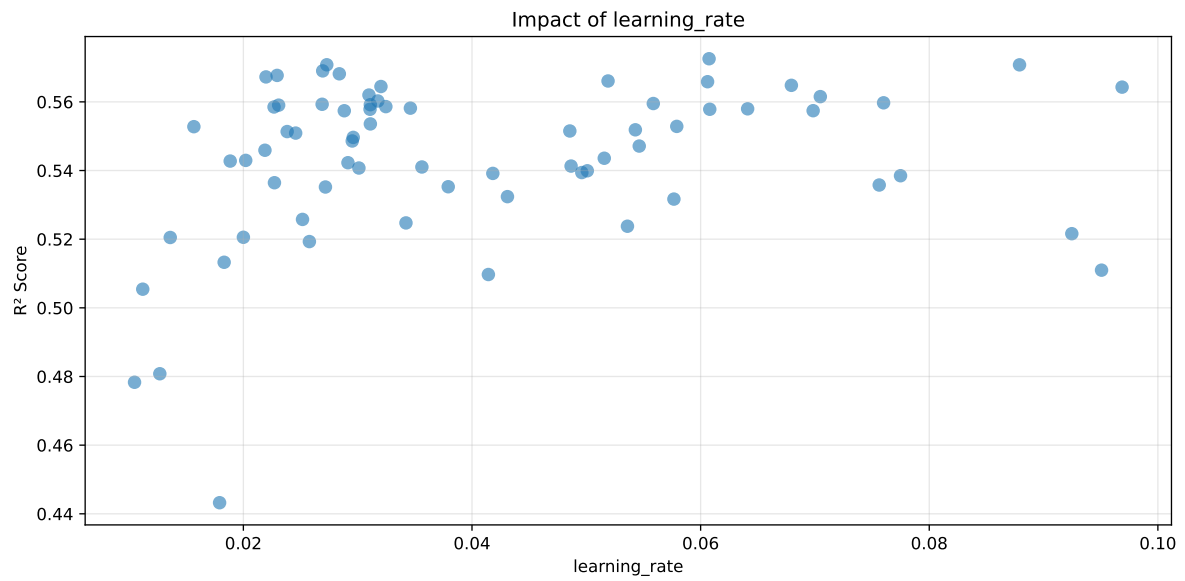
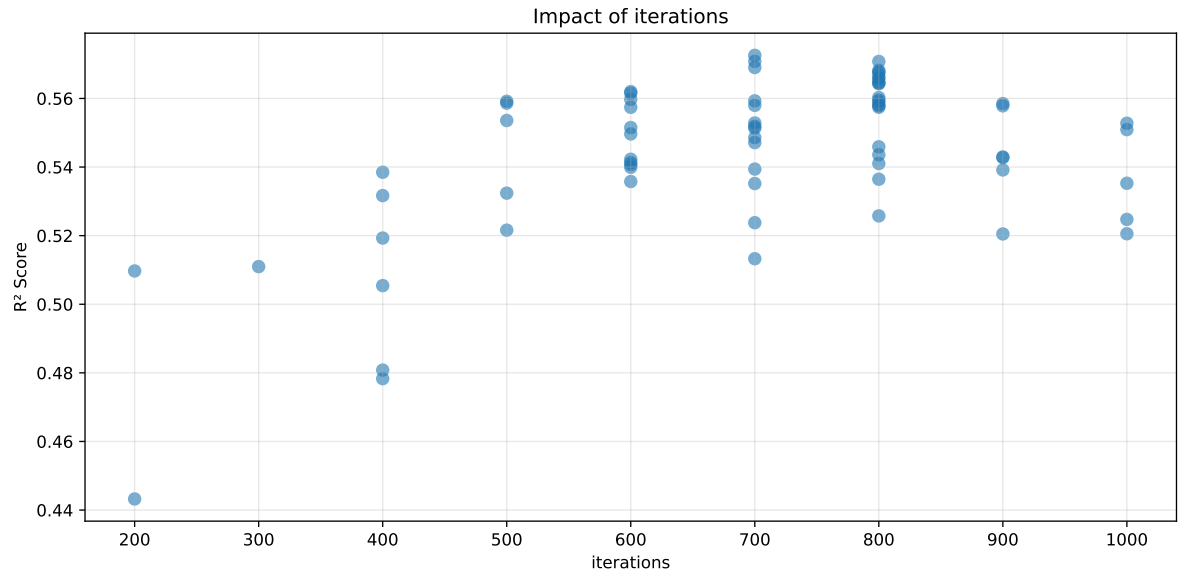
Len(group) : 1129, Len(training) : 791, Len(test) : 338

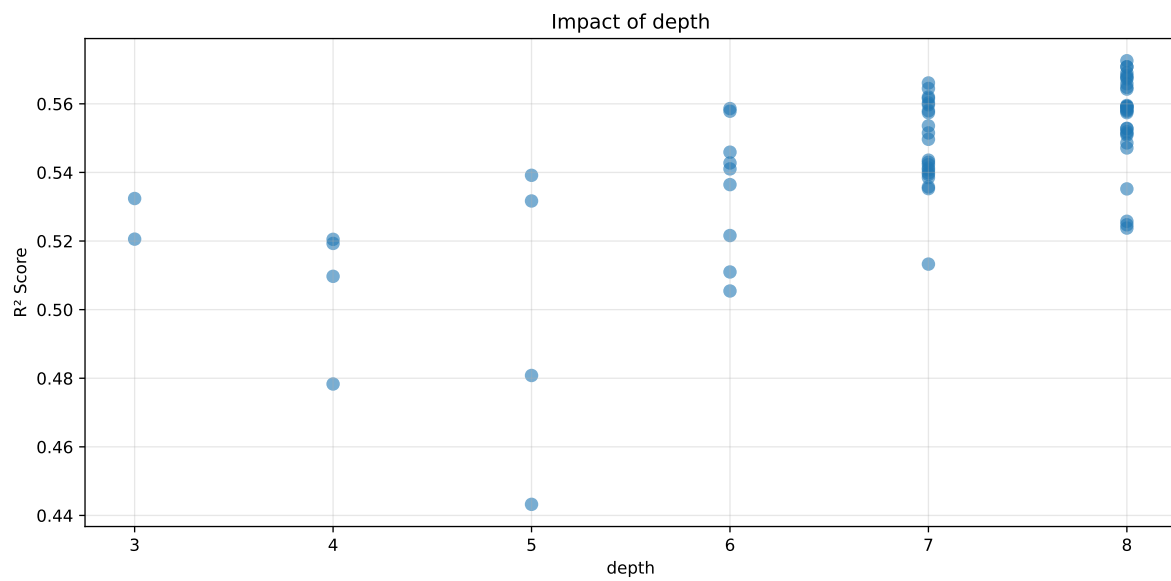
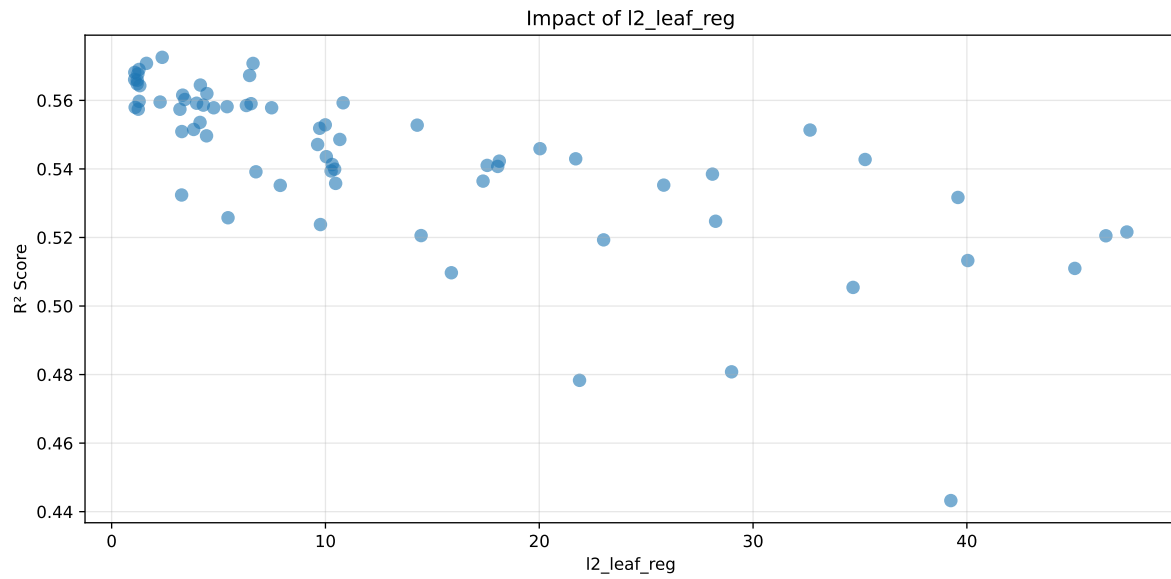
Distribution y_train vs y_test: y_train: mean=1.926, std=0.742, min=0.959, max=6.804
y_test: mean=1.945, std=0.805, min=1.070, max=7.822

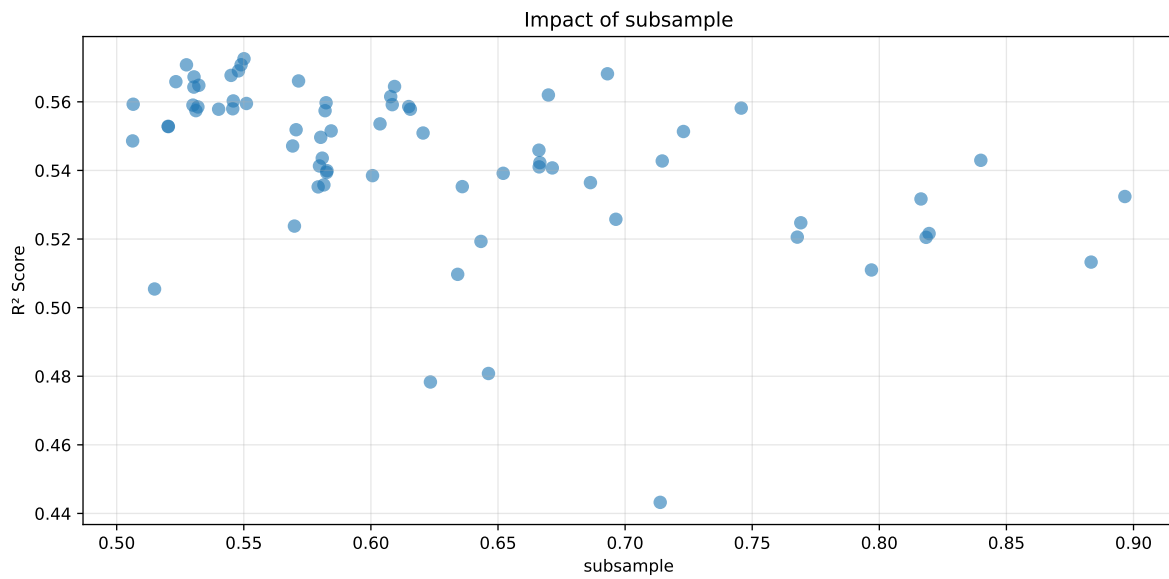
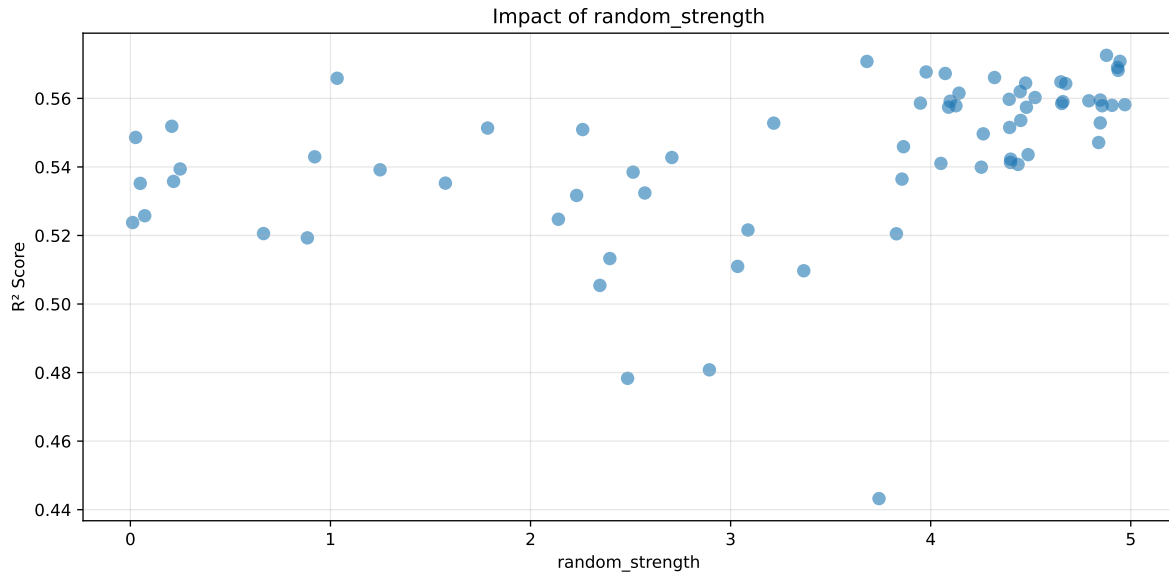
===== TOP Importance FEATURES =====
feature importance r_umidity 46.884237 r_THI 18.325315 r_CO2 16.513411 r_NH3 9.941104
r_temperature 8.335934



Paramètres avec >5% d'importance : ['iterations', 'learning_rate', 'l2_leaf_reg', 'depth', 'random_strength', 'subsample']

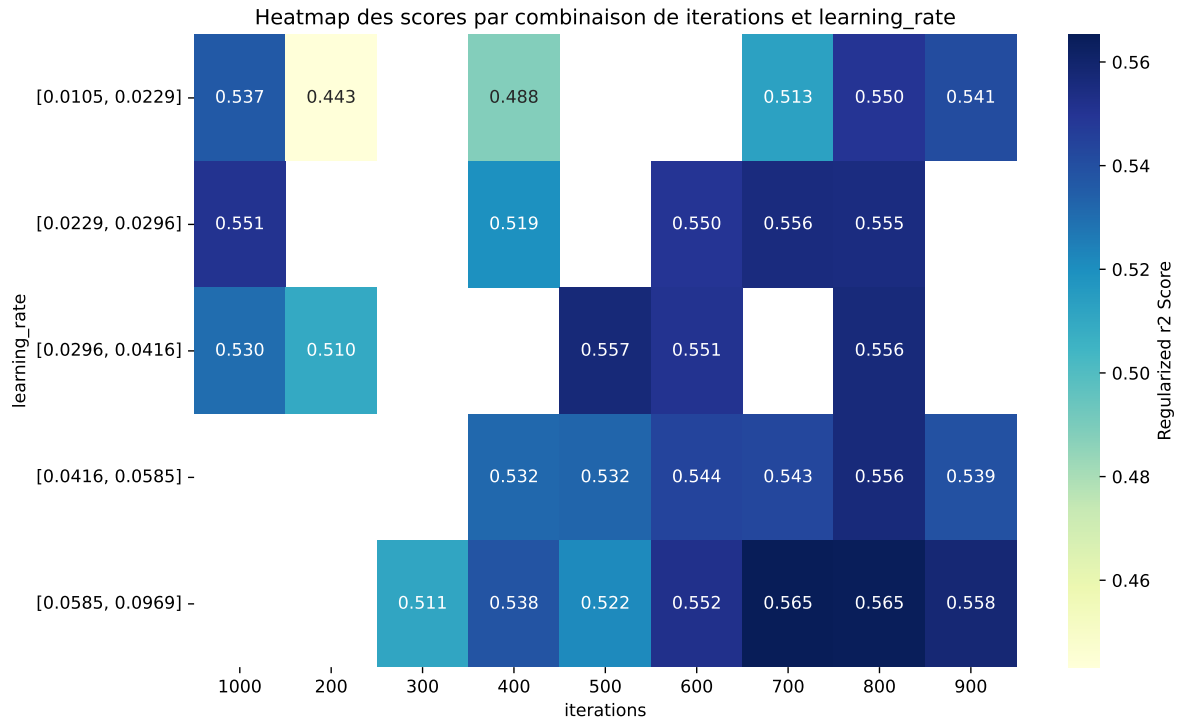






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5171	0.4432	0.47	0.0401	0.2218	0.4811	0.0379	1.0855	1.1666
2	0.574	0.4783	0.5242	0.0411	0.1914	0.5312	0.0529	1.1107	1.2
3	0.5868	0.4808	0.524	0.0398	0.1628	0.5391	0.0583	1.1213	1.2205
4	0.6564	0.5054	0.557	0.037	0.0525	0.5696	0.0642	1.127	1.2987
5	0.6688	0.5097	0.5775	0.0311	0.0324	0.5805	0.0708	1.1388	1.3121
6	0.7112	0.5205	0.5907	0.0328	-0.0517	0.5979	0.0774	1.1487	1.3664
7	0.717	0.5193	0.5896	0.0298	-0.0739	0.5936	0.0743	1.1431	1.3808
8	0.7435	0.5133	0.5787	0.0359	-0.1775	0.5889	0.0756	1.1474	1.4486
9	0.7725	0.511	0.5929	0.0365	-0.2735	0.6036	0.0926	1.1813	1.5118
10	0.8036	0.5317	0.6012	0.0419	-0.284	0.5849	0.0532	1.1001	1.5114
---	---	---	---	---	---	---	---	---	---
70	0.998	0.5238	0.6326	0.032	-0.8989	0.6568	0.133	1.2539	1.9054

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 5, 'reg__learning_rate': 0.017920300458213082, 'reg__l2_leaf_reg': 39.250215122984834, 'reg__bagging_temperature': 1.1207574549719268, 'reg__random_strength': 3.7416810002561665, 'reg__subsample': 0.7138081877424786, 'reg__min_data_in_leaf': 32, 'reg__leaf_estimation_iterations': 3}

Rank 2: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.010479658509559536,

'reg__l2_leaf_reg': 21.883715761423833, 'reg__bagging_temperature': 3.92736417586284, 'reg__random_strength': 2.4850731402281587, 'reg__subsample': 0.6234108901059254, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 10}

Rank 3: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.01269100265063459, 'reg__l2_leaf_reg': 28.994922622162623, 'reg__bagging_temperature': 0.3390876674848986, 'reg__random_strength': 2.8939557719346913, 'reg__subsample': 0.6462489028397308, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 3}

Rank 4: {'reg__iterations': 400, 'reg__depth': 6, 'reg__learning_rate': 0.011198416904707057, 'reg__l2_leaf_reg': 34.67664597049361, 'reg__bagging_temperature': 3.187167051509714, 'reg__random_strength': 2.3465957994863818, 'reg__subsample': 0.5148950474168098, 'reg__min_data_in_leaf': 39, 'reg__leaf_estimation_iterations': 8}

Rank 5: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.041433878624407325, 'reg__l2_leaf_reg': 15.889217617810216, 'reg__bagging_temperature': 0.3197364760732191, 'reg__random_strength': 3.365618435212721, 'reg__subsample': 0.6341415839689246, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 3}

Rank 6: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.01360351886279718, 'reg__l2_leaf_reg': 46.501719822668946, 'reg__bagging_temperature': 4.138805274075997, 'reg__random_strength': 3.8286786584933736, 'reg__subsample': 0.8184165016361025, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 4}

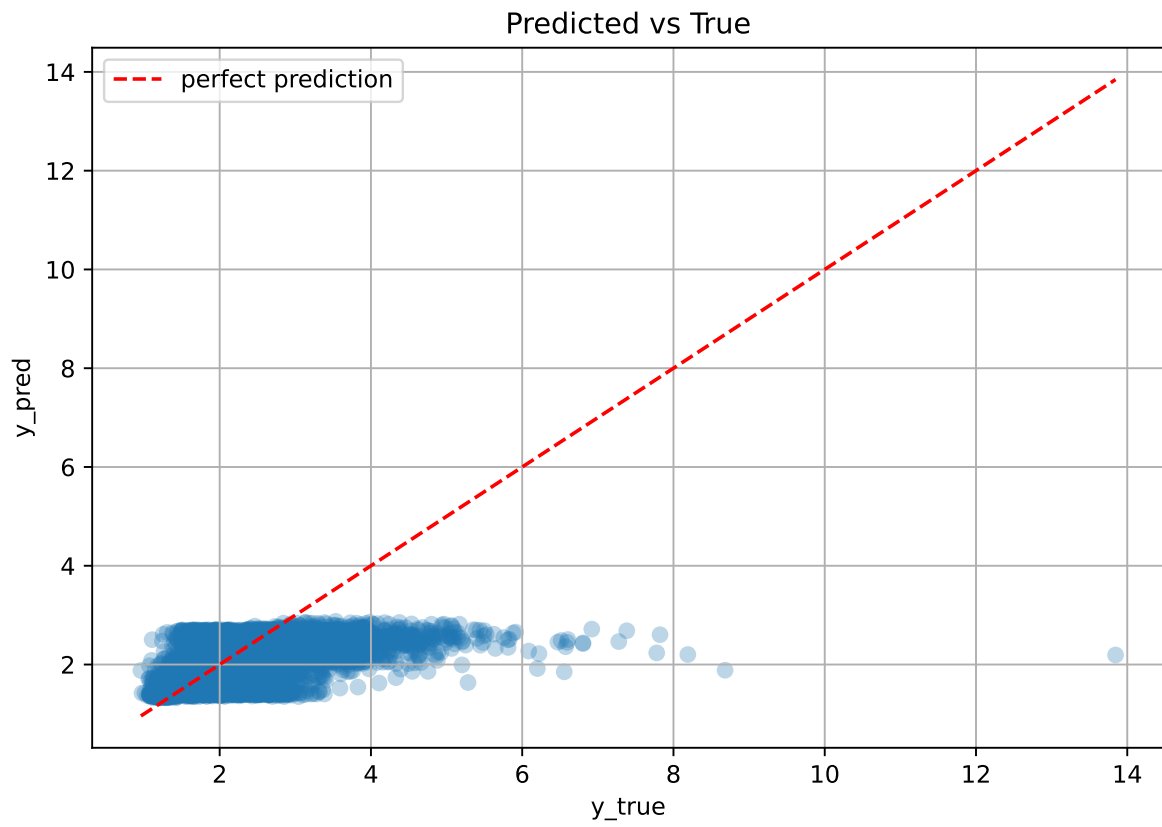
Rank 7: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.025775394600093517, 'reg__l2_leaf_reg': 23.009946617275784, 'reg__bagging_temperature': 2.508443615283892, 'reg__random_strength': 0.8844928341064517, 'reg__subsample': 0.6432962948659586, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 2}

Rank 8: {'reg__iterations': 700, 'reg__depth': 7, 'reg__learning_rate': 0.018318870180084573, 'reg__l2_leaf_reg': 40.043329027439526, 'reg__bagging_temperature': 0.2512090844076842, 'reg__random_strength': 2.3963589028196095, 'reg__subsample': 0.8833149702020104, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 2}

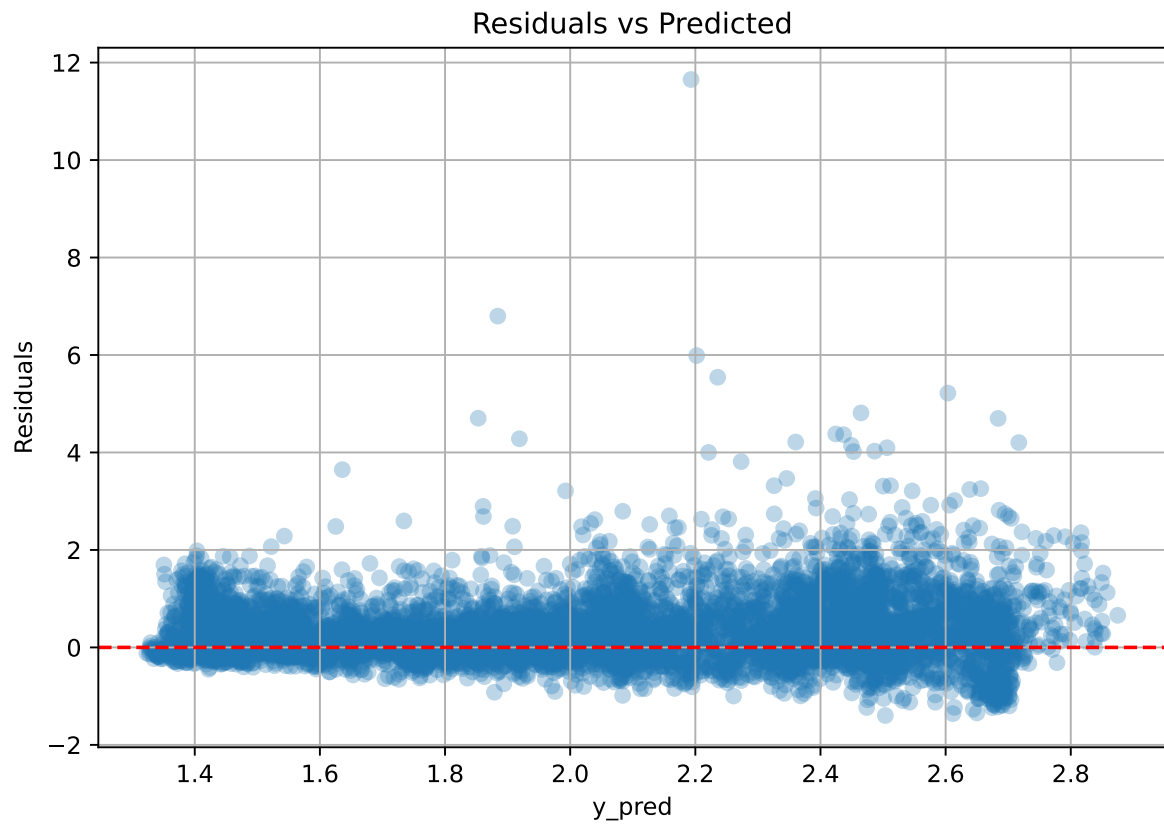
Rank 9: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.09507162222467294, 'reg__l2_leaf_reg': 45.04888176275445, 'reg__bagging_temperature': 0.20408972901329425, 'reg__random_strength': 3.03521938950055, 'reg__subsample': 0.7969103547194021, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1}

Rank 10: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.05766331799882717, 'reg__l2_leaf_reg': 39.58280791621068, 'reg__bagging_temperature': 3.3486109946517955, 'reg__random_strength': 2.230324232984579, 'reg__subsample': 0.8163902557289611, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 2}

2.5.14 Scatter



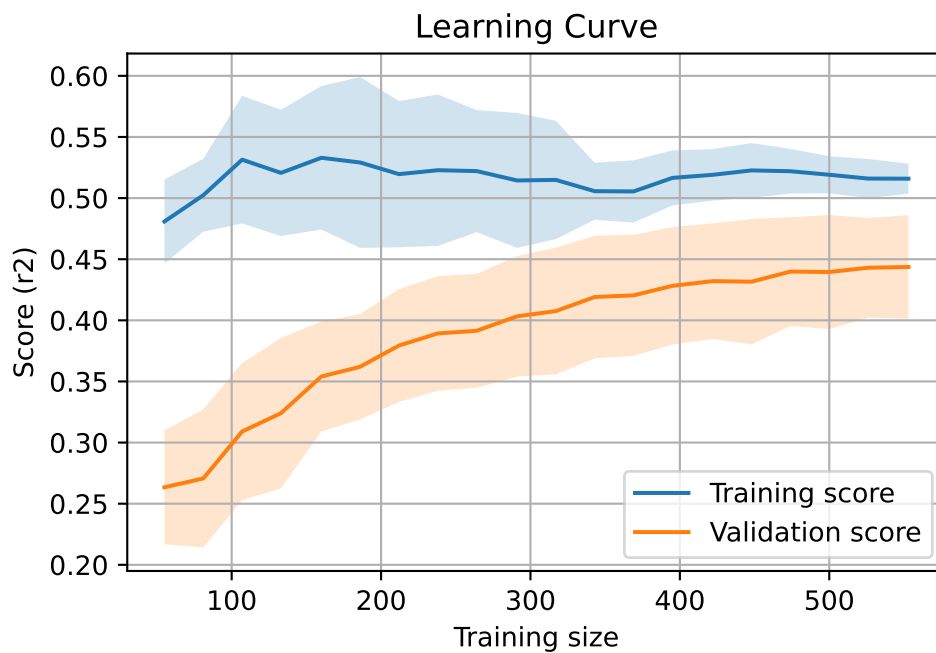
2.5.15 Scatter residuals



2.5.16 Learning curve — Group: 2025_Sep_2

`Len(group)` : 1129, `Len(training)` : 791,

`Len(training_real)` : 554, `Len(test_of_training)` : 237, `Len(test)` : 338



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	2025_Jun_1	1110	475	1585	0.347	0.417	0.042	0.394	0.396	-0.021	0.95	0.833	nan
Linear	2025_Jun_2	1111	476	1587	0.218	0.206	0.022	0.216	0.217	0.011	1.054	1.061	nan
Linear	2025_Sep_1	2707	1160	3867	0.426	0.433	0.022	0.449	0.45	0.017	1.039	0.984	nan
Linear	2025_Sep_2	791	338	1129	0.377	0.351	0.022	0.374	0.374	0.024	1.067	1.075	nan
Linear	MEAN	5719	2449	8168	0.342	0.352	0.027	0.358	0.359	0.008	1.028	0.988	nan
LGBM	2025_Jun_1	1110	475	1585	0.602	0.584	0.035	0.579	0.599	0.015	1.026	1.03	0.5841
LGBM	2025_Jun_2	1111	476	1587	0.4	0.362	0.019	0.354	0.397	0.036	1.098	1.105	0.2482
LGBM	2025_Sep_1	2707	1160	3867	0.583	0.556	0.031	0.559	0.578	0.022	1.04	1.048	0.5563
LGBM	2025_Sep_2	791	338	1129	0.384	0.344	0.032	0.358	0.387	0.043	1.124	1.116	0.2244
LGBM	MEAN	5719	2449	8168	0.492	0.462	0.029	0.462	0.49	0.029	1.072	1.075	0.403
CatBoost	2025_Jun_1	1110	475	1585	0.522	0.557	0.026	0.532	0.558	0.001	1.002	0.938	0.5568
CatBoost	2025_Jun_2	1111	476	1587	0.603	0.496	0.014	0.526	0.55	0.054	1.11	1.214	0.1771
CatBoost	2025_Sep_1	2707	1160	3867	0.591	0.576	0.028	0.588	0.596	0.02	1.035	1.026	0.5763
CatBoost	2025_Sep_2	791	338	1129	0.517	0.443	0.04	0.47	0.481	0.038	1.086	1.167	0.2218
CatBoost	MEAN	5719	2449	8168	0.558	0.518	0.027	0.529	0.547	0.028	1.058	1.086	0.383

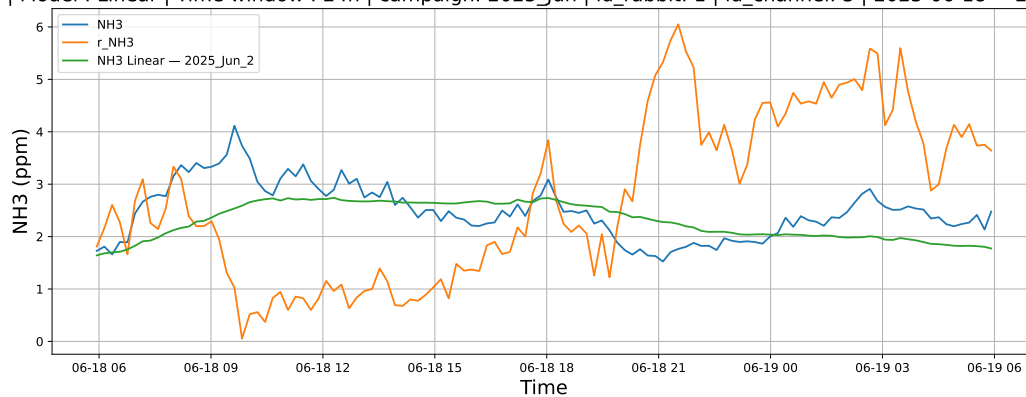
3 Plot

3.1 Standalone plots

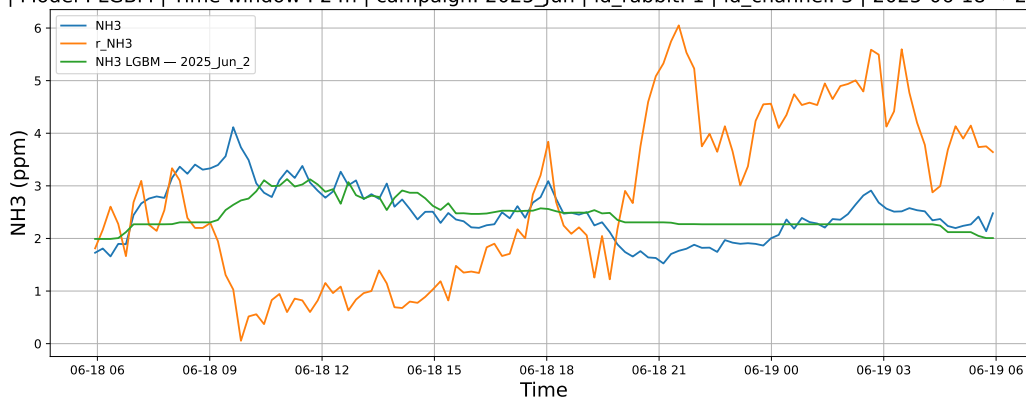
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

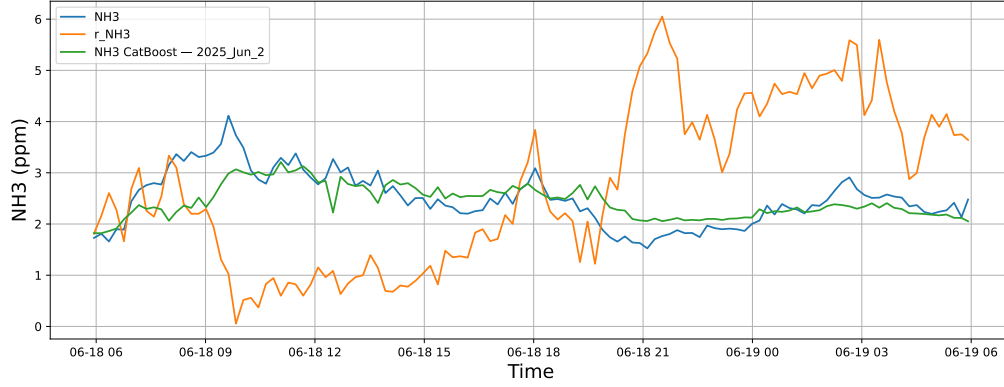
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

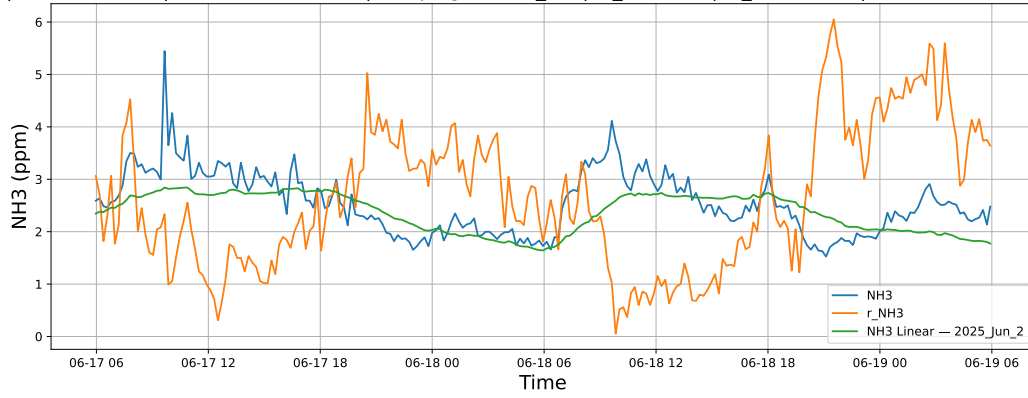


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

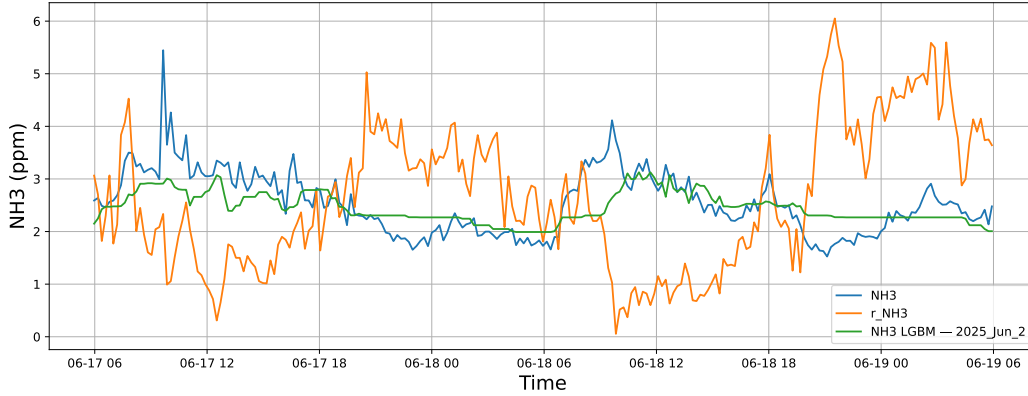


3.1.1.2 Time window : 48

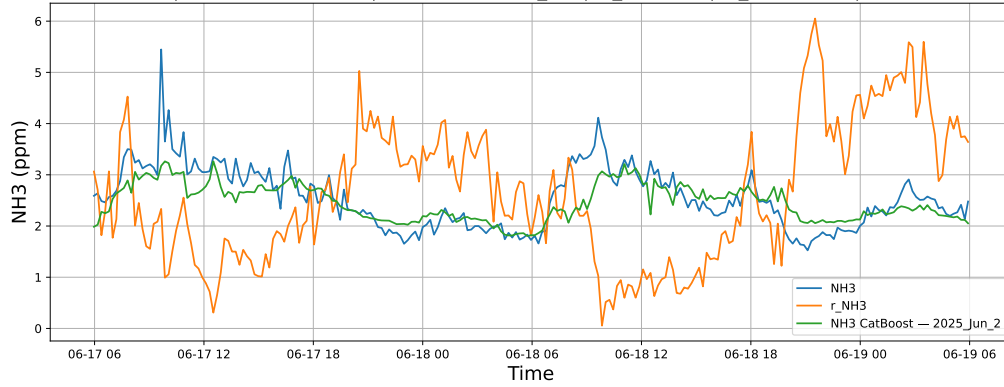
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

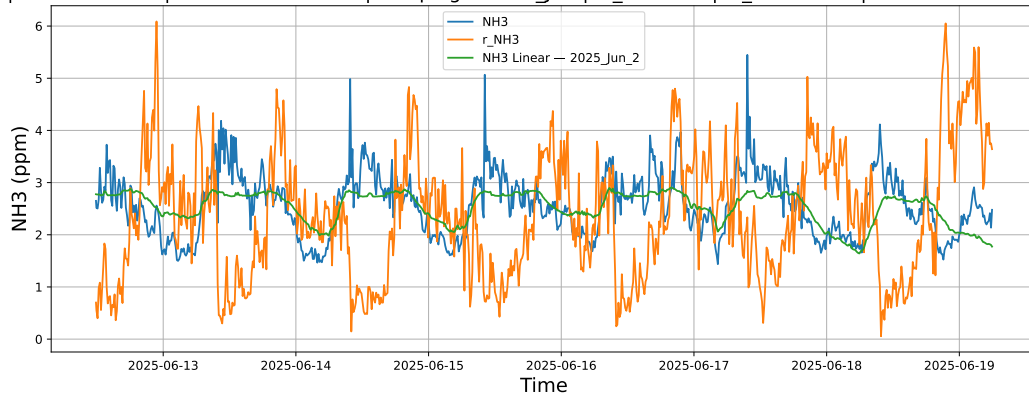


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

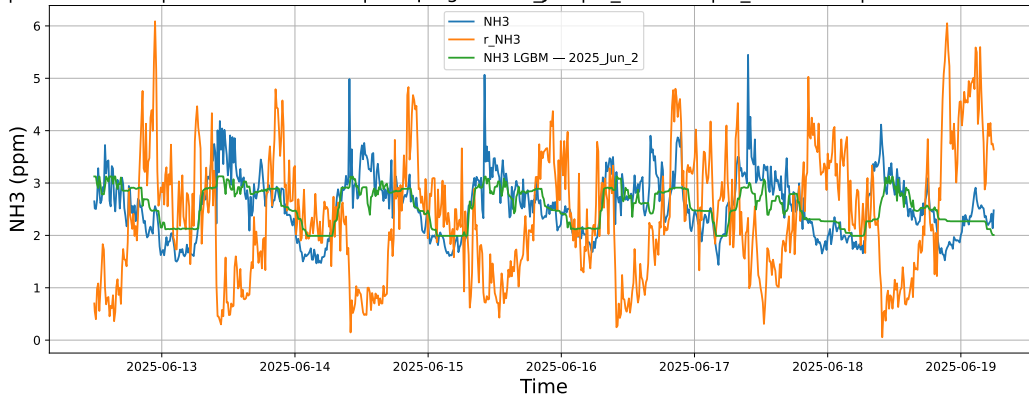


3.1.1.3 Time window : ALL

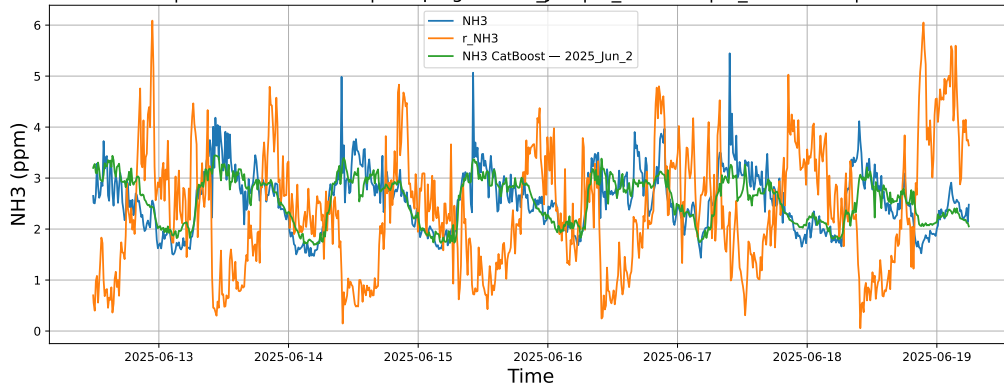
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



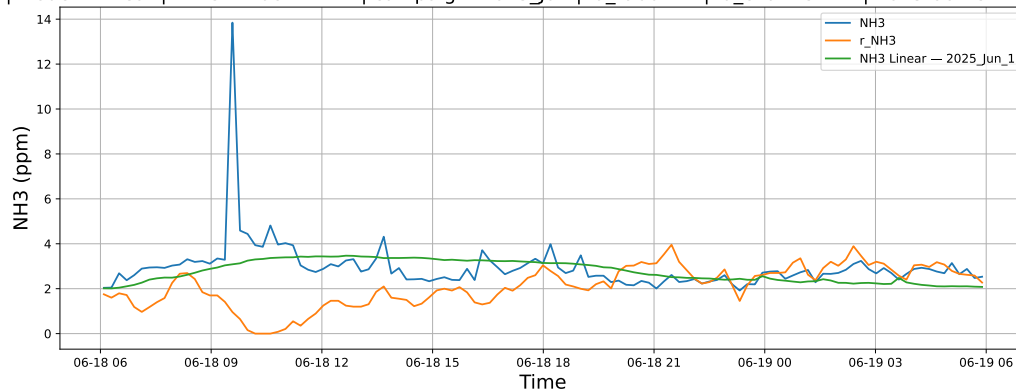
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



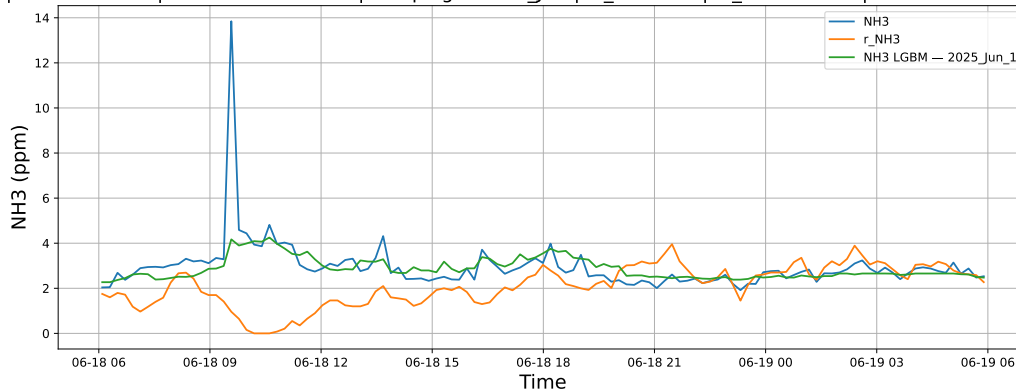
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

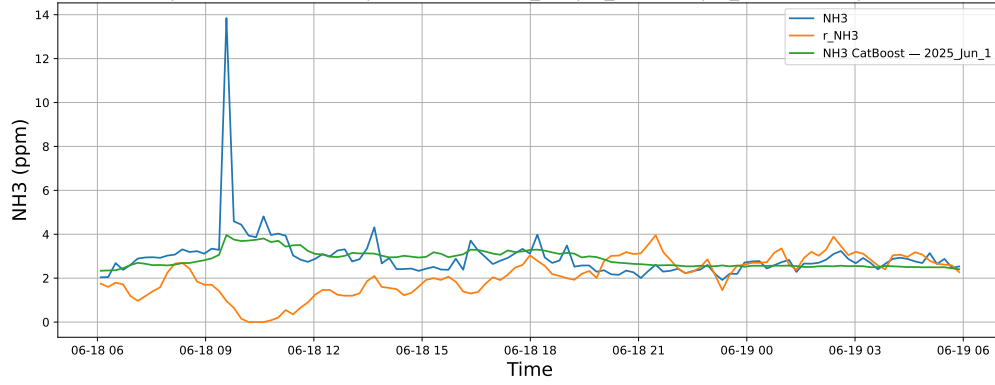
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

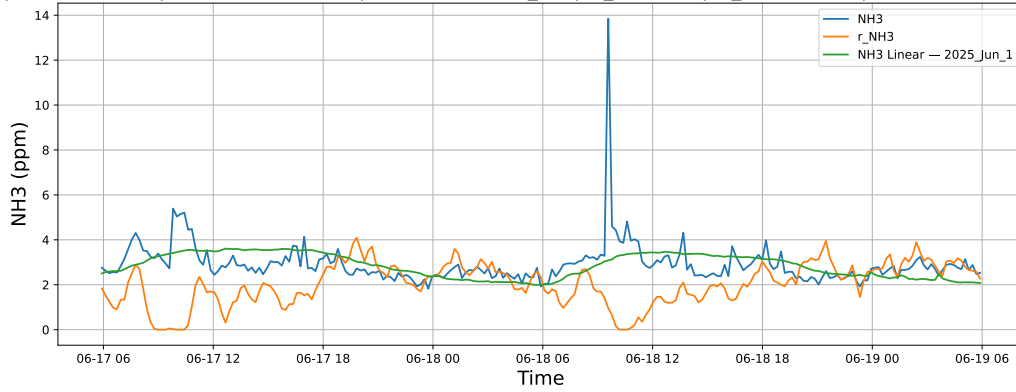


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

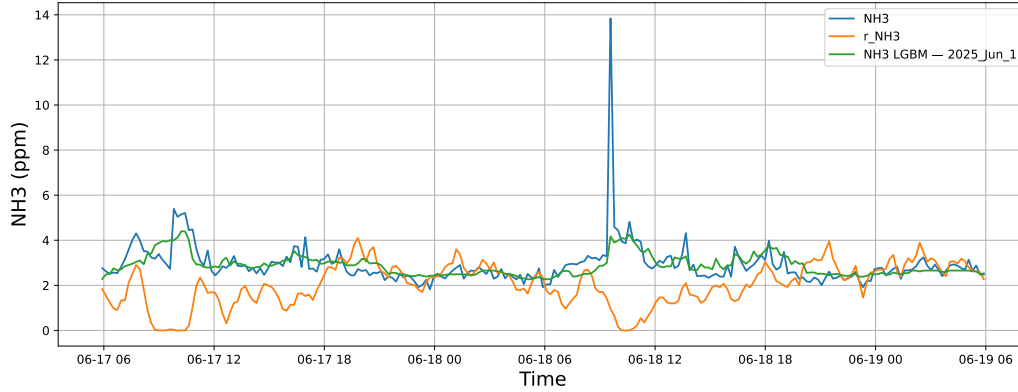


3.1.2.2 Time window : 48

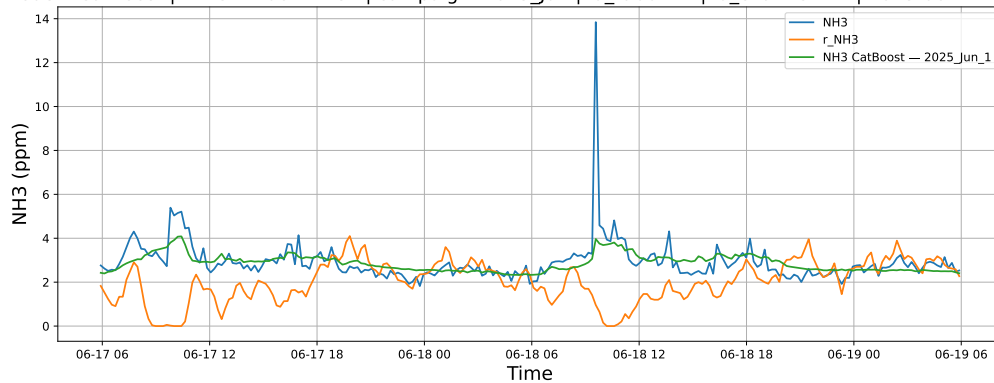
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

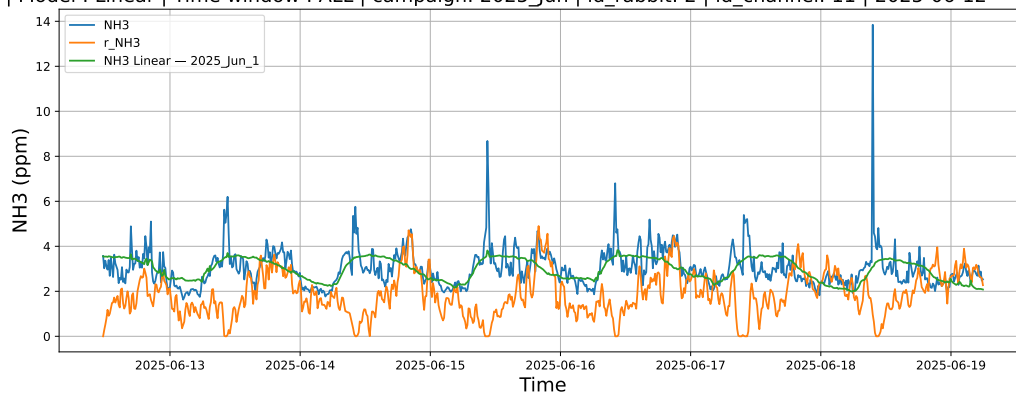


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

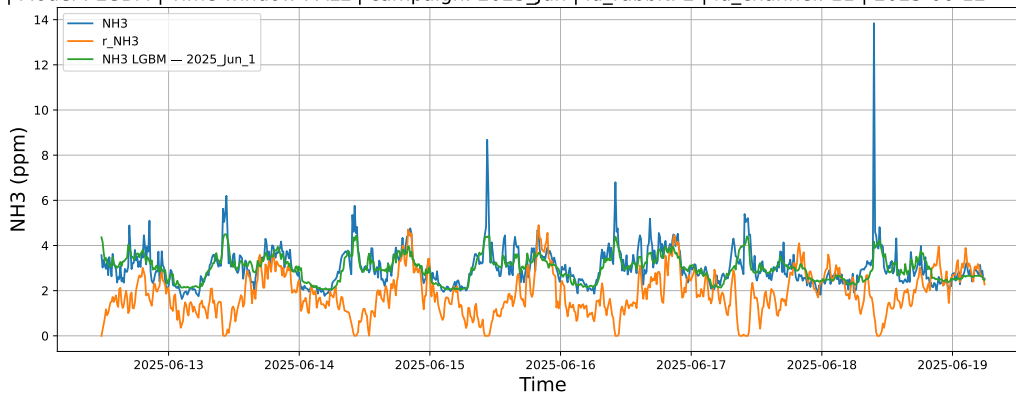


3.1.2.3 Time window : ALL

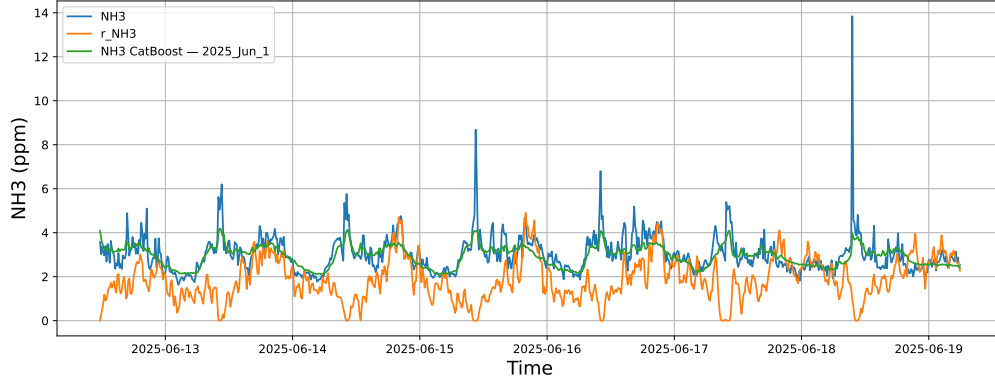
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



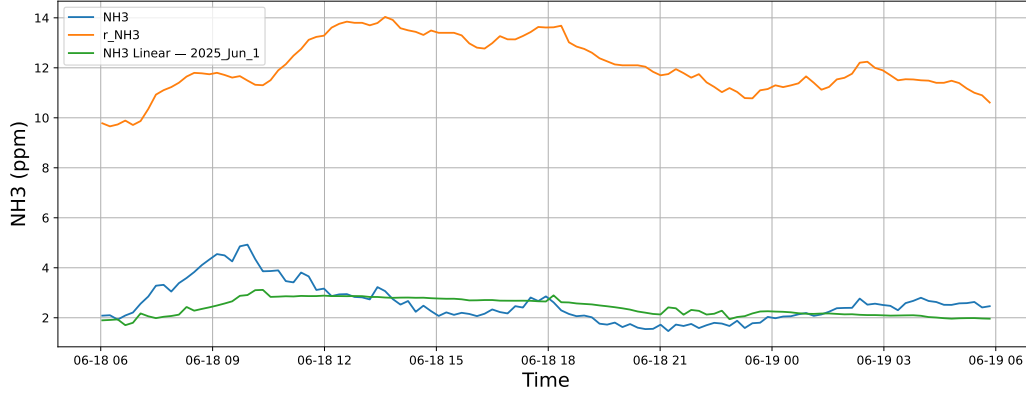
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



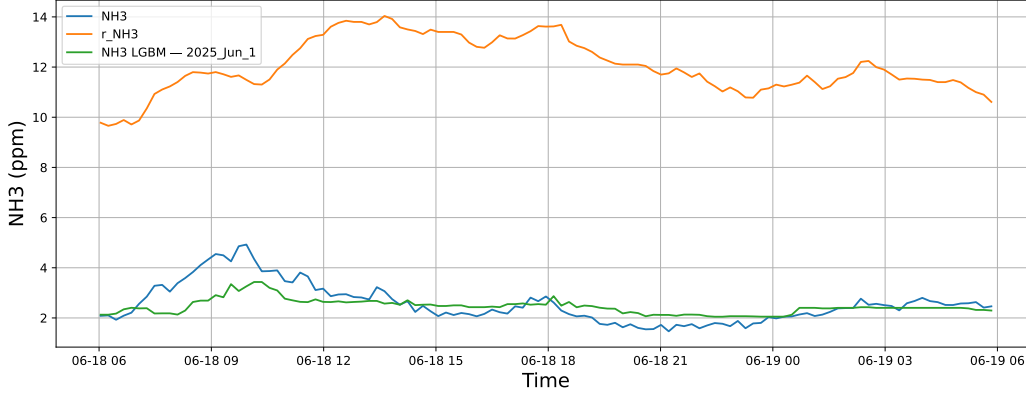
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

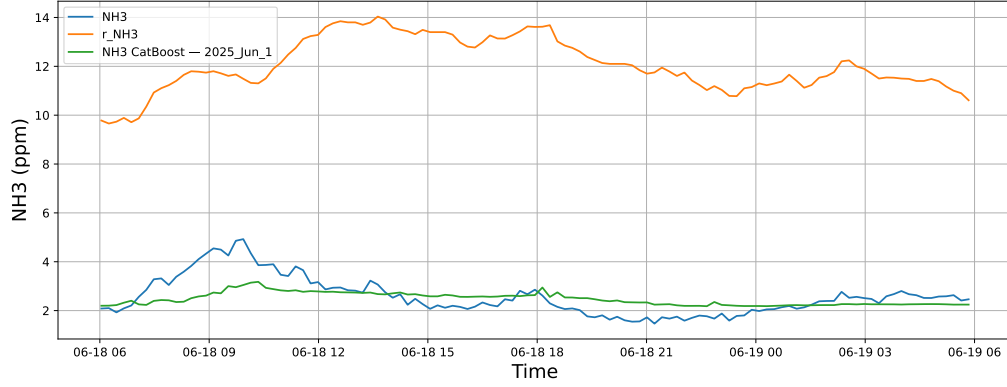
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

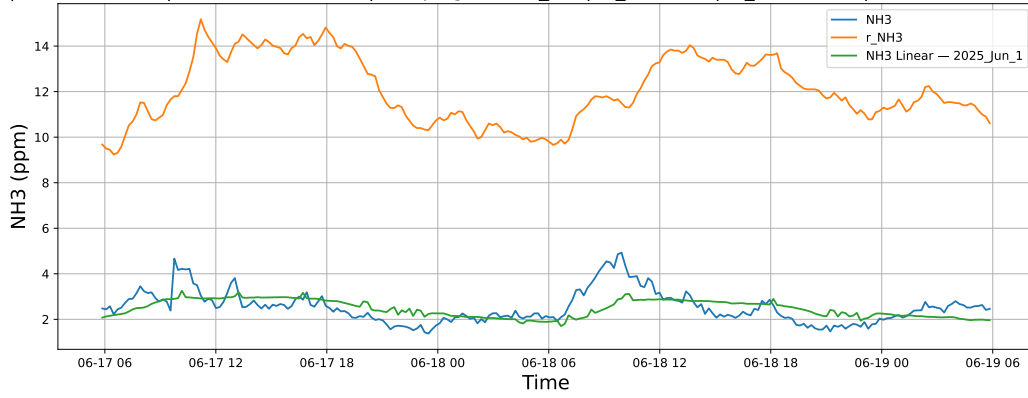


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

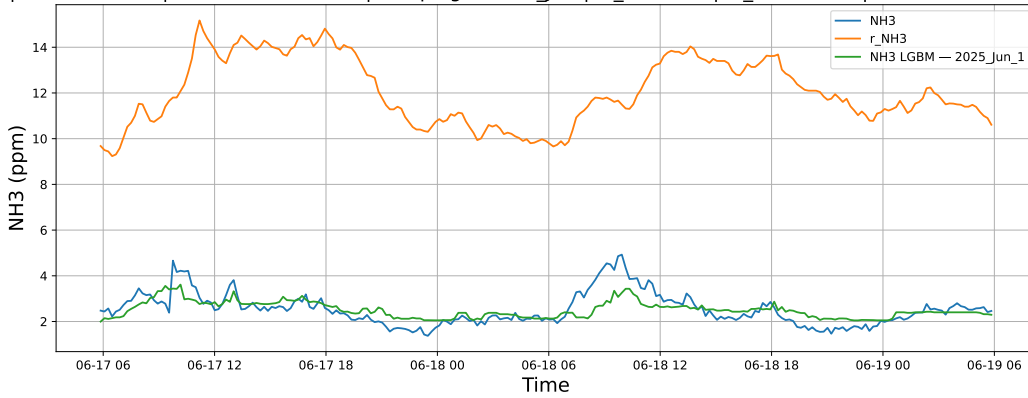


3.1.3.2 Time window : 48

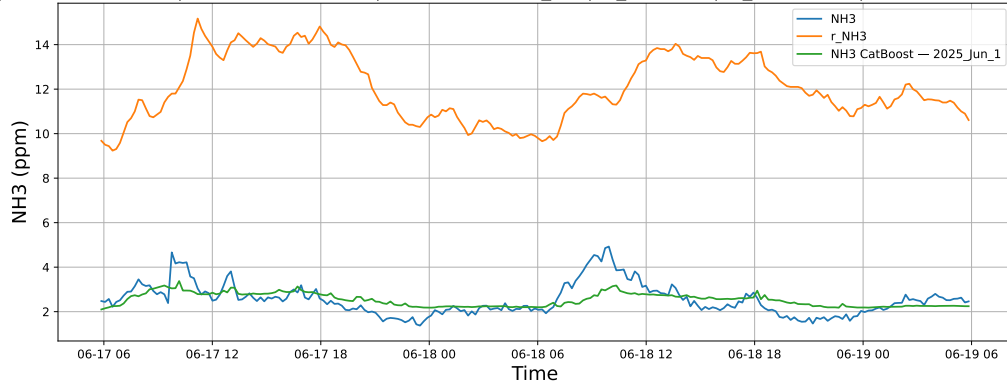
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

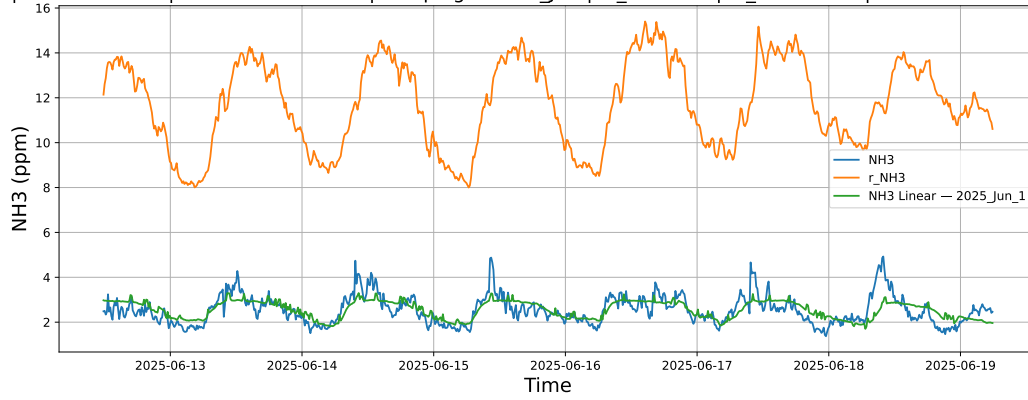


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

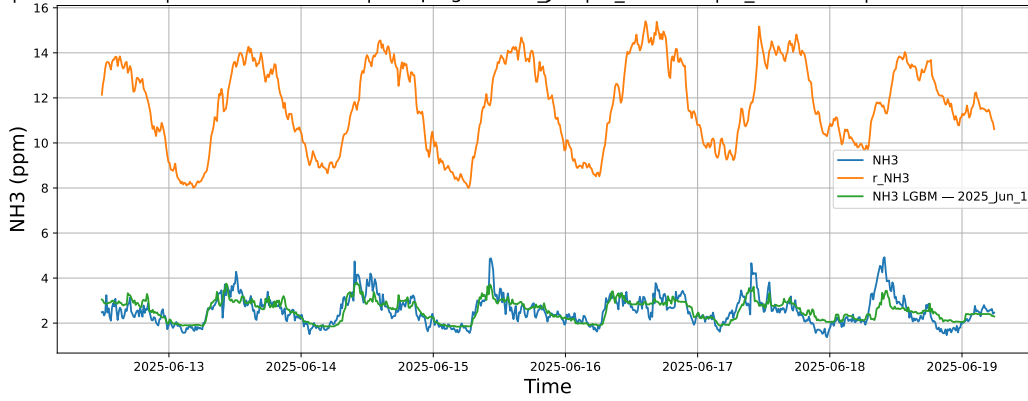


3.1.3.3 Time window : ALL

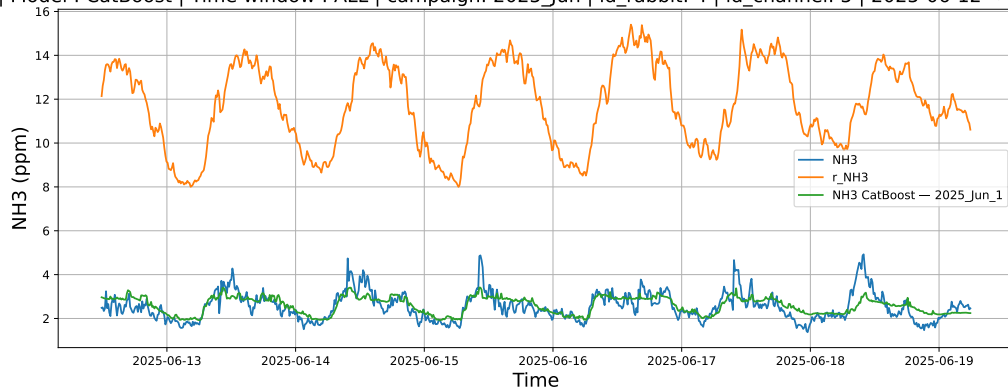
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



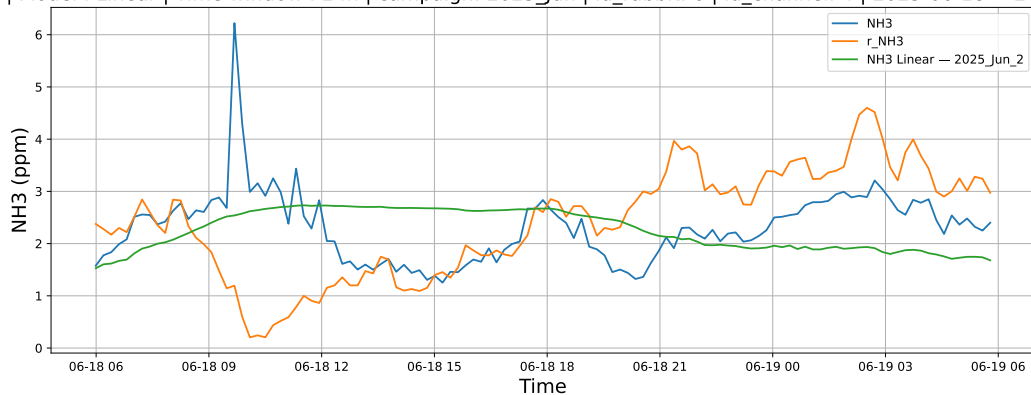
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



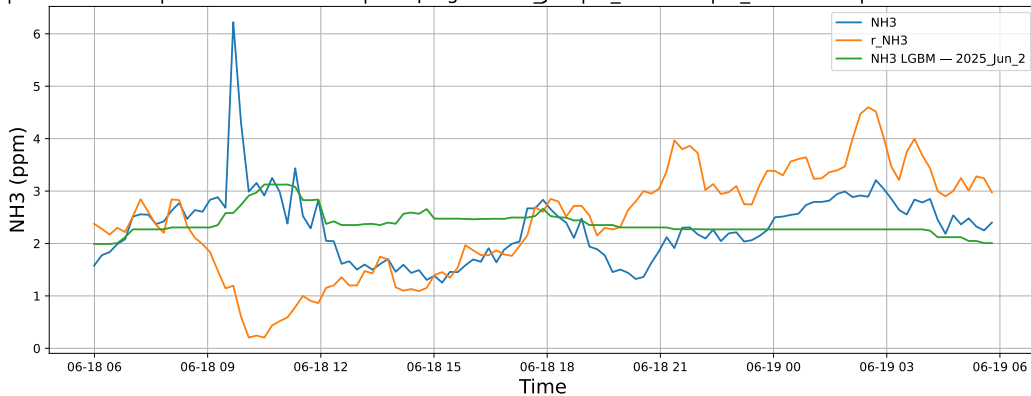
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

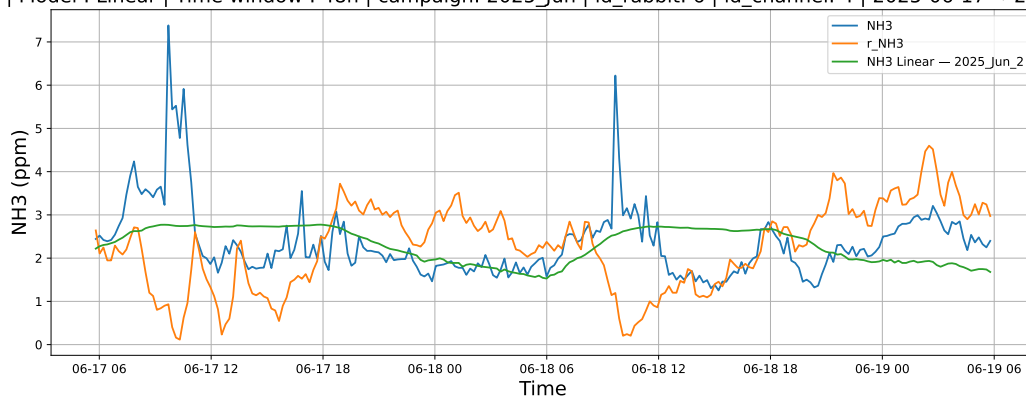


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

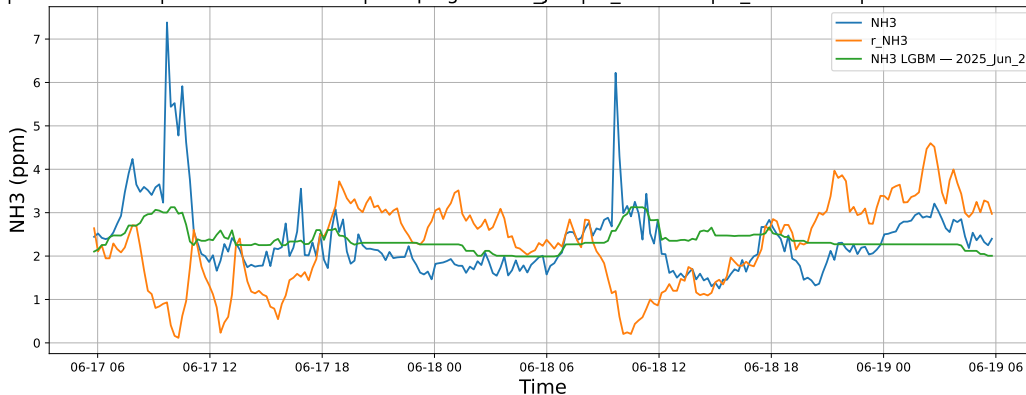


3.1.4.2 Time window : 48

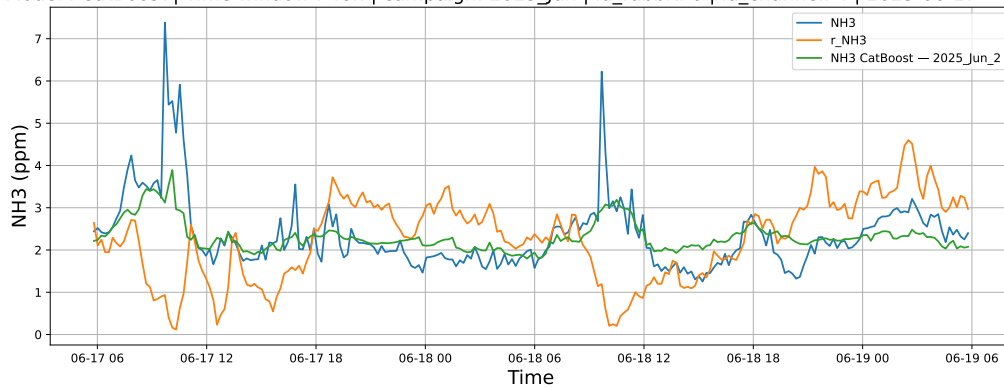
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

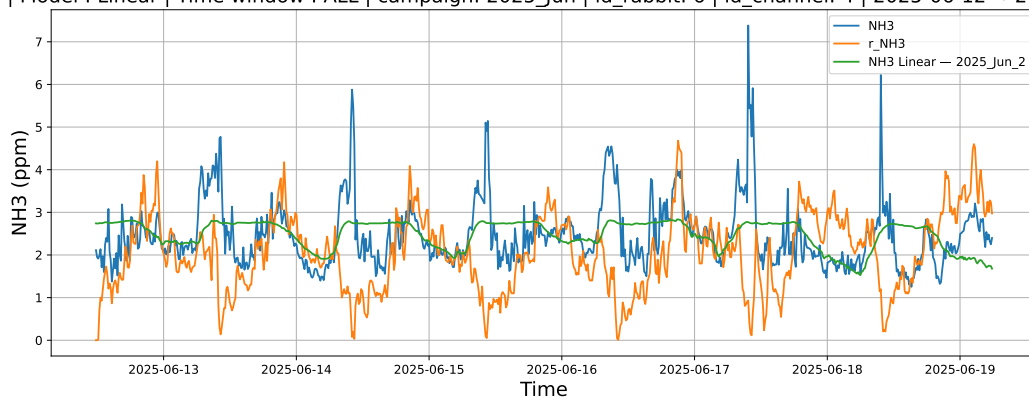


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

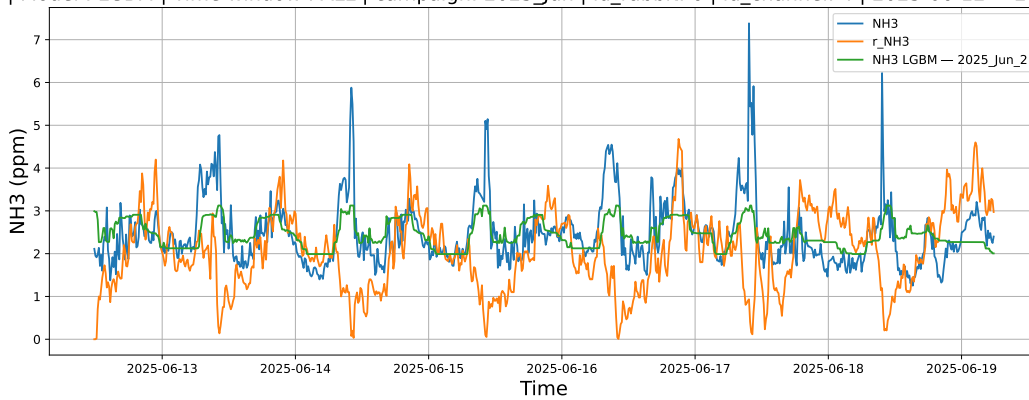


3.1.4.3 Time window : ALL

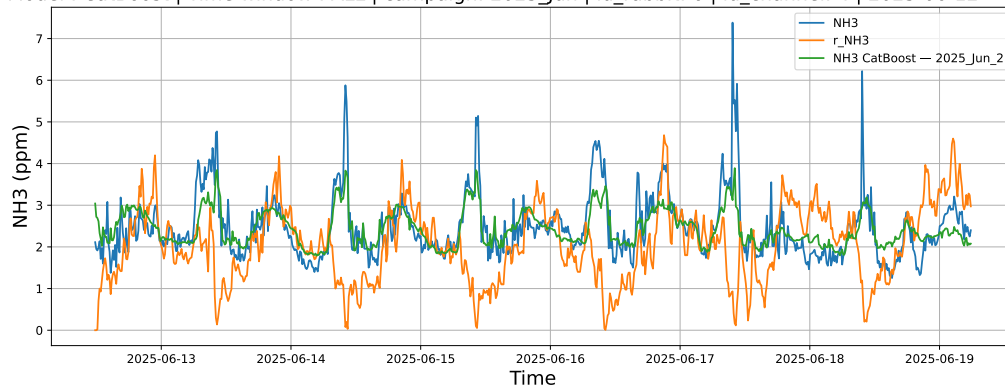
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



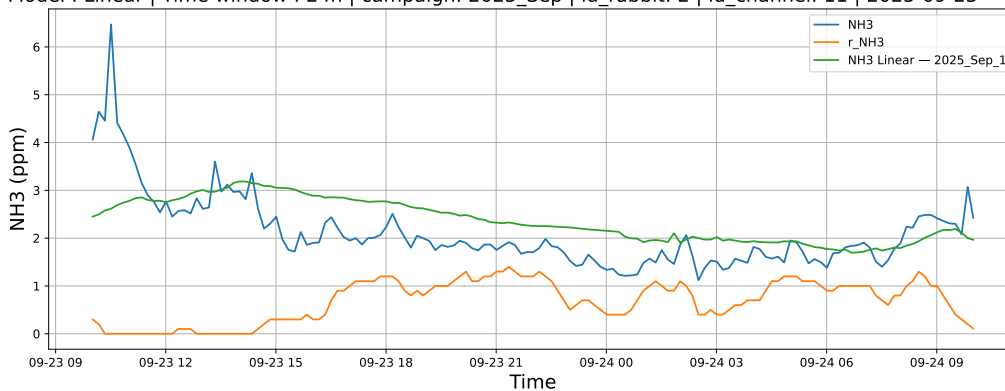
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



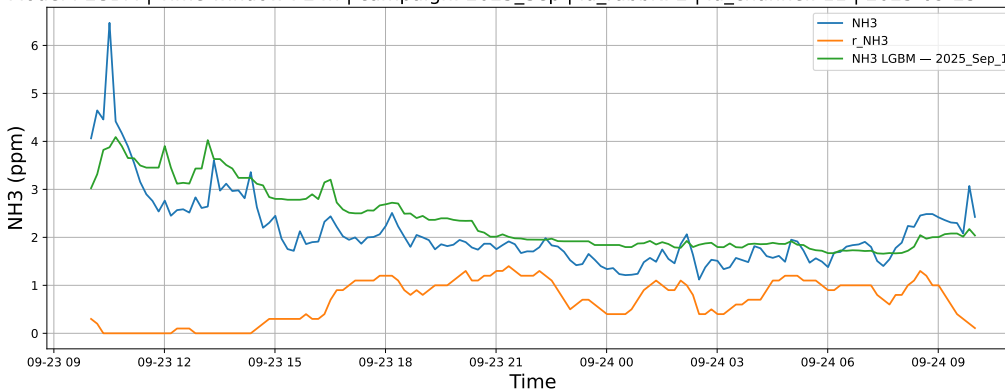
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

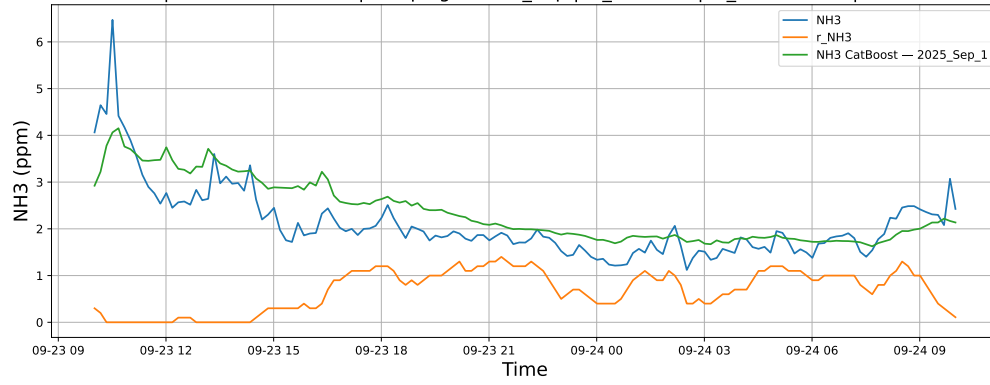
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

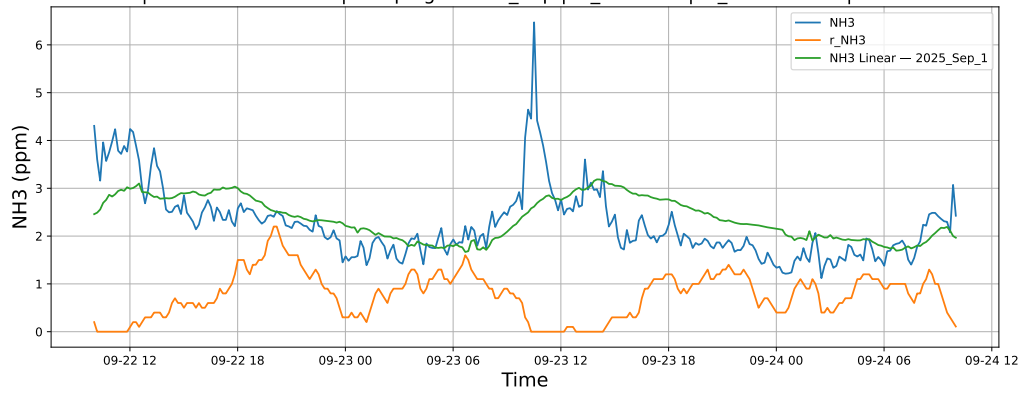


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

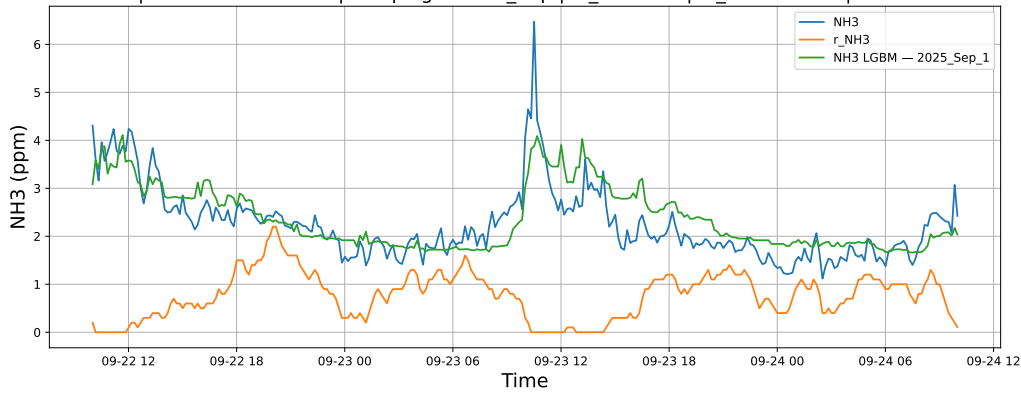


3.1.5.2 Time window : 48

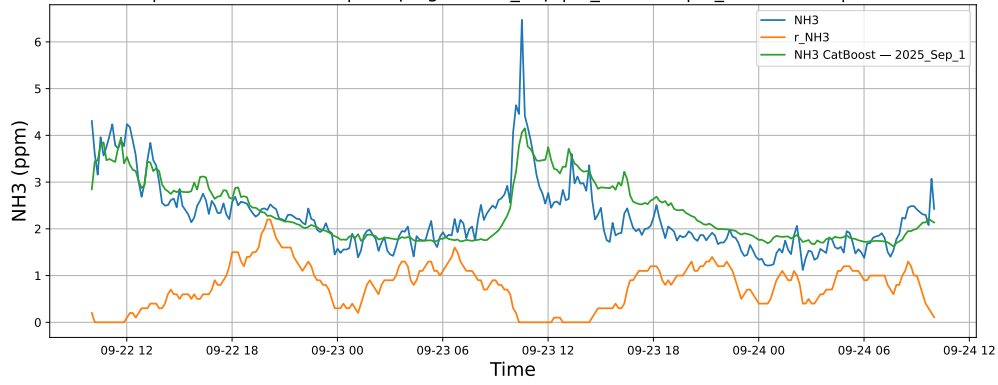
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

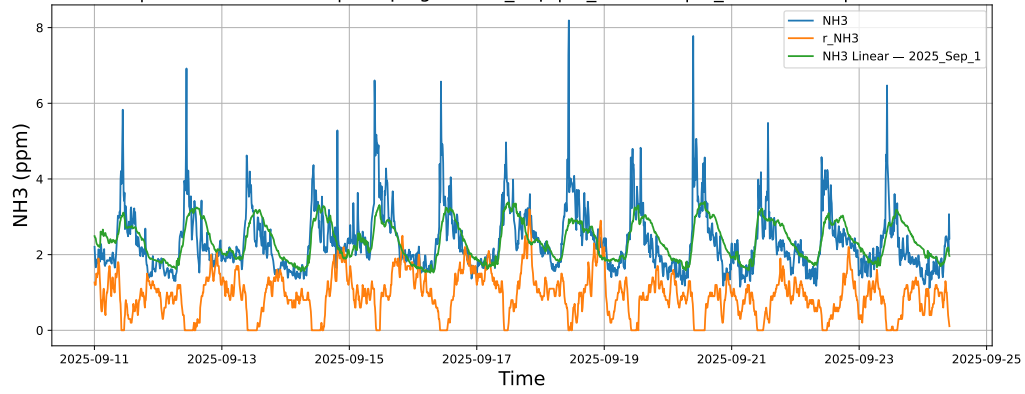


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

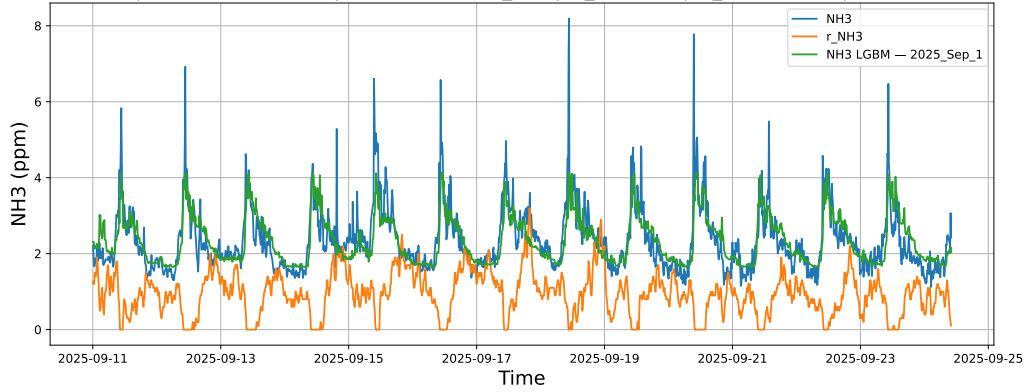


3.1.5.3 Time window : ALL

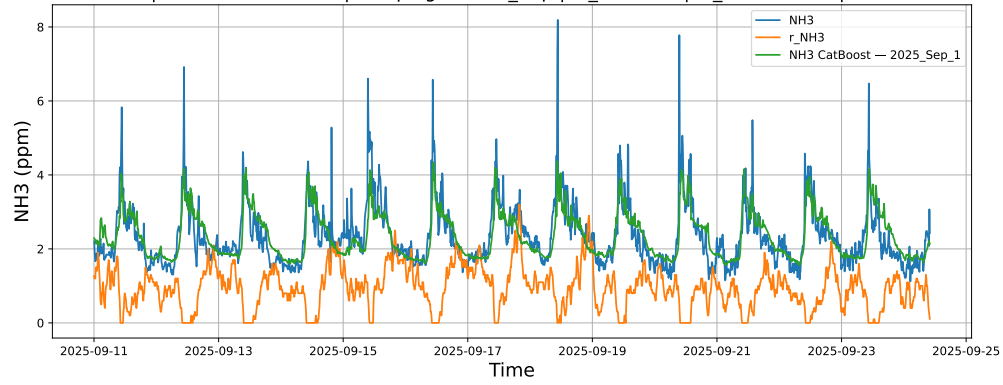
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



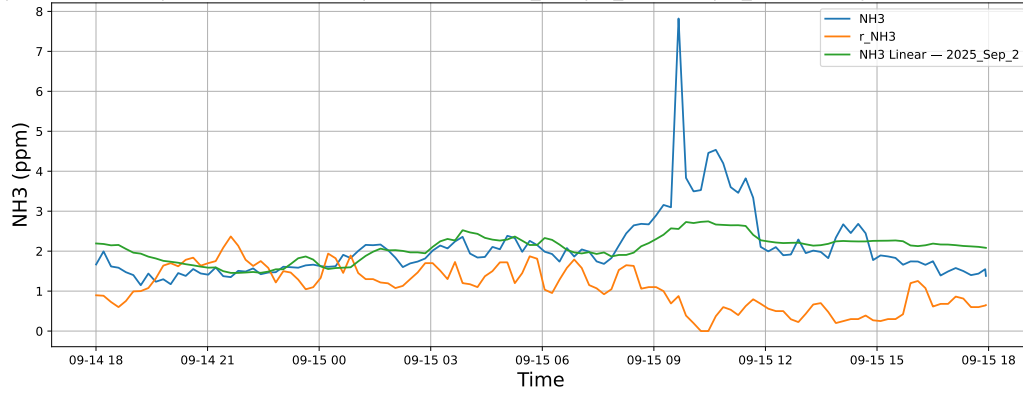
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



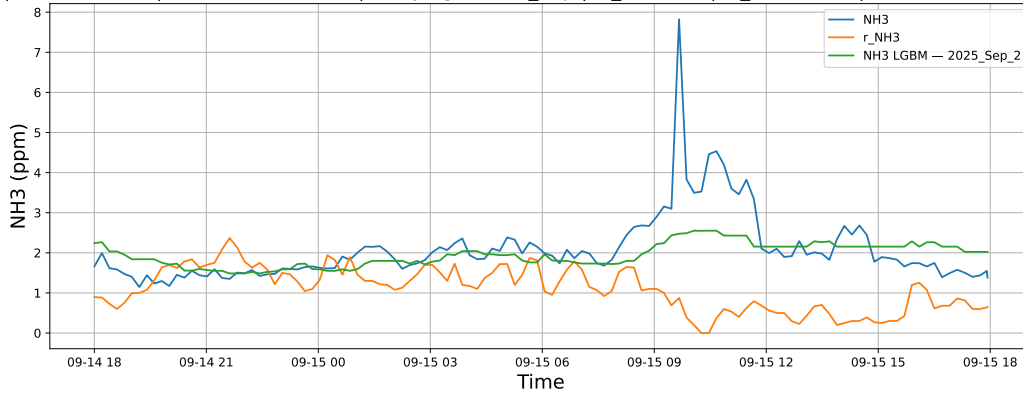
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

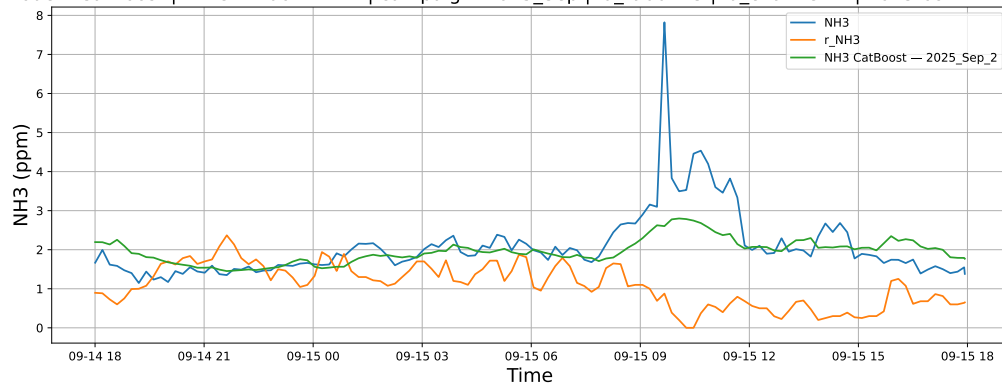
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

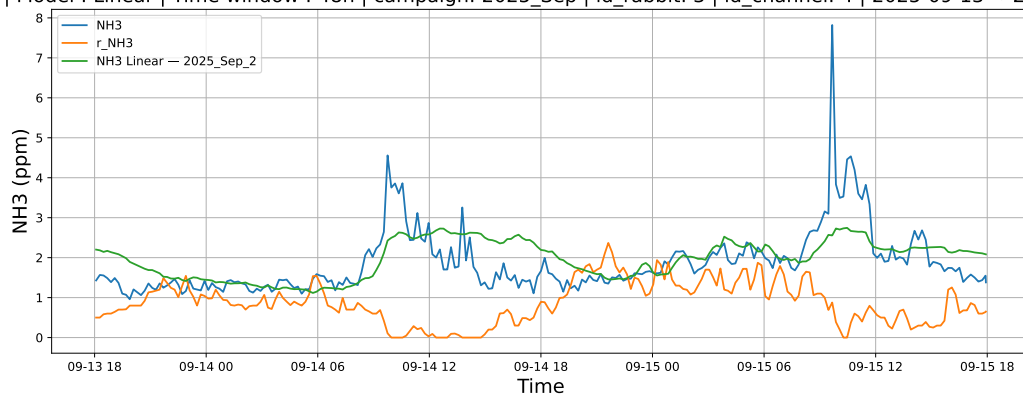


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

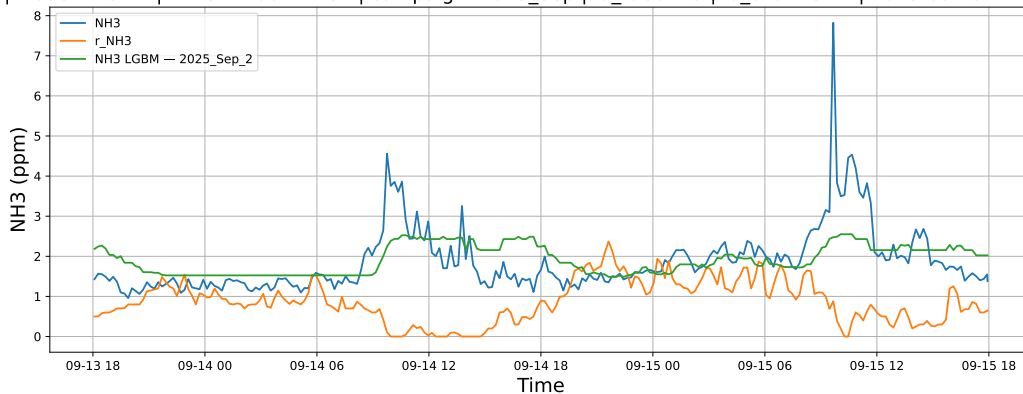


3.1.6.2 Time window : 48

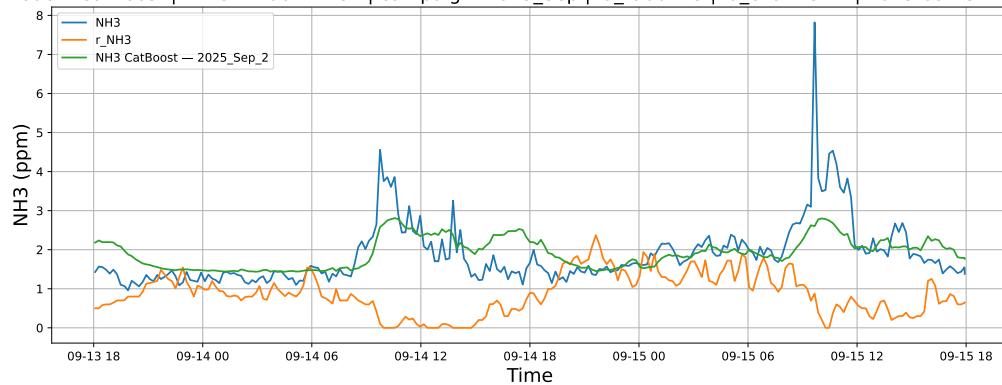
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

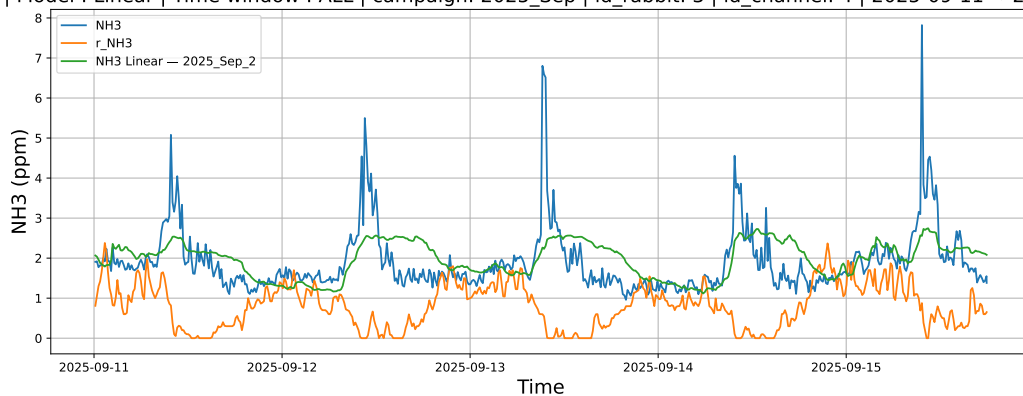


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

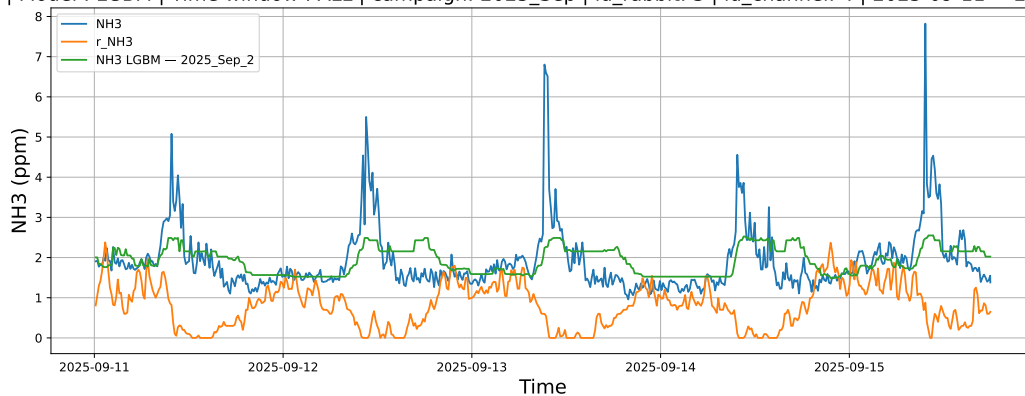


3.1.6.3 Time window : ALL

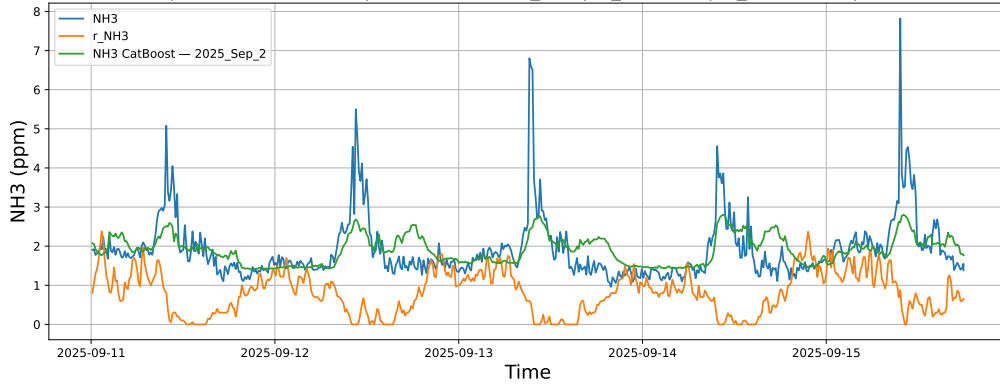
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



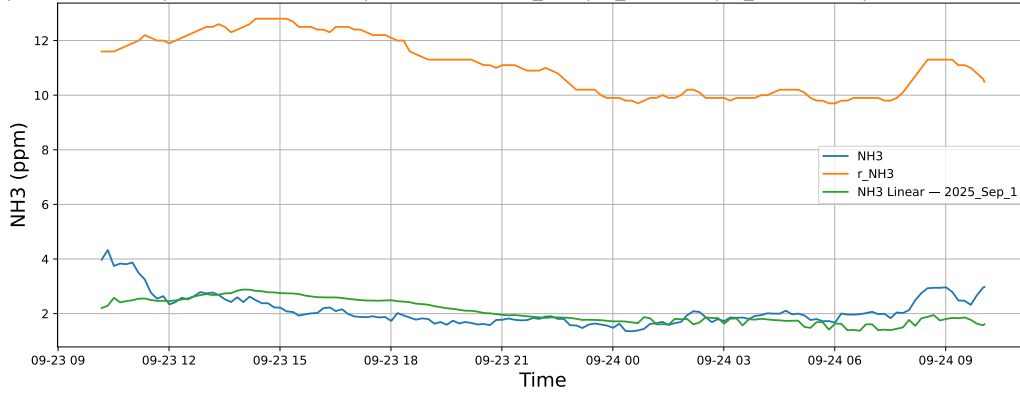
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



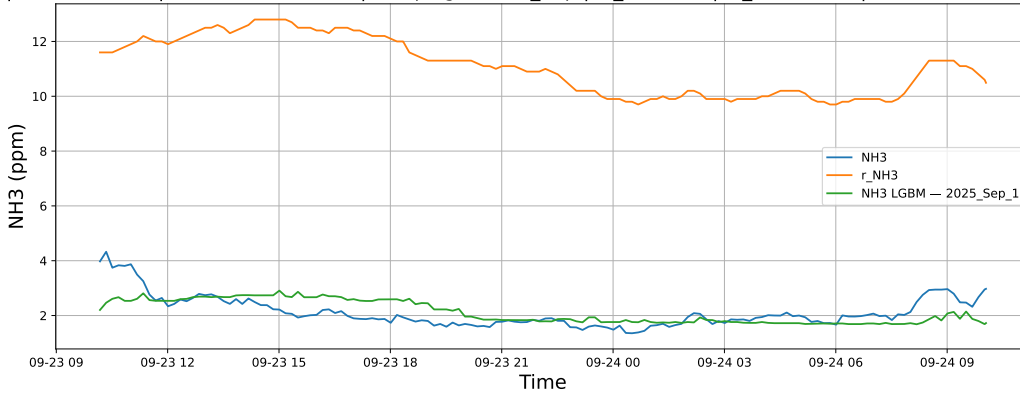
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

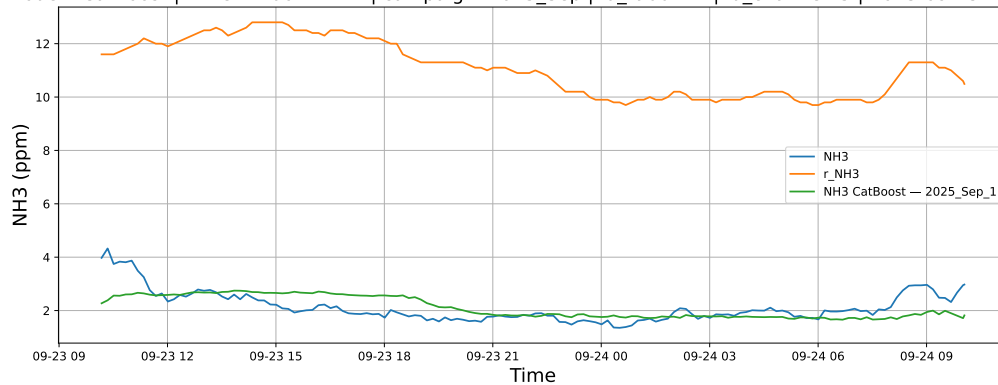
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

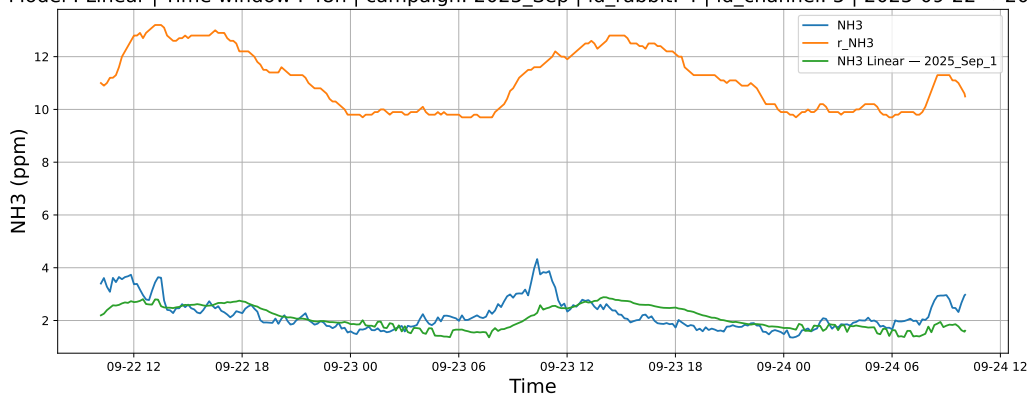


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

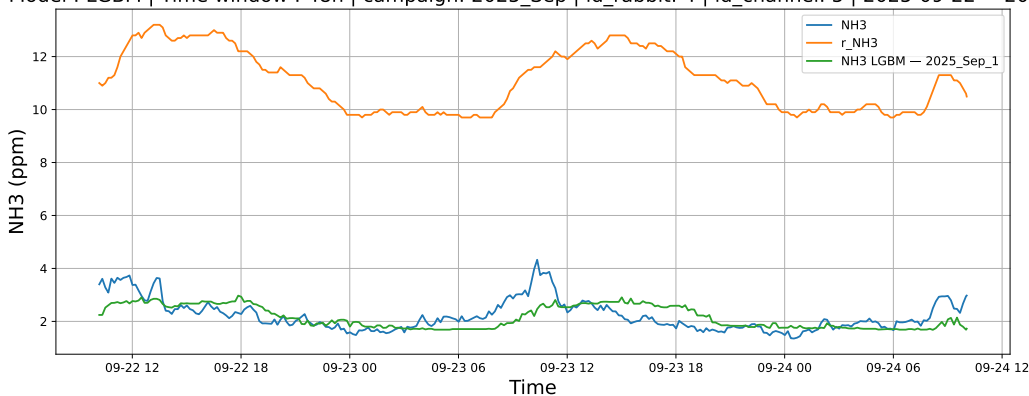


3.1.7.2 Time window : 48

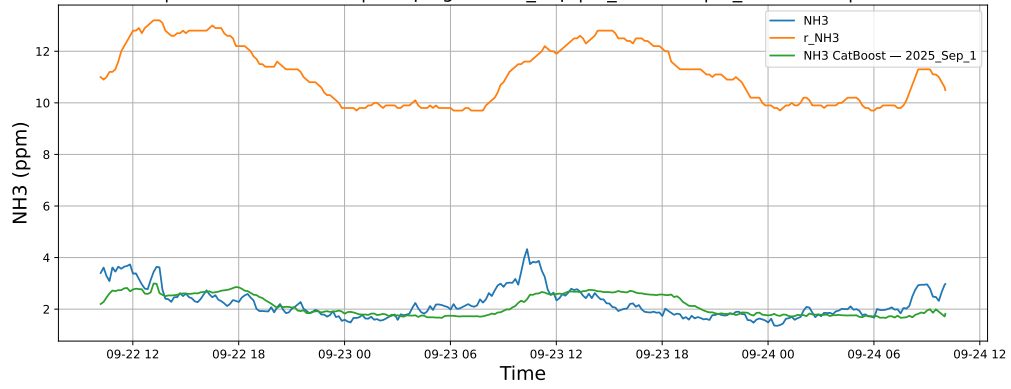
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

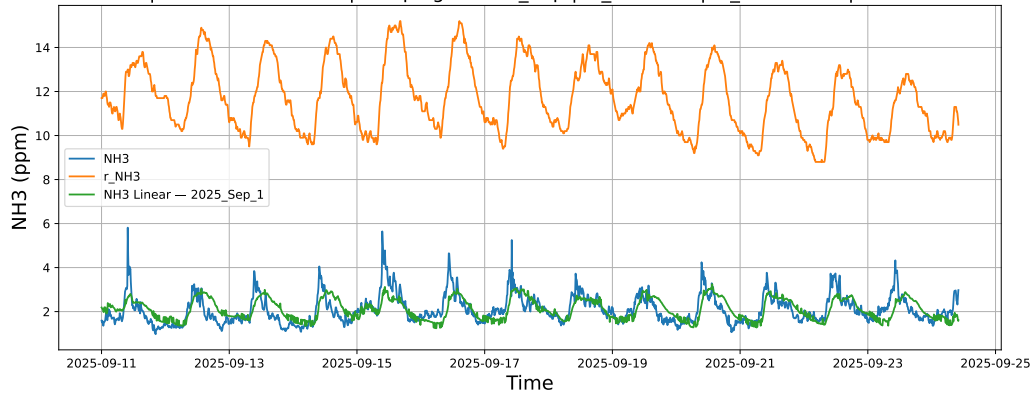


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

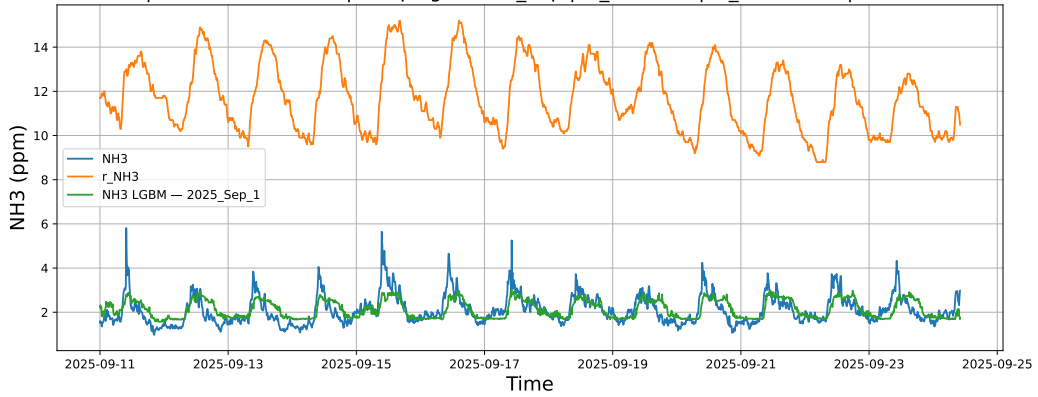


3.1.7.3 Time window : ALL

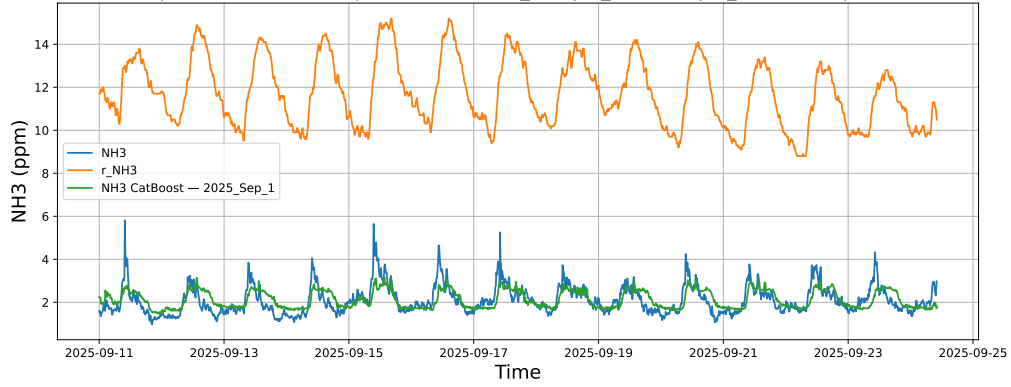
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



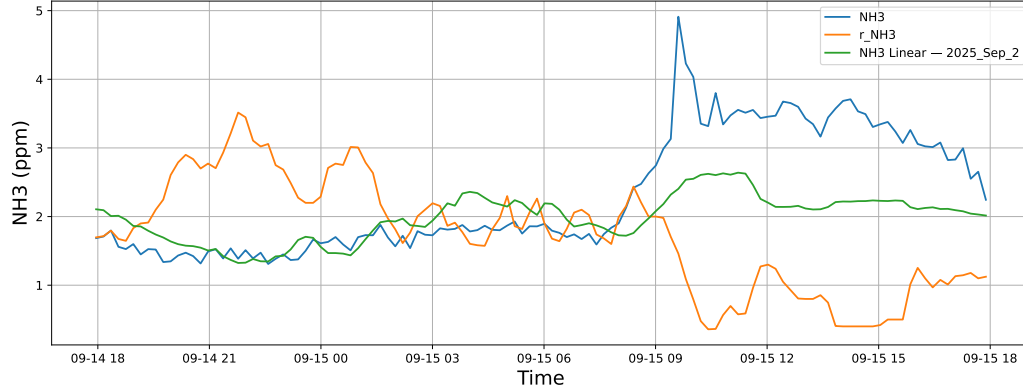
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



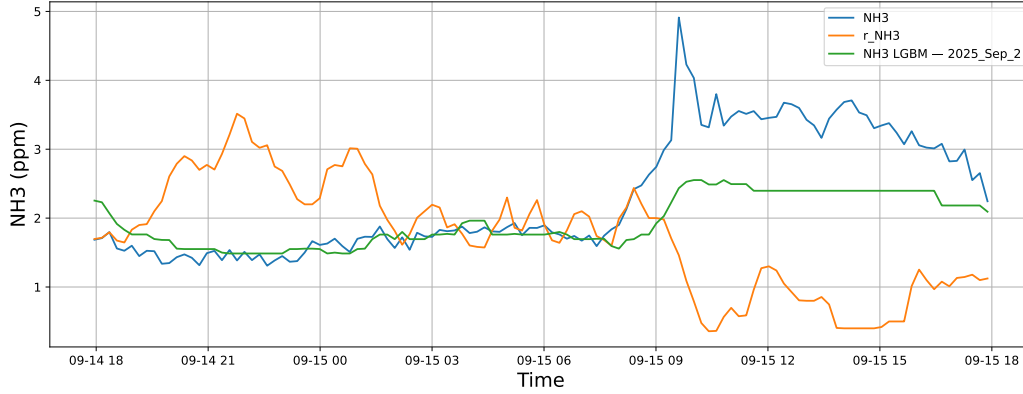
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

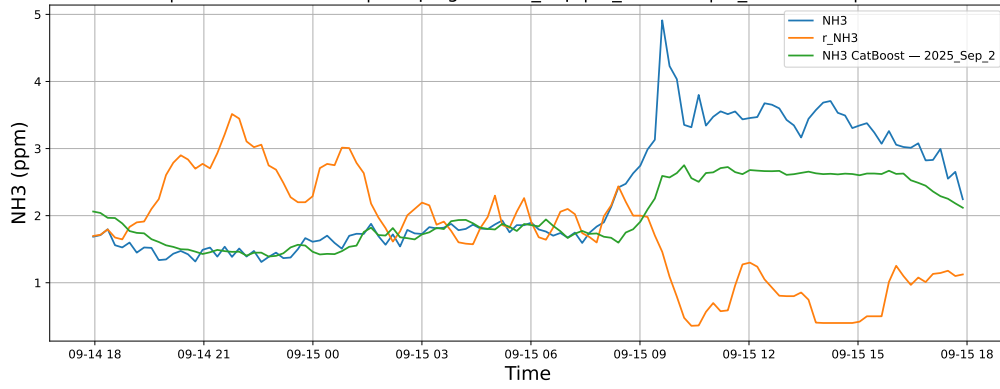
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

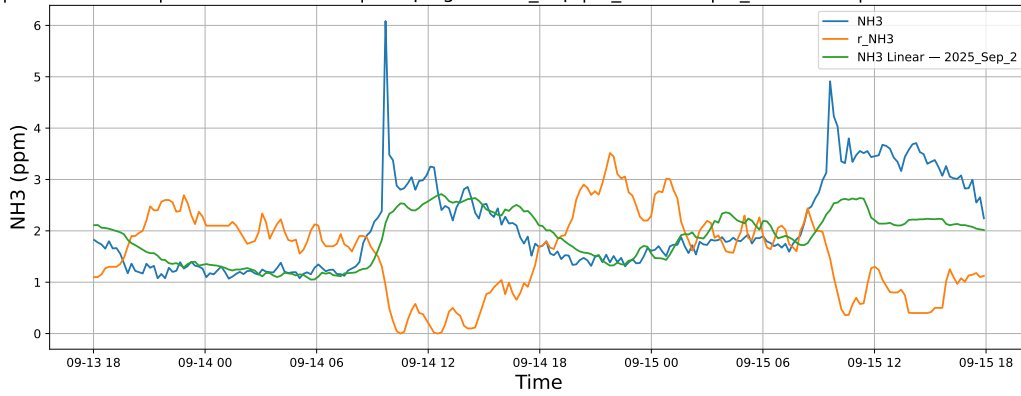


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

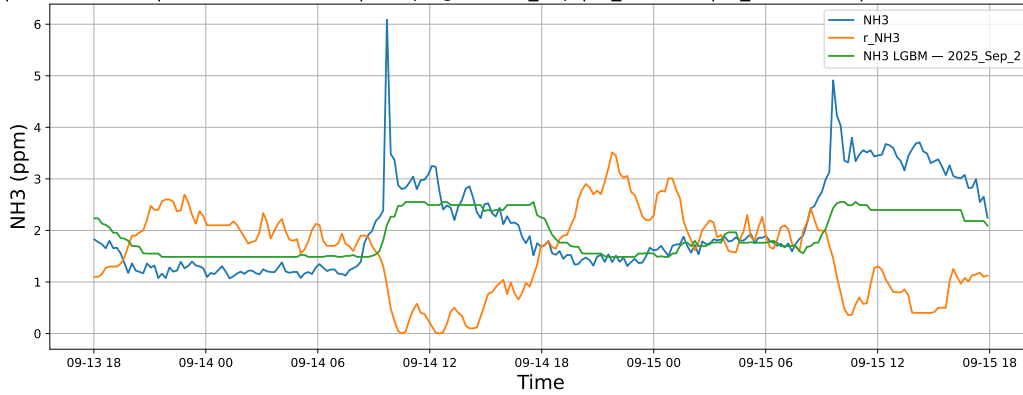


3.1.8.2 Time window : 48

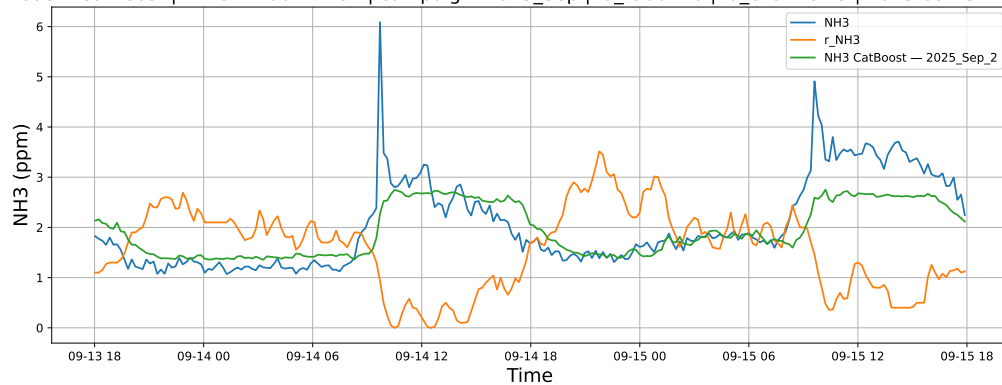
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

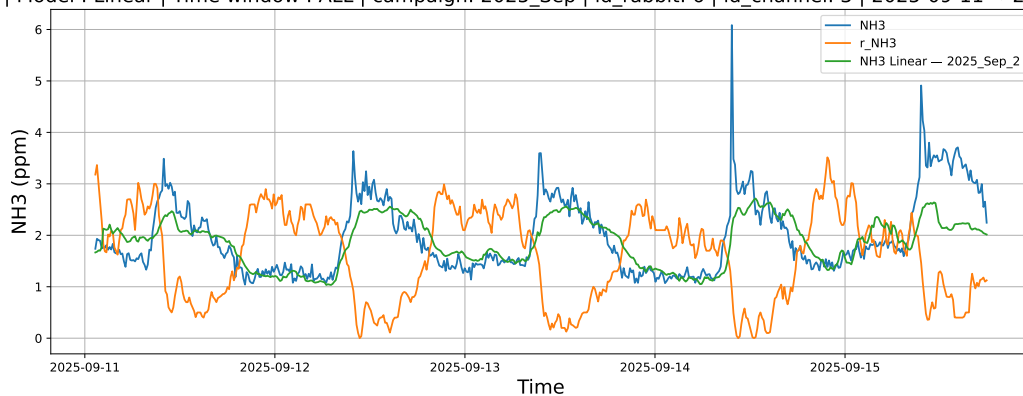


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

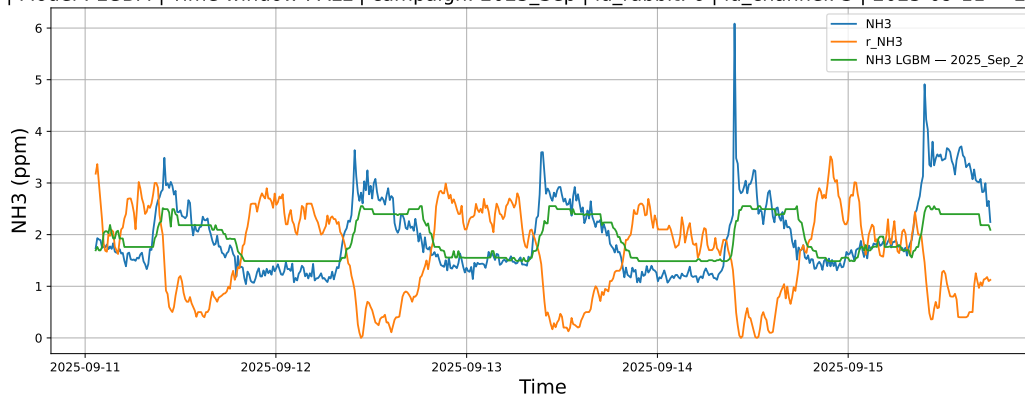


3.1.8.3 Time window : ALL

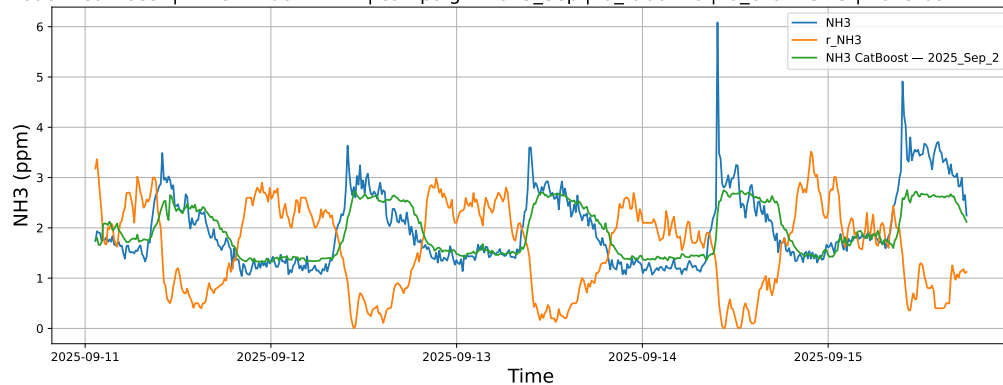
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

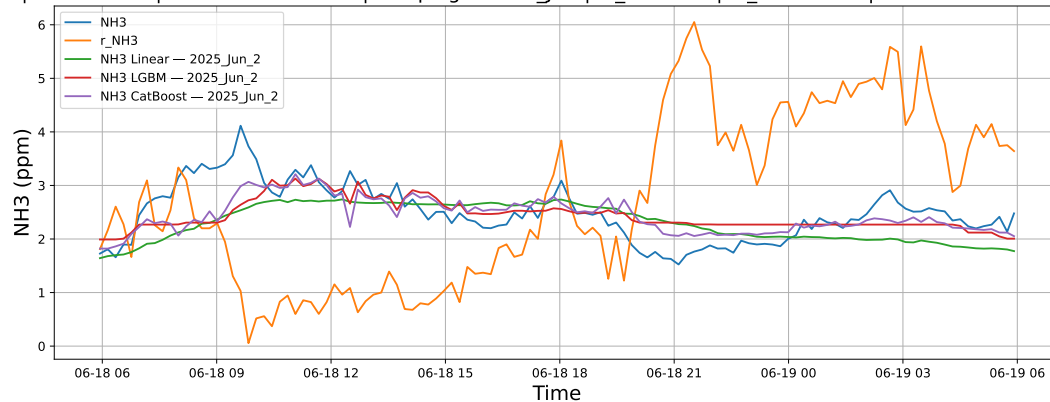


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

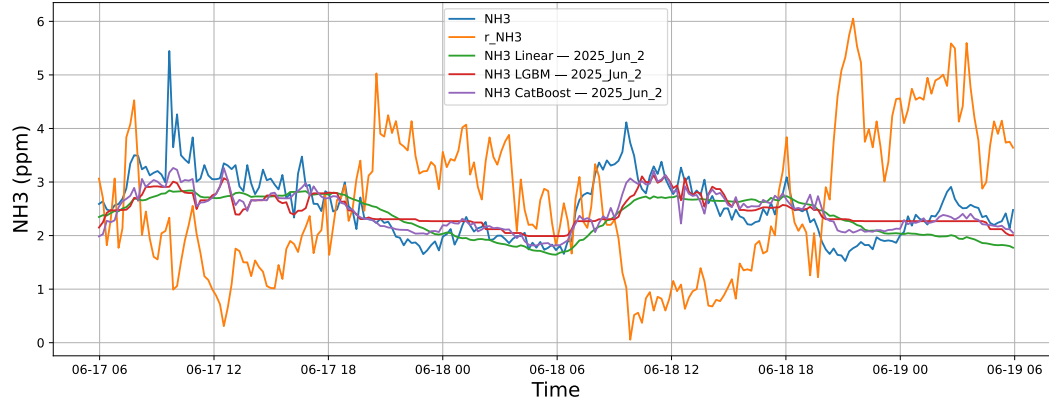
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



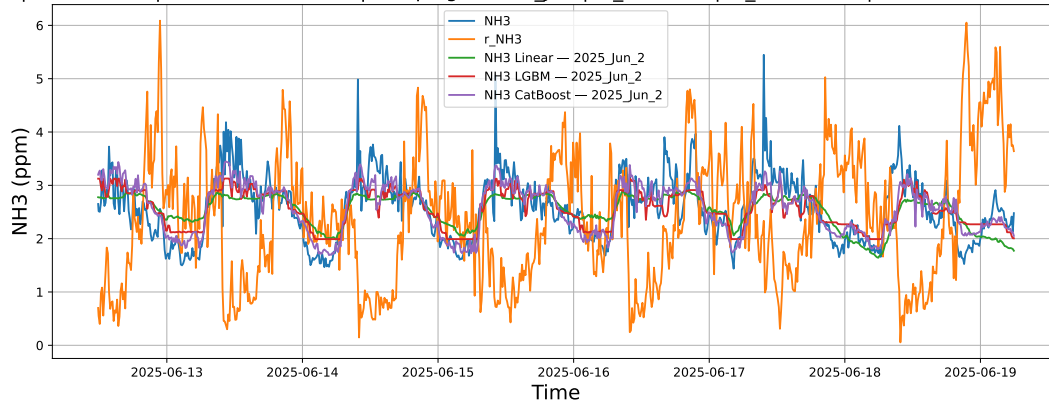
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

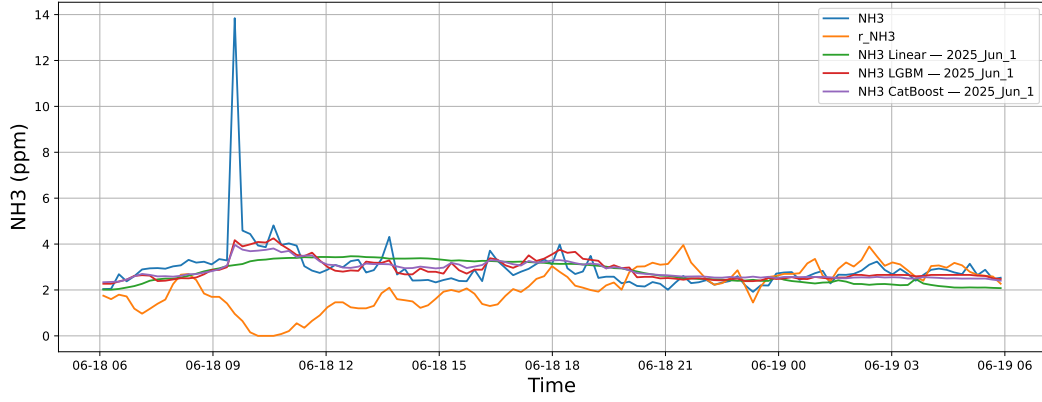
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

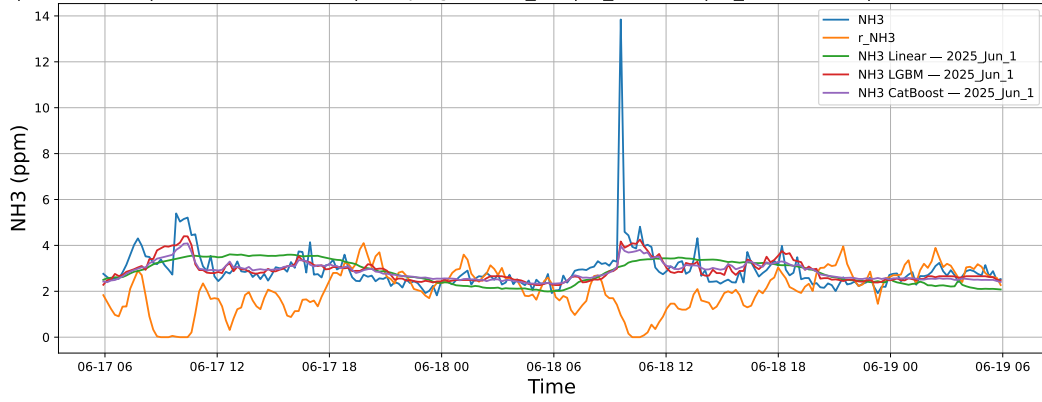
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



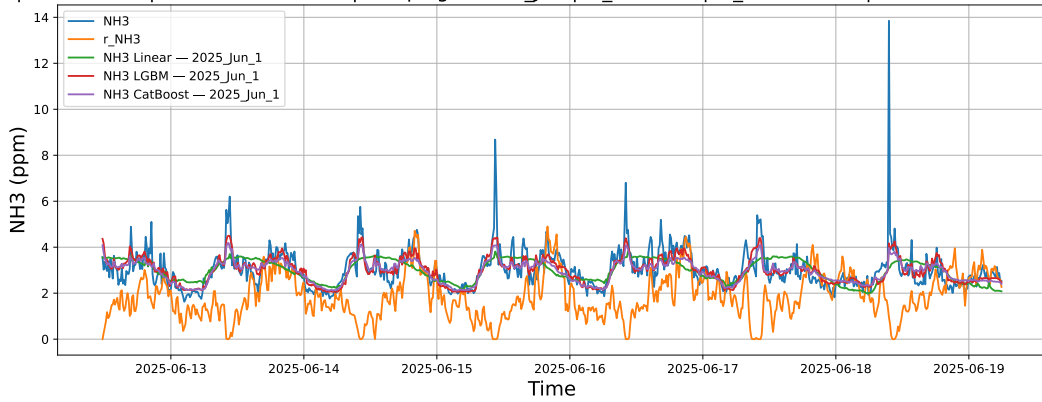
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

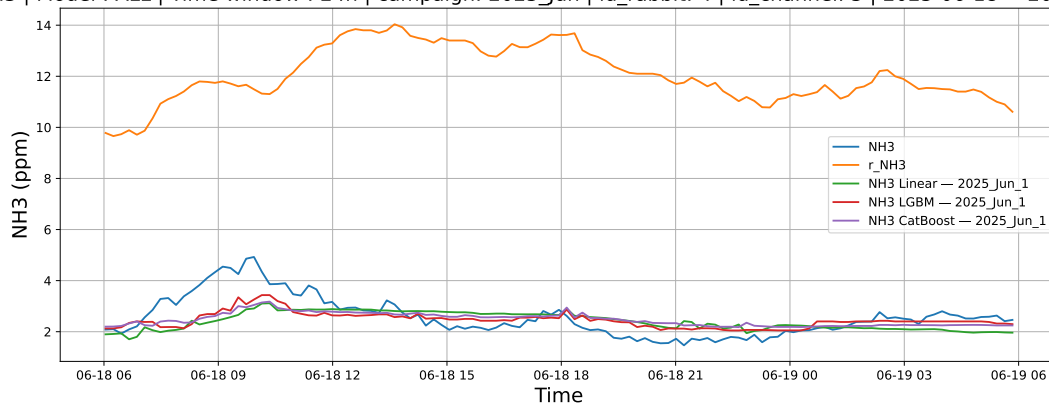
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

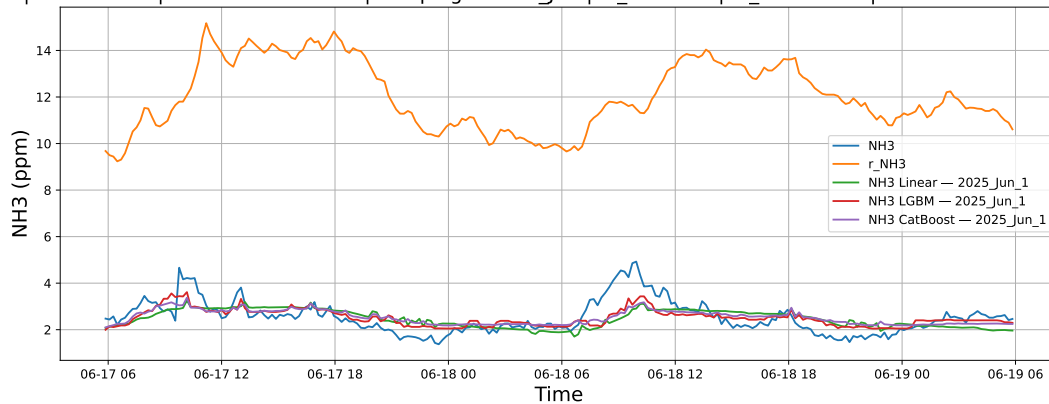
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



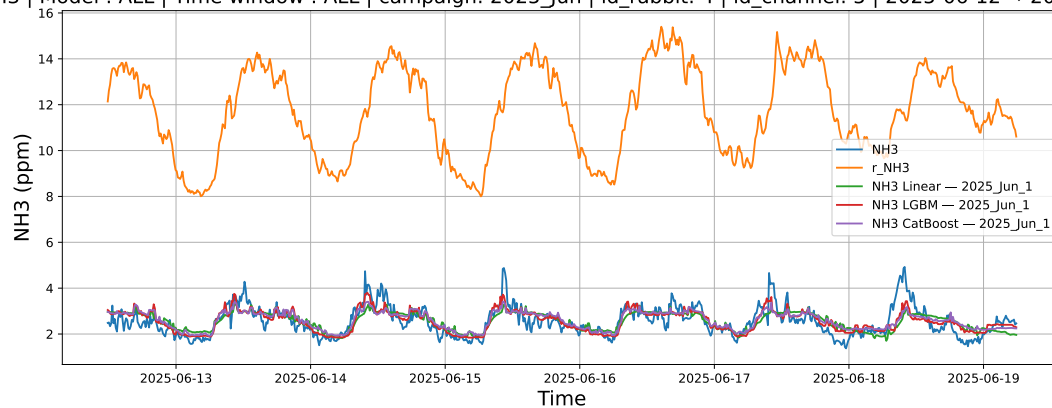
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

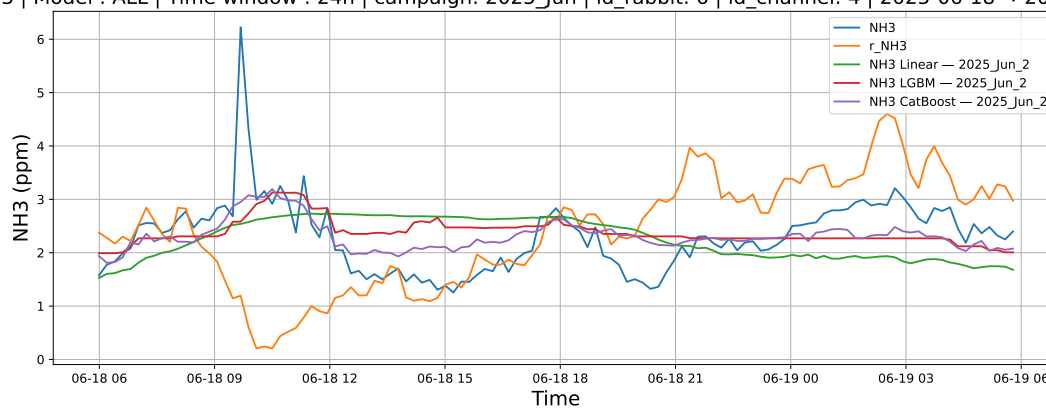
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

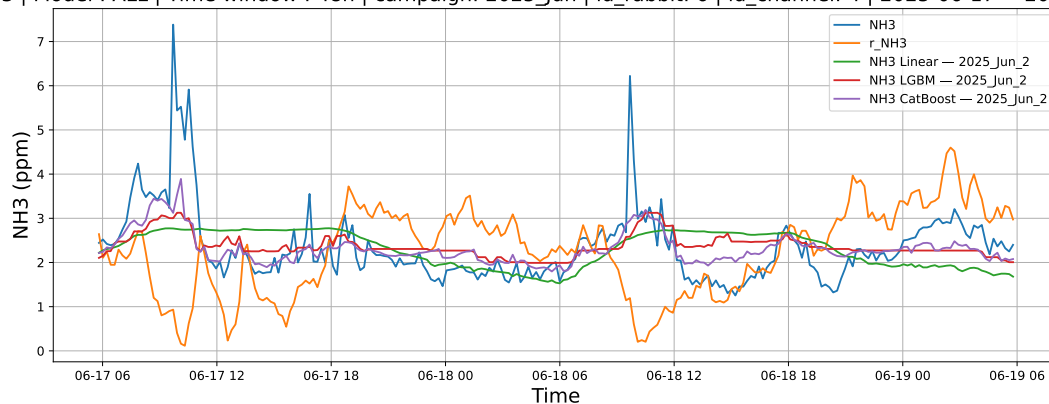
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



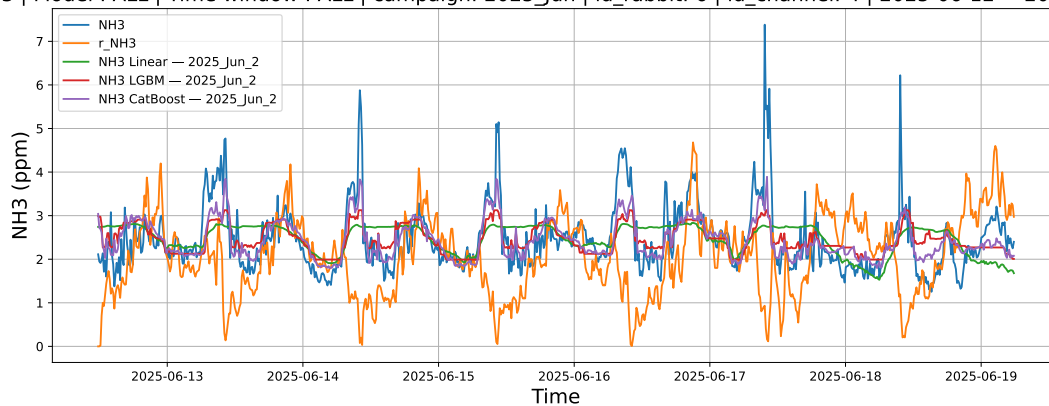
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

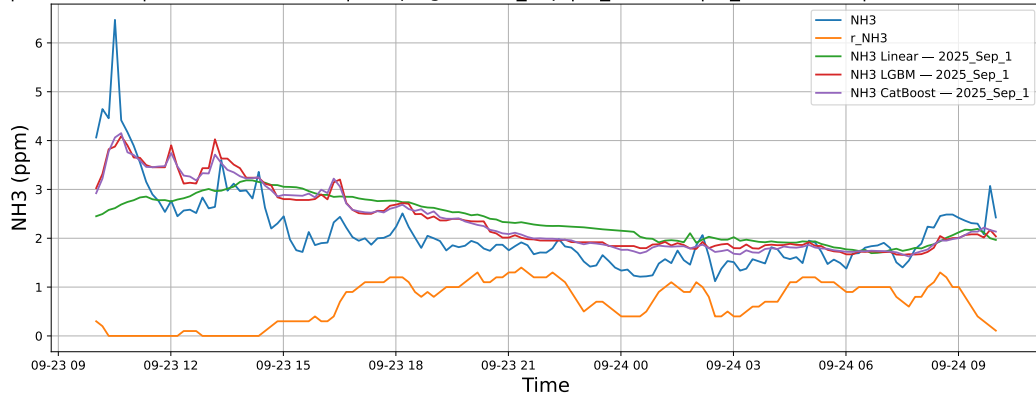
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

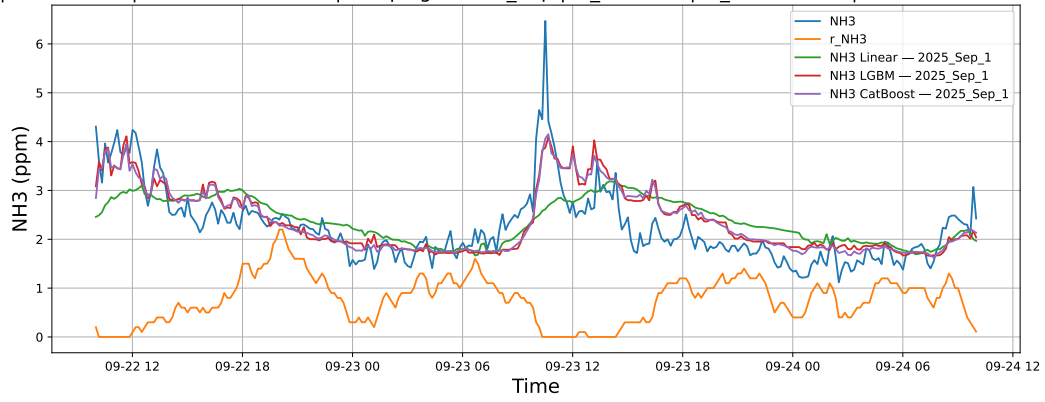
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



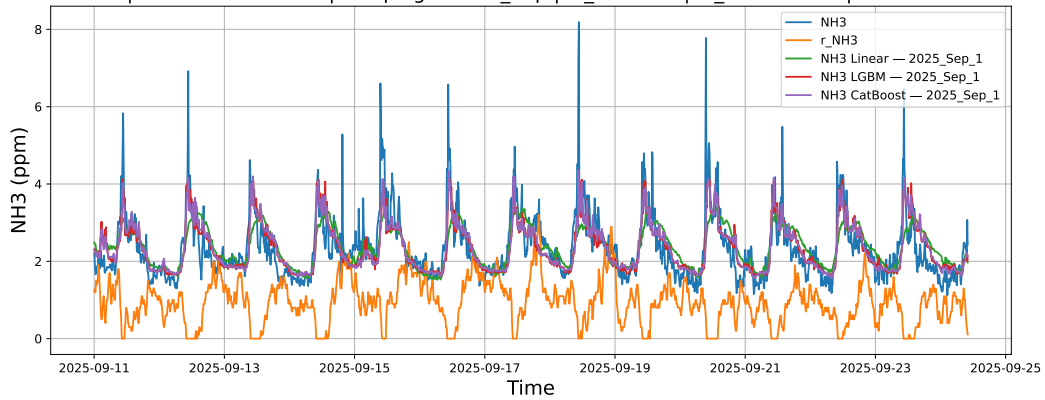
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

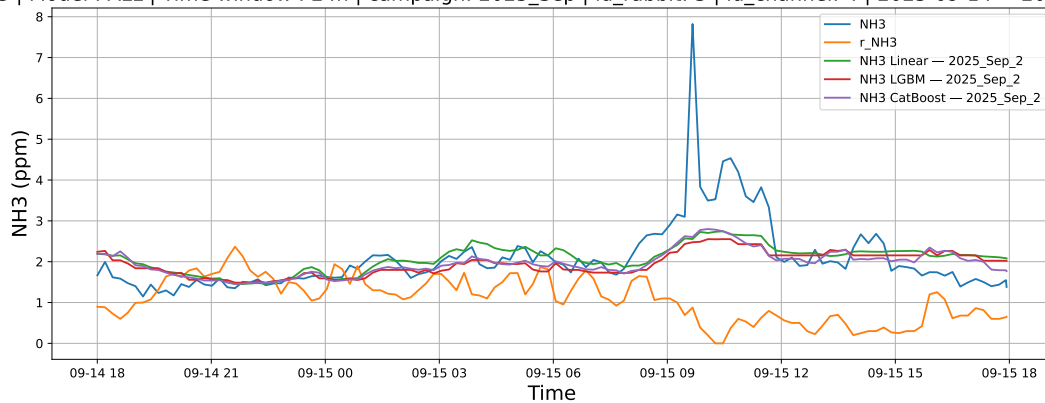
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

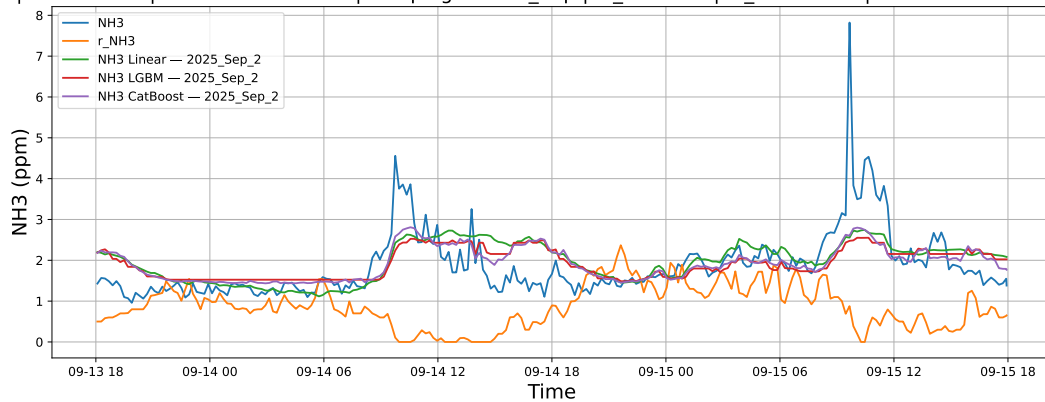
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



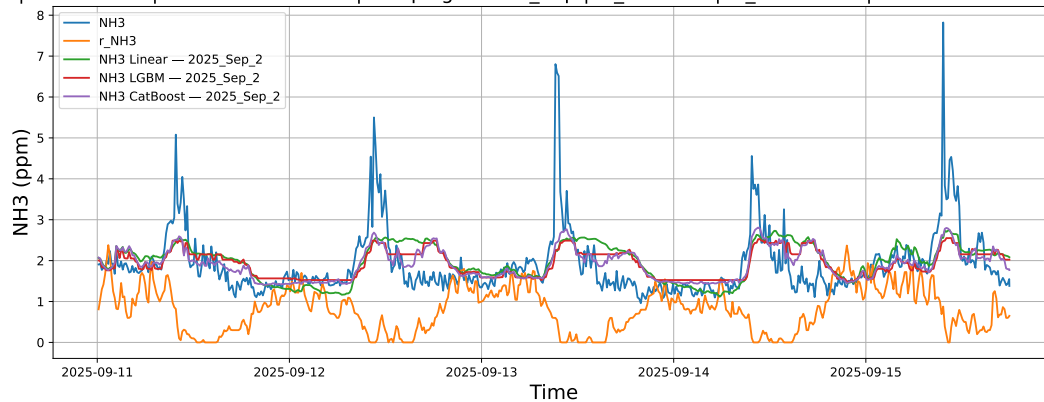
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

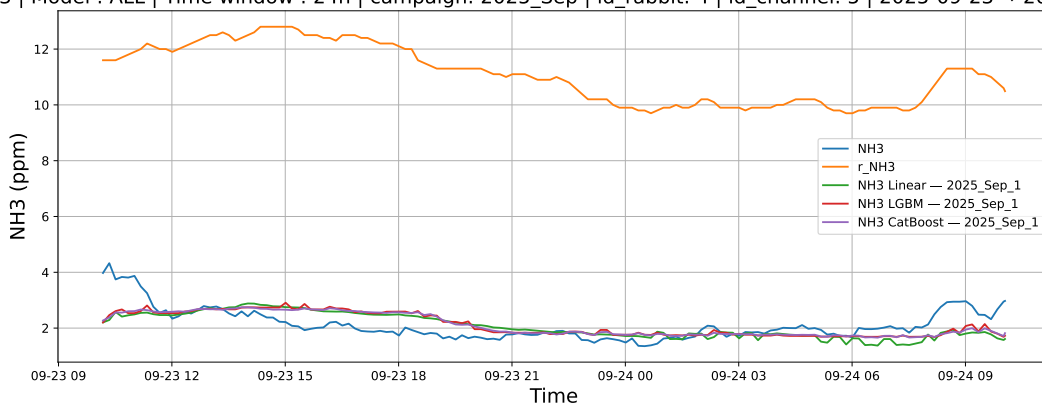
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

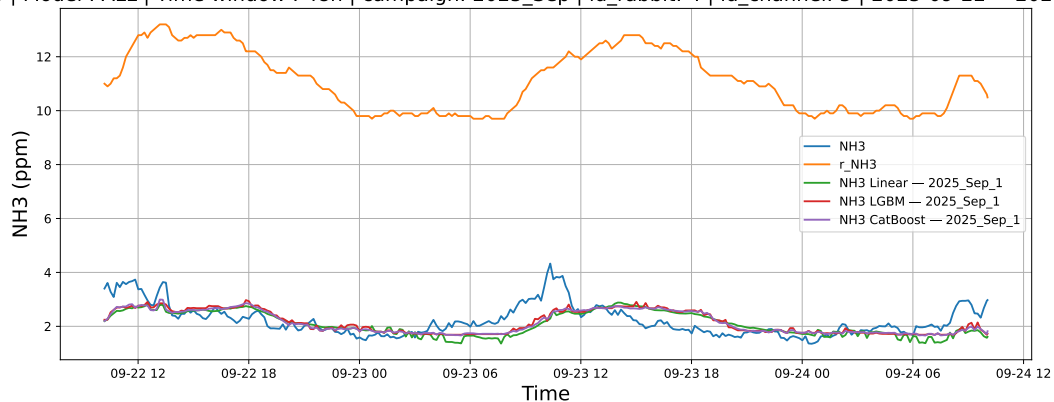
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



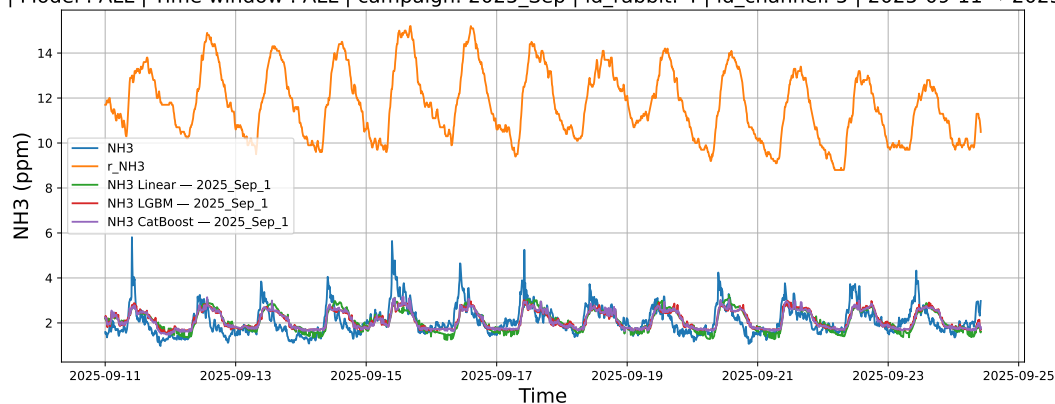
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

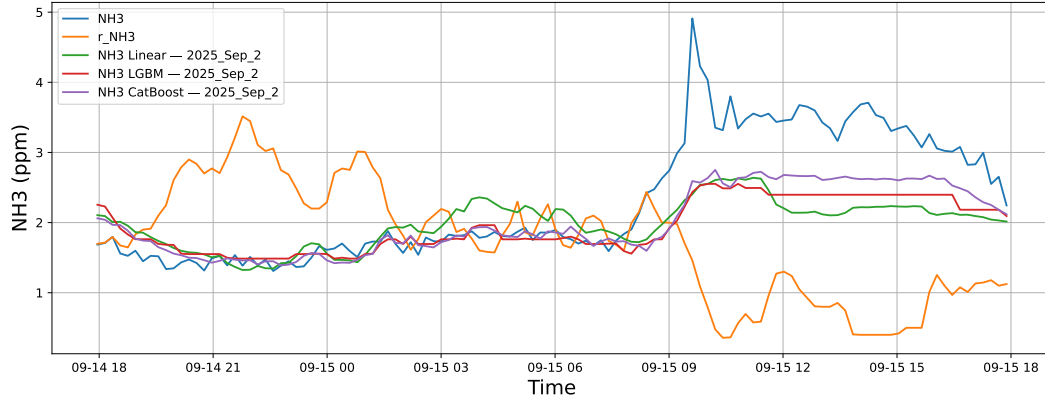
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

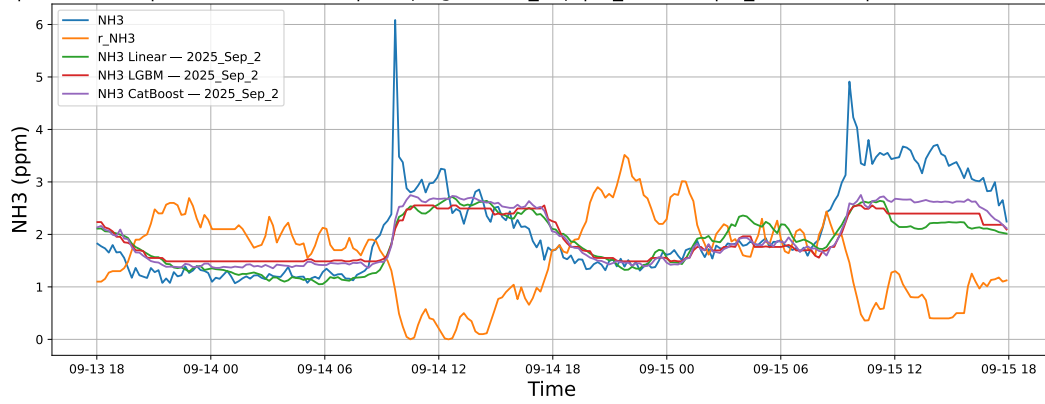
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

